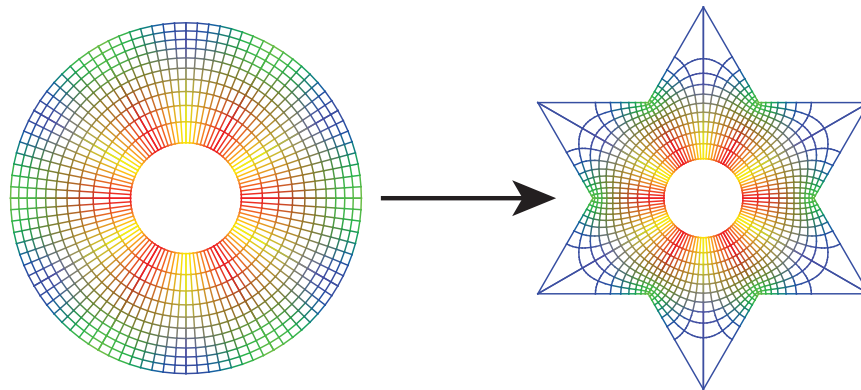


Complex Analysis

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1 Holomorphic Functions

1.1 REVIEW OF COMPLEX NUMBERS

I had an exhausting dream, he said the moment I sat down by him one morning. I dreamed that I was a negative number, and I couldn't figure out my square root.

It's possible, I said. Wait until you learn the imaginary number.

Mommy, I'm not stupid, he said. I know imaginary numbers, but I don't like to deal with that troublesome i .

YIYUN LI, *Where Reasons End*

The set $\mathbb{C}$ of complex numbers consists of all pairs $(a, b) \in \mathbb{R}^2$ with addition defined by

$$(a, b) + (a', b') = (a + a', b + b')$$

and multiplication by

$$(a, b) \cdot (a', b') = (aa' - bb', ab' + a'b) .$$

We set $i = (0, 1)$ and write suggestively

$$a + bi = (a, b) .$$

Then $i^2 = -1$. Moreover, real part, imaginary part, conjugate and norm are defined as

$$\operatorname{Re}(a + bi) = a$$

$$\operatorname{Im}(a + bi) = b$$

$$\overline{a + bi} = a - bi$$

$$|a + bi| = \sqrt{(a + bi)(\overline{a + bi})} = \sqrt{a^2 + b^2}$$

The norm is *multiplicative*

$$|z \cdot z'| = |z| \cdot |z'| ,$$

which allows to compute the reciprocal or multiplicative inverse of a complex number as

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2} .$$

With all this, $\mathbb{C}$ becomes a commutative field, and we can define elementary complex functions:

1. $f(z) = z + a$ is a translation by a .
2. $f(z) = az$ scales z by a with center the origin.
3. $f(z) = e^{it}z$ rotates z by the angle t .

We also have *polynomials*

$$p(z) = a_n z^n + \dots + a_1 z + a_0$$

and rational functions

$$q(z) = \frac{a_m z^m + \dots + a_1 z + a_0}{b_n z^n + \dots + b_1 z + b_0}$$

1.2 HOLOMORPHIC FUNCTIONS

Definition 1.2.1 Let $U \subset \mathbb{C}$ be an open set, and $f : U \rightarrow \mathbb{C}$ a complex-valued function. We say that f is holomorphic at $z_0 \in U$ if

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists. We say that f is holomorphic in U if it is holomorphic at every point of U . The set of all holomorphic functions in U is denoted by $O(U)$.

Constants and the identity function are trivially holomorphic. Our first two goals will be to construct examples and to determine how holomorphic functions are more special than real differentiable functions.

This definition is formally analogous to the definition of the differentiability of a real-valued function at a point $x_0 \in \mathbb{R}$. We therefore expect all elementary properties of the derivative to hold, with literally identical proofs.

In particular:

Theorem 1.2.2 The sum, product, and quotient of holomorphic functions are again complex differentiable whenever defined.

Proof. Clear.



This shows that all polynomials and all rational functions are holomorphic.

Example 1.2.1

1. $f(z) = az$ is a rotation by $\arg a$ and scaling by $|a|$.
2. $f(z) = z + b$ is a translation.
3. $f(z) = 1/z$ is an inversion at the unit circle followed by a reflection at the real axis. It maps lines and circles to lines or circles.

1.3 MÖBIUS TRANSFORMATIONS

An important special class of rational functions is the class of *Möbius transformations*

$$f(z) = \frac{az + b}{cz + d}$$

They compose like 2×2 matrices (as follows without computation by letting 2×2 matrices act on the complex projective line $\mathbb{CP}^1 = \hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$). We also note that

$$f(z) = \frac{c - b}{c - a} \cdot \frac{z - a}{z - b}$$

is the unique Möbius transformation that maps $a \mapsto 0$, $b \mapsto \infty$, and $c \mapsto 1$. Thus Möbius transformations act three-point transitively on $\mathbb{C}$.

Moreover,

$$\begin{aligned} \frac{az + b}{cz + d} &= \frac{\frac{a}{c}(cz + d) + b - \frac{a}{c}d}{cz + d} \\ &= \frac{a}{c} + \frac{b - \frac{a}{c}d}{cz + d} \end{aligned}$$

shows that any Möbius transformation is a composition of translations, scalings, and reciprocals.

An important subset of the Möbius transformations map the unit disk bijectively onto itself. To see this, we need the following

Lemma 1.3.1 *Let $|z| \leq 1$ and $|w| \leq 1$. Then*

$$\left| \frac{z - w}{1 - \bar{w}z} \right| \leq 1$$

with equality if and only if $|z| = 1$ or $|w| = 1$.

Proof. Exercise. ◻

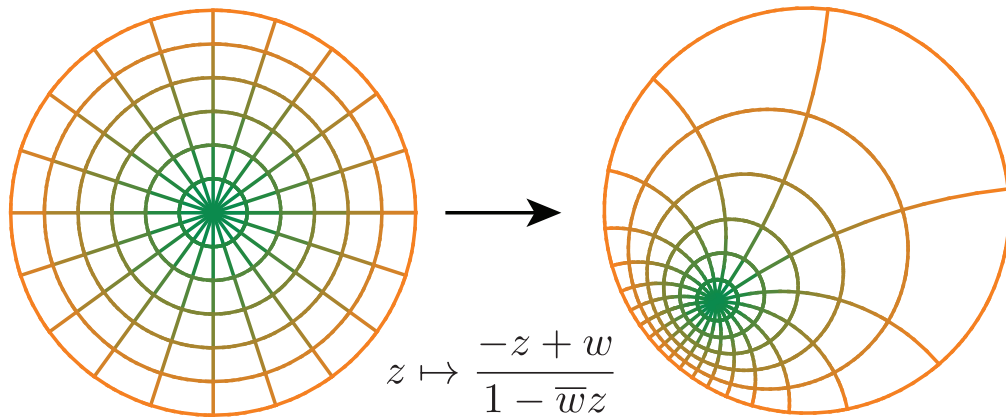


Figure 1.1 Holomorphic automorphism of the disk

Corollary 1.3.2 *For $|w| < 1$, the Möbius transformation*

$$\mu_{w,\alpha} : z \mapsto e^{i\alpha} \frac{z - w}{1 - \bar{w}z}$$

is a biholomorphic map of the unit disk $\mathbb{D}$ with $w \mapsto 0$ and inverse

$$z \mapsto \mu_{e^{i\alpha}\bar{w}, -\alpha}(z) = \frac{z + e^{i\alpha}w}{e^{i\alpha} + \bar{w}z}.$$

Proof. By the lemma, both functions map $\mathbb{D}$ into $\mathbb{D}$, and it is easy to check that their composition is the identity. ◻

1.4 INVERSION

The geometry of the reciprocal $z \mapsto 1/z$ is intimately related to the *inversion* (or *reflection*) at the unit circle

$$I(z) = \frac{z}{|z|^2} = \frac{1}{\bar{z}};$$

they only differ by a conjugation, which is the reflection at the real axis. We claim

Theorem 1.4.1 *The inversion (and thus the reciprocal as well) maps*

1. *lines through 0 to lines through 0;*
2. *lines not through 0 to circles through 0;*
3. *circles not through 0 to circles not through 0.*

Proof. The case of a line through 0 is trivial.

We begin with the special case of the vertical line through 1. By Thales, the angle at q is a right angle. Because the triangles $\Delta(0, 1, q')$ and $\Delta(0, q, 1)$ are similar, we have $|q| \cdot |q'| = 1$ so that q lies on the line $\operatorname{Re} z = 1$ if and only if q' lies on the circle with center at $1/2$ and radius $1/2$.

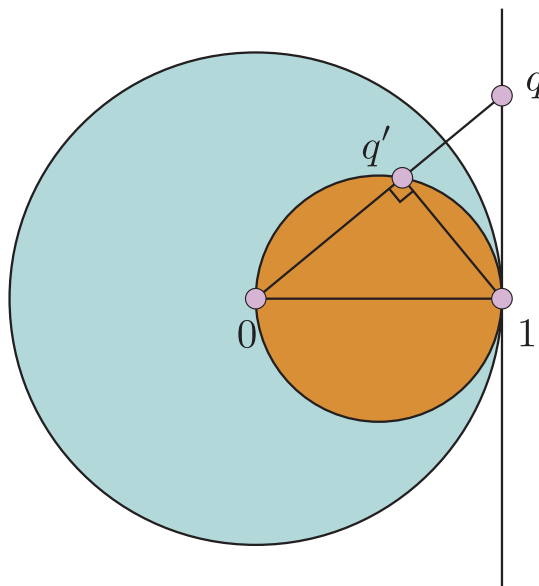


Figure 1.2 Inverting line to circle

Note that this implies that any line that touches the unit circle is mapped by I to a circle touching the unit circle at the same point as the line, as I is equivariant with respect to rotations.

Next we show that the inversion takes any vertical line to a circle. To see this, let $R(z) = r \cdot z$ for a fixed positive real number. Then $I = R \circ I \circ R$. Given a line L not through 0, find r so that $R(L)$ touches the circle. Then $I(L) = R \circ I \circ R(L)$ is indeed a circle, by the previous step.

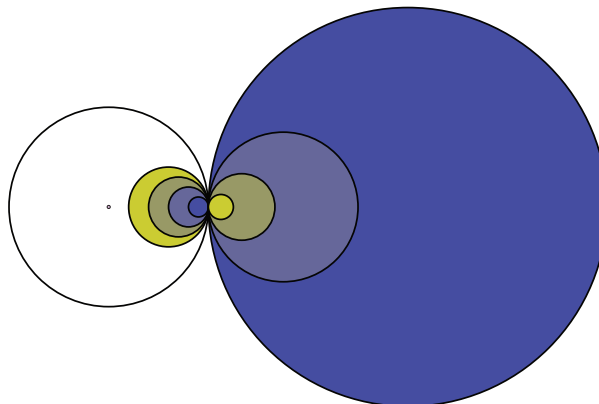


Figure 1.3 Inversion of a line

Now let C be a circle that touches the unit circle in 1. We will argue with $1/z$ for simplicity. The map $z \mapsto 1/(z-1)$ maps C to a line, because it first translates the circle so that it passes through 0, and then inverts. Thus $z \mapsto 1 + 1/(z-1)$ maps C to a line as well, and

$$z \mapsto \frac{1}{1 + \frac{1}{z-1}} = 1 - \frac{1}{z}$$

maps C to a circle. It follows that also $1/z$ maps C to a circle.

To see the last identity, rewrite it as

$$z = \frac{1}{1 - \frac{1}{1 - \frac{1}{1-z}}}.$$

As Möbius transformations are determined by the values at three distinct points and

$$\phi(z) = \frac{1}{1-z}.$$

cyclically permutes $0 \mapsto 1 \mapsto \infty \mapsto 0$, we have $\phi^{(3)}(z) = z$, which is the claimed identity.



1.5 CHAIN RULES

We note that all Möbius transformations preserve angles. Let's prove this in general.

Lemma 1.5.1 *Let f be holomorphic in a domain U , and let $\gamma : (0, 1) \rightarrow U$ be a differentiable curve. Then the image curve $f(\gamma(t))$ has tangent vectors $f'(\gamma(t)) \cdot \gamma'(t)$. Note that the product here is a complex multiplication.*

Proof. First suppose that $\gamma'(t_0) \neq 0$. Then near t_0 also $\gamma'(t) \neq 0$, and we can compute

$$\begin{aligned} \lim_{t \rightarrow t_0} \frac{f(\gamma(t_0)) - f(\gamma(t))}{t_0 - t} &= \lim_{t \rightarrow t_0} \frac{f(\gamma(t_0)) - f(\gamma(t))}{\gamma(t_0) - \gamma(t)} \cdot \frac{\gamma(t_0) - \gamma(t)}{t_0 - t} \\ &= f'(\gamma(t_0)) \cdot \gamma'(t_0) \end{aligned}$$

Next suppose that $\gamma'(t_0) = 0$. As f is differentiable at $w_0 = \gamma(t_0)$, the difference quotient

$$\left| \frac{f(w_0) - f(w)}{w_0 - w} \right| \leq C$$

near w_0 for some constant C . Thus in this case

$$\begin{aligned} \left| \frac{f(\gamma(t_0)) - f(\gamma(t))}{t_0 - t} \right| &\leq C \left| \frac{\gamma(t_0) - \gamma(t)}{t_0 - t} \right| \\ &\rightarrow 0 \quad \text{for } t \rightarrow t_0 \end{aligned}$$

as was to be shown. 

Corollary 1.5.2 *Let f be holomorphic at z_0 with $f'(z_0) \neq 0$. Then f preserves the angles between curves that pass through z_0 .*

Corollary 1.5.3 *Let g be holomorphic at z_0 , and f holomorphic at $g(z_0)$. Then $f \circ g$ is holomorphic at z_0 , and the chain rule*

$$(f \circ g)'(z_0) = f'(g(z_0)) \cdot g'(z_0)$$

holds.

Proof. Same proof! 

1.6 CAUCHY-RIEMANN EQUATIONS

Lemma 1.6.1 (Cauchy-Riemann Equations) *Let*

$$f(x + iy) = u(x, y) + iv(x, y)$$

be complex differentiable. Then

$$u_x = v_y \quad \text{and} \quad u_y = -v_x .$$

Proof. Let $h \in \mathbb{R}$. Taking the limit $h \rightarrow 0$ in

$$\frac{f(z + h) - f(z)}{h} = \frac{u(x + h, y) - u(x, y)}{h} + i \frac{v(x + h, y) - v(x, y)}{h}$$

shows that

$$f'(z) = u_x(x, y) + iv_x(x, y) .$$

On the other hand, taking the limit $h \rightarrow 0$ in

$$\frac{f(z + ih) - f(z)}{ih} = \frac{v(x, y + h) - v(x, y)}{h} - i \frac{u(x, y + h) - u(x, y)}{h}$$

shows that

$$f'(z) = v_y(x, y) - iu_y(x, y) .$$

Comparing both right-hand sides proves the claim. 

These two equations are called the *Cauchy-Riemann equations*.

Note that if f' is continuous, so are u_x , u_y , v_x , and v_y .

We now prove a converse:

Lemma 1.6.2 Suppose that $u(x, y)$ and $v(x, y)$ are two functions in a domain U that satisfy the **Cauchy-Riemann equations 1.6.1** and such that the map

$$(x, y) \mapsto (u(x, y), v(x, y))$$

is (real) differentiable. Then $f(x + iy) = u(x, y) + iv(x, y)$ is holomorphic.

Proof. Write

$$u(x + h, y + k) = u(x, y) + u_x h + u_y k + R(h, k)$$

$$v(x + h, y + k) = v(x, y) + v_x h + v_y k + S(h, k)$$

with real valued functions R and S . Write $\Delta = h + ik$. By assumption,

$$\lim_{\Delta \rightarrow 0} \frac{R(h, k)}{|\Delta|} = 0 \quad \text{and} \quad \lim_{\Delta \rightarrow 0} \frac{S(h, k)}{|\Delta|} = 0 .$$

Thus

$$\begin{aligned} f(x + \Delta) &= f(z) + u_x h + iv_x h + u_y k + iv_y k + R + iS \\ &= f(z) + (u_x + iv_x) \Delta + R + iS , \end{aligned}$$

so that f' exists. 

We can now see clearly that the derivatives of holomorphic functions are rather special:

$$df = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

by the **Cauchy-Riemann equations 1.6.1**. The latter matrix is a multiple of a rotation matrix. We also note in passing a useful formula for the Jacobian of f :

$$\det df|_z = |f'(z)|^2 .$$

Corollary 1.6.3 Let f be continuously differentiable with $f'(z_0) \neq 0$. Then f is near z_0 invertible, and $g = f^{-1}$ is holomorphic near $f(z_0)$ with

$$g'(w) = \frac{1}{f'(g(w))} .$$

Proof. To see this, we use the fact that df is invertible at z_0 and apply the (real) inverse function theorem to obtain an inverse g of f defined near $f(z_0)$. Then g is differentiable and satisfies the **Cauchy-Riemann equations 1.6.1**, hence is holomorphic. The formula follows from the chain rule. 

Exercise 1.6.4 Let $f \in O(\mathbb{D})$ and assume that $|f|$ is constant. Show that f is constant.

1.7 PATH INTEGRALS

An important tool in analysis to construct new functions is integration. For instance, the logarithm can be introduced as an antiderivative of $1/x$, and the exponential function as its inverse. Our goal is to introduce an integral for holomorphic functions that serves as an antiderivative.

Definition 1.7.1 Let $\gamma : [a, b] \rightarrow U$ be a piecewise continuously differentiable curve, and f holomorphic in U . Define the path integral

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \cdot \gamma'(t) dt .$$

A motivation behind this definition is that the right-hand side is the limit of complex Riemann sums

$$\sum_{k=1}^N f(\zeta_k) (z_k - z_{k-1})$$

for a partition $a = t_0 < \dots < t_N = b$ of $[a, b]$, setting $z_k = \gamma(t_k)$ and $\zeta_k = \gamma(\tau_k)$ for a choice of intermediate points $\tau_k \in [t_{k-1}, t_k]$. We first verify that this integral depends only on the trace of the path:

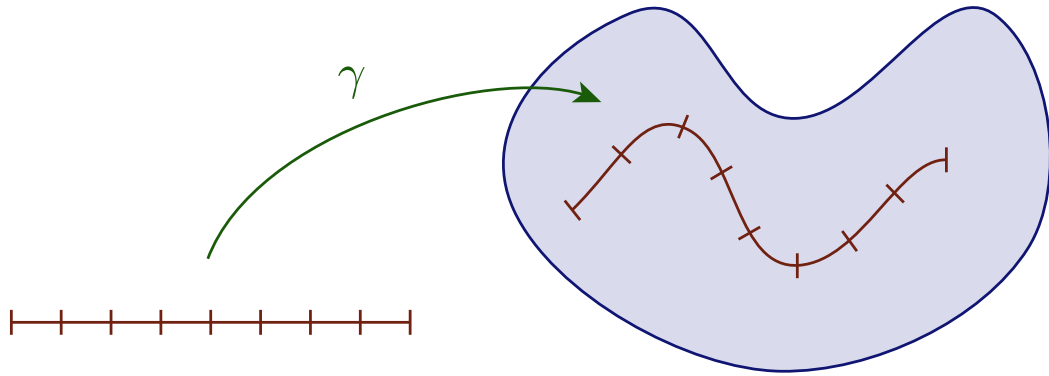


Figure 1.4 Path Integrals

Lemma 1.7.2 Let $\phi : [\tilde{a}, \tilde{b}] \rightarrow [a, b]$ be continuously differentiable, and $\tilde{\gamma} = \gamma \circ \phi$. Then

$$\int_{\gamma} f(z) dz = \int_{\tilde{\gamma}} f(z) dz .$$

Proof. We use the complex chain rule and the real transformation formula:

$$\begin{aligned}
\int_{\tilde{\gamma}} f(z) dz &= \int_{\tilde{a}}^{\tilde{b}} f(\tilde{\gamma}(t)) \cdot \tilde{\gamma}'(t) dt \\
&= \int_{\tilde{a}}^{\tilde{b}} f(\gamma(\phi(t))) \cdot \gamma'(\phi(t)) \phi'(t) dt \\
&= \int_a^b f(\gamma(u)) \cdot \gamma'(u) du \\
&= \int_{\gamma} f(z) dz .
\end{aligned}$$

Example 1.7.1 For $k \neq -1$,

$$\begin{aligned}
\int_{\partial \mathbb{D}} z^k dz &= \int_0^{2\pi} e^{kit} i e^{it} dt \\
&= i \int_0^{2\pi} e^{(k+1)it} dt \\
&= \frac{1}{k+1} e^{(k+1)it} \Big|_0^{2\pi} dt \\
&= 0 .
\end{aligned}$$

On the other hand, for $k = -1$ we get

$$\begin{aligned}
\int_{\partial \mathbb{D}} \frac{dz}{z} &= \int_0^{2\pi} i dt \\
&= 2\pi i .
\end{aligned}$$

More generally: Suppose f has a holomorphic antiderivative F in U , i.e. $F'(z) = f(z)$. Suppose that $\gamma : [a, b] \rightarrow U$ is a path. Then

$$\int_{\gamma} f(z) dz = F(b) - F(a) .$$

While this is expected from the real analogy, it is also surprising because it shows that the integral of a function with an antiderivative depends only on the endpoints of the path, not on the path itself. We are aiming to show this for general holomorphic functions without assuming the existence of an antiderivative. At the moment, we see that $1/z$ has no antiderivative in any domain containing the unit circle.

We will frequently need to estimate path integrals. A simple but essential tool is the following lemma:

Lemma 1.7.3 *Let f be continuous along a curve $\gamma : [a, b] \rightarrow \mathbb{C}$, and assume that $|f(\gamma(t))| \leq M$ for all $t \in [a, b]$. Then*

$$\left| \int_{\gamma} f(z) dz \right| \leq M \cdot \text{length}(\gamma) .$$

Proof.

$$\begin{aligned}
\left| \int_{\gamma} f(z) dz \right| &= \left| \int_0^1 f(\gamma(t)) \cdot \gamma'(t) dt \right| \\
&\leq \int_0^1 |f(\gamma(t))| \cdot |\gamma'(t)| dt \\
&\leq M \int_0^1 |\gamma'(t)| dt \\
&\leq M \cdot \text{length}(\gamma)
\end{aligned}$$

1.8 HOMOTOPY

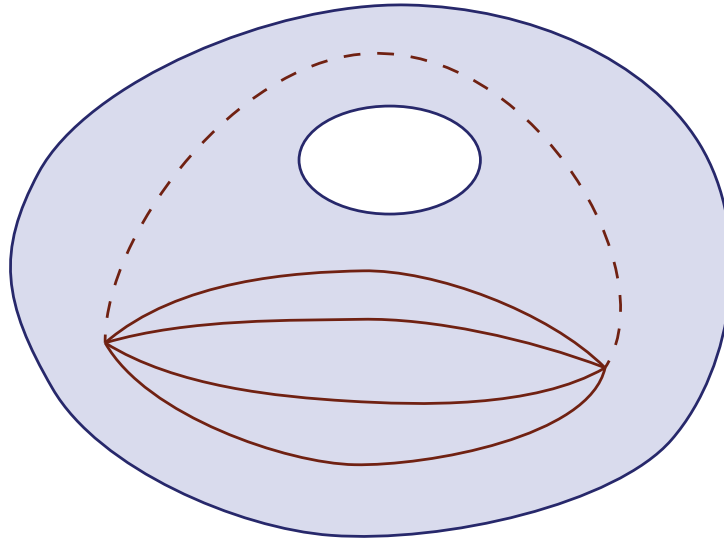


Figure 1.5 Homotopy

Definition 1.8.1 A homotopy of curves γ_t in U is a continuous map $\gamma : [a, b] \times [0, 1] \rightarrow U$ with $\gamma_t(s) = \gamma(s, t)$. If $\gamma(s, 0)$ and $\gamma(s, 1)$ are constant, the homotopy is said to fix endpoints.

We would like to show that for a homotopy γ_t with fixed endpoints, the integrals $\int_{\gamma_t} f(z) dz$ are constant in t .

Theorem 1.8.2 Let U be a domain, f holomorphic in U with a continuous derivative, and γ_t a continuously differentiable homotopy of curves in U . Assume that either the endpoints are fixed, or the curves γ_t are closed. Then

$$\frac{d}{dt} \int_{\gamma_t} f(z) dz = 0.$$

Proof. The key step in the computation below is that we can exchange second derivatives:

$$\begin{aligned}
\frac{d}{dt} \int_{\gamma_t} f(z) dz &= \frac{d}{dt} \int_a^b f(\gamma_t(s)) \gamma'_t(s) ds \\
&= \int_a^b f'(\gamma_t(s)) \dot{\gamma}_t(s) \gamma'_t(s) + f(\gamma_t(s)) \dot{\gamma}'_t(s) ds \\
&= \int_a^b \frac{d}{ds} (f(\gamma_t(s)) \dot{\gamma}_t(s)) ds \\
&= f(\gamma_t(s)) \dot{\gamma}_t(s) \Big|_{s=a}^b \\
&= 0
\end{aligned}$$

We note that we need rather strong assumptions on f . We will soon see that this is not really necessary.

1.9 ANTIDERIVATIVES

Now we can construct new functions by integration. For instance, if f is holomorphic (with continuous derivative) in a star-shaped domain, we can define $F(z) = \int^z f(w) dw$. Our next goal will be to show that F is also holomorphic.

Lemma 1.9.1 *Let f be continuous in U , and assume that*

$$\int_{\gamma} f(z) dz = 0$$

for all closed curves in U . Then f has an antiderivative in U .

Proof. By assumption, the function $F(z) = \int^z f(w) dw$ is well defined. We need to show that it is holomorphic, and its derivative is $f(z)$. Let z be close to z_0 . Then

$$\begin{aligned}
\frac{F(z) - F(z_0)}{z - z_0} &= \frac{1}{z - z_0} \int_{z_0}^z f(w) dw \\
&= \frac{1}{z - z_0} \int_0^1 f(z_0 + t(z - z_0)) \cdot (z - z_0) dt \\
&= \int_0^1 f(z_0 + t(z - z_0)) dt.
\end{aligned}$$

Taking the limit $z \rightarrow z_0$ implies $F'(z_0) = f(z_0)$. 

Note that this lemma still holds if we assume that U is star-shaped with respect to $a \in U$, and $\int_{\gamma} f(z) dz = 0$ for all triangular paths γ in U that have one vertex at a . Thus:

Corollary 1.9.2 *Let U be star-shaped, and f continuous in U . Assume that $\int_{\gamma} f(z) dz = 0$ for all triangular paths γ in U . Then f has an antiderivative in U .*

The example of $f(z) = 1/z$ in any domain containing 0 shows that some assumption about U is needed to guarantee the existence of an antiderivative.

1.10 GOURSAT'S THEOREM

We will now remove the assumption about the continuity of the derivative and the path independence of the integral of holomorphic functions:

Theorem 1.10.1 (Goursat) *Let f be holomorphic in U , $\Delta \subset U$ a triangle. Then*

$$\int_{\partial\Delta} f(z) dz = 0$$

Proof. We already know that if f has an antiderivative, the claim is true for any closed curve in U . The first idea is to write

$$f(z) = f(z_0) + (z - z_0) f'(z_0) + (z - z_0) g(z)$$

where g is continuous in U with $g(z_0) = 0$. By the remark above, we just need to show that

$$\int_{\partial\Delta} (z - z_0) g(z) dz = 0,$$

or

$$\left| \int_{\partial\Delta} (z - z_0) g(z) dz \right| < \epsilon$$

for any given $\epsilon > 0$. Note that this gets easier to prove if the triangles are smaller and closer to z_0 . This suggests that we subdivide Δ into four (congruent) smaller triangles Δ_1^k as indicated.

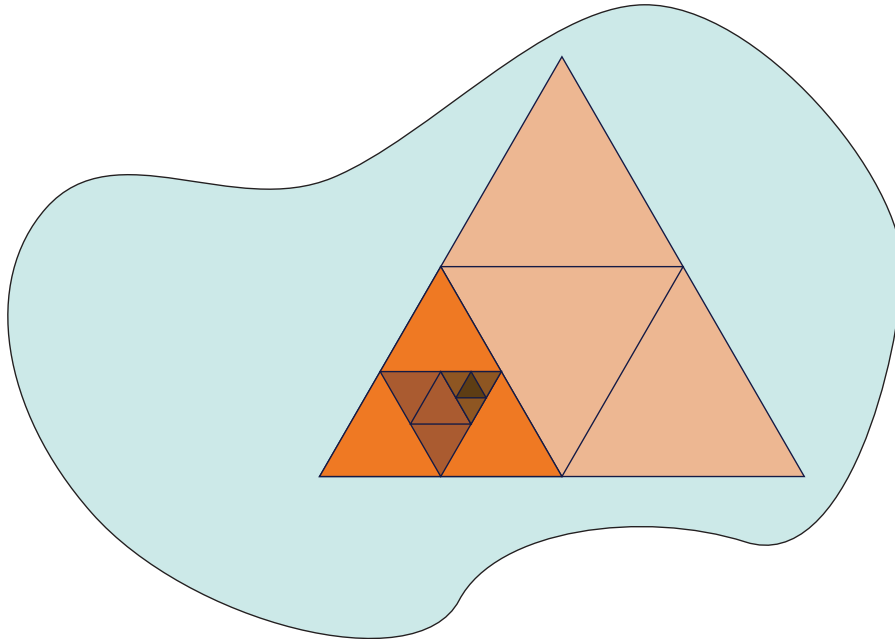


Figure 1.6 Divide and Conquer

Then

$$\begin{aligned}
 \left| \int_{\partial\Delta} f(z) dz \right| &= \left| \sum_{k=1}^4 \int_{\partial\Delta_1^k} f(z) dz \right| \\
 &\leq 4 \max_k \left| \int_{\partial\Delta_1^k} f(z) dz \right| \\
 &= 4 \left| \int_{\partial\Delta_1} f(z) dz \right|,
 \end{aligned}$$

where we denote by Δ_1 the triangle where the integral is maximal. We now continue this subdivision repeatedly, finding a nested sequence of triangles

$$\Delta \supset \Delta_1 \supset \dots \supset \Delta_n \supset \dots$$

with

$$\bigcap_n \Delta_n = \{z_0\},$$

which defines our choice of z_0 . Note that

$$\text{length}(\partial\Delta_n) = \frac{1}{2^n} \text{length}(\partial\Delta)$$

and

$$\text{diam}(\Delta_n) = \frac{1}{2^n} \text{diam}(\Delta).$$

We continue now our estimate above by

$$\begin{aligned}
 \left| \int_{\partial\Delta} f(z) dz \right| &\leq 4 \left| \int_{\partial\Delta_1} f(z) dz \right| \\
 &\leq \dots \leq 4^n \left| \int_{\partial\Delta_n} (z - z_0) g(z) dz \right| \\
 &\leq 4^n \text{length}(\partial\Delta_n) \text{diam}(\Delta_n) \sup_{z \in \Delta_n} |g(z)| \\
 &\leq \text{length}(\partial\Delta) \text{diam}(\Delta) \sup_{z \in \Delta_n} |g(z)|
 \end{aligned}$$

But $\sup_{z \in \Delta_n} |g(z)|$ converges to $g(z_0) = 0$, as the triangles shrink to z_0 and g is continuous. 

We can immediately strengthen **Goursat's theorem 1.10.1**:

Corollary 1.10.2 *Let f be holomorphic in a star-shaped domain U . Then for any closed curve γ in U ,*

$$\int_{\gamma} f(z) dz = 0.$$

Proof. By **Goursat's theorem 1.10.1**

$$\int_{\partial\Delta} f(z) dz = 0$$

for all triangles. Thus f has an antiderivative, and thus $\int_{\gamma} f(z) dz = 0$ for all closed curves. 

2 Cauchy's Integral Theorem

2.1 CAUCHY'S INTEGRAL FORMULA

The next question we would like to address is what happens with path integrals when a function is not holomorphic at a point enclosed by the path. The prototype computation is

$$\int_{\partial \mathbb{D}} \frac{1}{z} dz = 2\pi i ,$$

which we generalize as follows.

Theorem 2.1.1 (Cauchy Integral Formula for Disks) *Let f be holomorphic in a domain U , and assume that $\overline{D_r(z_0)} \subset U$. Then for any $a \in D_r(z_0)$,*

$$f(a) = \frac{1}{2\pi i} \int_{\partial D_r(z_0)} \frac{f(z)}{z-a} dz$$

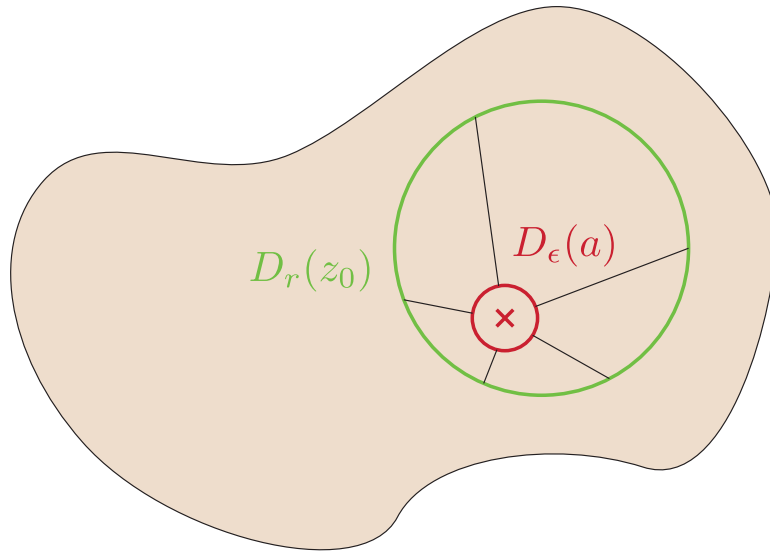


Figure 2.1 Localizing an integral

Proof. We first use a homotopy to localize the path near a , replacing the integral over $\partial D_r(z_0)$ by the integral over $\partial D_\epsilon(a)$.

There is a slight catch because the integrand is not holomorphic at a , and we only know the path independence for holomorphic integrands in star-shaped domains. But we can add segments back and forth between the two boundaries of the disks to the path. This doesn't affect the integral, because the new contributions cancel. It allows us to write the path as a sum of cycles, each lying in a star-shaped domain where the integrand is holomorphic.

Next, we rewrite the integrand so that one term becomes arbitrarily small and the other can be evaluated explicitly:

$$\begin{aligned} \int_{\partial D_r(z_0)} \frac{f(z)}{z-a} dz &= \int_{\partial D_\epsilon(a)} \frac{f(z)}{z-a} dz \\ &= \int_{\partial D_\epsilon(a)} \frac{f(z) - f(a)}{z-a} dz + \int_{\partial D_\epsilon(a)} \frac{f(a)}{z-a} dz \end{aligned}$$

For $\epsilon \rightarrow 0$, the first integrand is bounded, because we assume that f is holomorphic at a . Thus, the first integral converges to 0 for $\epsilon \rightarrow 0$. The second integral evaluates to $2\pi i \cdot f(a)$. This proves the claim. 

This result is quite remarkable. It means that if we know a holomorphic function in a disk is already determined by its values on the boundary of the disk.

2.2 CONSEQUENCES OF CAUCHY'S THEOREM

Corollary 2.2.1 (Mean Value Theorem) *Let f be holomorphic in a domain U , and assume that $\overline{D_r(z_0)} \subset U$. Then*

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt$$

Another class of holomorphic functions can be constructed as power series. Recall from Analysis (we will reprove this later, more generally):

Theorem 2.2.2 *Suppose $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ converges for $|z - z_0| < r$. Then f is holomorphic, and can be differentiated infinitely often. Moreover,*

$$a_n = \frac{f^{(n)}(z_0)}{n!}.$$

We would like to prove a converse: Holomorphic functions have convergent power series expansions. In particular, they are *infinitely* often differentiable. The key is to generalize the proof of the mean value theorem in order to recover *all* of the a_n .

Theorem 2.2.3 (Power Series Expansion) *Let $f : U \rightarrow \mathbb{C}$ be holomorphic, and let $\overline{D_r(z_0)} \subset U$. Define*

$$a_n = \frac{1}{2\pi} \int_{\partial D_r(z_0)} \frac{f(z)}{(z - z_0)^{n+1}} dz.$$

Then, for $z \in D_r(z_0)$,

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n.$$

Proof. Without loss, let's assume that $z_0 = 0$. To motivate the proof, let's illegally assume that f has a power series representation. To *find* an integral formula for the coefficients, differentiate

$$f(z) = \frac{1}{2\pi i} \int_{|w|=r} \frac{f(w)}{w - z} dw$$

to obtain

$$\frac{f^{(n)}(z)}{n!} = \frac{1}{2\pi i} \int_{|w|=r} \frac{f(w)}{(w - z)^{n+1}} dw$$

and therefore

$$\begin{aligned}
 \sum_{n=0}^{\infty} \frac{f^n(0)}{n!} z^n &= \frac{1}{2\pi i} \sum_{n=0}^{\infty} \int_{|w|=r} \frac{f(w)}{w^{n+1}} z^n dw \\
 &= \frac{1}{2\pi i} \int_{|w|=r} \frac{f(w)}{w} \sum_{n=0}^{\infty} \left(\frac{z}{w}\right)^n dw \\
 &= \frac{1}{2\pi i} \int_{|w|=r} \frac{f(w)}{w} \frac{1}{1 - \frac{z}{w}} dw \\
 &= \frac{1}{2\pi i} \int_{|w|=r} \frac{f(w)}{w - z} dw \\
 &= f(z) .
 \end{aligned}$$

Now, reading the proof backwards shows that, in fact, we only needed to assume that f is holomorphic.

We end by presenting the elegant but not well-motivated standard proof: Assume without loss that $z_0 = 0$, and let $D = D_r(0)$ be a disk whose closure is contained in U . Then

$$\begin{aligned}
 f(z) &= \frac{1}{2\pi i} \int_{|w|=r} \frac{f(w)}{w - z} dw \\
 &= \frac{1}{2\pi i} \int_{|w|=r} \frac{f(w)}{w} \cdot \frac{1}{1 - \frac{z}{w}} dw \\
 &= \frac{1}{2\pi i} \int_{|w|=r} \frac{f(w)}{w} \cdot \sum_{n=0}^{\infty} \left(\frac{z}{w}\right)^n dw \\
 &= \frac{1}{2\pi i} \sum_{n=0}^{\infty} \left(\int_{|w|=r} \frac{f(w)}{w^{n+1}} dw \right) \cdot z^{n+1}
 \end{aligned}$$

As a corollary, we obtain explicit estimates for the coefficients:

Corollary 2.2.4 *Let $f : U \rightarrow \mathbb{C}$ be holomorphic and let $\overline{D_r(z_0)} \subset U$. In $D_r(z_0)$ write*

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n .$$

Suppose that

$$|f(z)| \leq M \quad \forall z \in \partial D_r(z_0) .$$

Then

$$|a_n| \leq \frac{M}{r^n} .$$

Proof.

$$\begin{aligned}
 |a_n| &= \frac{1}{2\pi} \left| \int_{\partial D_r(z_0)} \frac{f(w)}{(w - z_0)^{n+1}} dw \right| \\
 &\leq \frac{1}{2\pi} 2\pi r \frac{M}{r^{n+1}} \\
 &= \frac{M}{r^n}
 \end{aligned}$$

For a function f defined in a star-shaped domain U , we have now shown that the following are equivalent:

1. f is holomorphic in U ;
2. f is infinitely often complex differentiable in U ;
3. f has at every point of U a locally convergent power series expansion;
4. f is continuous in U and has an antiderivative;
5. $\int_{\Delta} f(z) dz = 0$ for all triangles contained in U .
6. $\int_{\Delta} f(z) dz = 0$ for all closed curves contained in U .

2.3 ENTIRE FUNCTIONS

Definition 2.3.1 An entire function is a function that is holomorphic in all of $\mathbb{C}$.

Theorem 2.3.2 (Liouville) Let f be entire and bounded. Then f is constant.

Proof. We have $|f(z)| \leq M$ for all D_r . Thus

$$a_n \leq \frac{M}{r^n}$$

for all r , so that $a_n = 0$ for all $n \geq 1$. 

Corollary 2.3.3 Every polynomial of degree $n \geq 1$ has a zero.

Proof. We argue by contradiction. Write

$$p(z) = z^n \left(a_n + \frac{a_{n-1}}{z} + \cdots + \frac{a_0}{z^n} \right)$$

with $a_n \neq 0$. From this, we see that $|p(z)| \geq 1$ for all $|z| > r$ for some large r . Therefore $\frac{1}{|p(z)|} \leq 1$ for $|z| > r$. As $p(z) \neq 0$ in D_r ,

$$m = \max_{|z| \leq r} \frac{1}{|p(z)|} < \infty.$$

Thus

$$\frac{1}{|p(z)|} \leq \min\{1, m\}$$

for all $z \in \mathbb{C}$, and $1/p(z)$ is a bounded entire function. By **Liouville's theorem 2.3.2** it is constant, contradicting that $n \geq 1$. 

The following theorem provides us with a compelling *topological* test for holomorphicity.

Theorem 2.3.4 (Morera) Let $f : U \rightarrow \mathbb{C}$ be continuous, and assume that

$$\int_{\gamma} f(z) dz = 0$$

for all triangles in U . Then f is holomorphic in U .

Proof. Without loss assume that $U = D_r$. Then

$$F(z) = \int_0^z f(w) dw$$

is an antiderivative of f . Hence F is holomorphic, and thus f as well.



2.4 CONVERGENCE OF SEQUENCES OF HOLOMORPHIC FUNCTIONS

Definition 2.4.1 Let f_n be a sequence of complex-valued functions in a domain U . We say that f_n converges locally uniformly to a complex valued function f in U if for every $z_0 \in U$ there is a neighborhood $D \subset U$ of z_0 such that f_n converges uniformly to f in D , i.e. such that for every $\epsilon > 0$ there is an index n_0 such that for all $n \geq n_0$ and all $z \in D$ we have

$$|f(z) - f_n(z)| < \epsilon .$$

In Real Analysis, we learn that if f_n converges to f uniformly and the functions f_n are continuous, then so is f . This also holds obviously for locally uniform convergence. However, it is not generally true that a uniformly convergent sequence of differentiable functions converges to a differentiable function. However, we will show that the locally uniform limit of holomorphic functions is again holomorphic. To see this, we need suitable estimates for the derivatives of a holomorphic function. These are given by the *Cauchy estimates*, which strengthen our **previous inequalities 2.2.4**:

Theorem 2.4.2 (Cauchy Estimates) Let $\overline{D_r(z_0)} \subset \overline{D_R(z_0)} \subset U$, and let $f \in O(U)$. Then

$$|f^{(n)}(z)| \leq C_n \max_{w \in \partial D_R(z_0)} |f(w)|$$

for all $z \in D_r(z_0)$, where

$$C_n = \frac{Rn!}{(R-r)^{n+1}}$$

is independent of f .

Observe that we are estimating derivatives of a function in a small disk by its values on the boundary of a larger disk. This is a very typical situation.

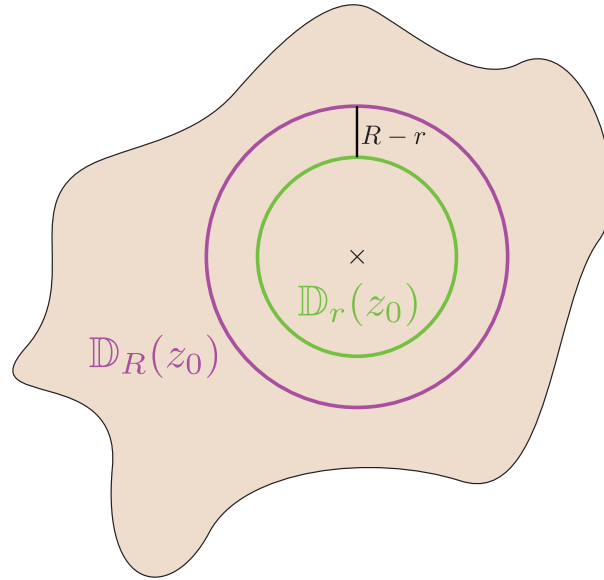


Figure 2.2 Losing is winning

Proof. By **Cauchy's theorem 2.1.1**,

$$f(z) = \frac{1}{2\pi i} \int_{\partial D_R(z_0)} \frac{f(w)}{w - z} dw$$

for all $z \in D_r(z_0)$. Differentiating yields

$$f^n(z) = \frac{n!}{2\pi i} \int_{\partial D_R(z_0)} \frac{f(w)}{(w - z)^{n+1}} dw$$

so that

$$|f^n(z)| \leq n! \frac{R}{(R - r)^{n+1}} \max_{w \in \partial D_R(z_0)} |f(w)|.$$

Corollary 2.4.3 Let $\overline{D_r(z_0)} \subset \overline{D_R(z_0)} \subset U$, and let $f \in O(U)$. Then

$$|f(z_1) - f(z_2)| \leq C_1 |z_1 - z_2| \max_{w \in \partial D_R(z_0)} |f(w)|$$

for all $z_1, z_2 \in D_r(z_0)$.

Proof. Write

$$|f(z_1) - f(z_2)| = \left| \int_{z_2}^{z_1} f'(w) dw \right|$$

and apply the **Cauchy inequality 2.4.2**. ◻

Theorem 2.4.4 (Weierstrass) Let $f_n \in O(U)$, and assume that $f_n \rightarrow f$ locally uniformly. Then $f \in O(U)$, and all derivatives $f_n^{(k)} \rightarrow f^{(k)}$ locally uniformly.

Proof. From Real Analysis, we know that f is continuous in U . Let $\bar{\Delta} \subset U$ be a triangle. Then

$$\begin{aligned} \int_{\partial\Delta} f(z) dz &= \int_{\partial\Delta} \lim_{n \rightarrow \infty} f_n(z) dz \\ &= \lim_{n \rightarrow \infty} \int_{\partial\Delta} f_n(z) dz \\ &= 0 \quad (\text{Goursat 1.10.1}) . \end{aligned}$$

By **Morera's theorem 2.3.4**, f is holomorphic in U .

To prove the claim about the convergence of the derivatives, we apply the **Cauchy inequalities 2.4.2** to $f_n - f$. For all $z \in D_r(z_0)$,

$$|f_n^{(k)}(z) - f^{(k)}(z)| \leq C_k \max_{w \in \partial D_R(z_0)} |f_n(w) - f(w)| .$$

For $n \rightarrow \infty$, the right-hand side converges to 0, and we are done. 

Exercise 2.4.5 Prove **Weierstrass Theorem 2.4.4** without using **Morera's Theorem 2.3.4**.

We can turn $O(U)$ into a topological vector space by introducing a family of semi-norms: For each compact disk $\bar{D} \subset U$ and each index k , define

$$|f|_{D,k} = \max_{z \in \bar{D}} |f^{(k)}(z)| .$$

These semi-norms generate a topology of locally uniform convergence. The corresponding uniform structure is complete by **Weierstrass' theorem 2.4.4**.

2.5 ZEROES

By the inverse function theorem, we know that if z_0 is a point with $f'(z_0) \neq 0$, the holomorphic function f maps a neighborhood of f bijectively and conformally onto a disk. We are now going to analyze the situation when $f'(z_0) = 0$. Without loss, we can assume that $f(z_0) = 0$, so that we will be looking at holomorphic functions with zeroes of higher order. Our goal is to find a normal form for them. In preparation, we make sure that we can extract square roots:

Lemma 2.5.1 Let $0 \neq w_0 \in \mathbb{C}$. Then there is a neighborhood V of w_0 and a holomorphic function $f \in O(V)$ such that $f(z)^k = z$.

Proof. Consider $g(z) = z^k$ and find z_0 with $z_0^k = w_0$. As $w_0 \neq 0$, $g'(z_0) \neq 0$. By the inverse function theorem, there is a neighborhood V of w_0 and a holomorphic function $f: V \rightarrow \mathbb{C}$ such that $f(z)^k = z$.

We can also do this more explicitly: Write $w_0 = r_0 e^{it_0}$, and let $V = \{w = r e^{it} : r > 0 \text{ and } t_0 - \pi < t < t_0 + \pi\}$. Then define

$$f(w) = r^{1/k} e^{it/k}$$

and check the **Cauchy-Riemann equations 1.6.1**. 

Definition 2.5.2 Let

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

near z_0 with

$$a_0 = \dots = a_{k-1} = 0$$

but $a_k \neq 0$. Then we say that f has a zero of order k at z_0 .

Example 2.5.1 The function $\sin(z)$ has a zero of order 1 at 0, as $\sin(0) = 0$ and $\sin'(0) = \cos(0) = 1 \neq 0$.

Note that if f has a simple zero at z_0 , then f^k has a zero of order k at z_0 . Here is a converse:

Theorem 2.5.3 Suppose that $f \in O(U)$ has a zero of order k at z_0 . Then there is a holomorphic function $g \in O(D_r(z_0))$ with a simple zero at z_0 such that

$$f(z) = g(z)^k.$$

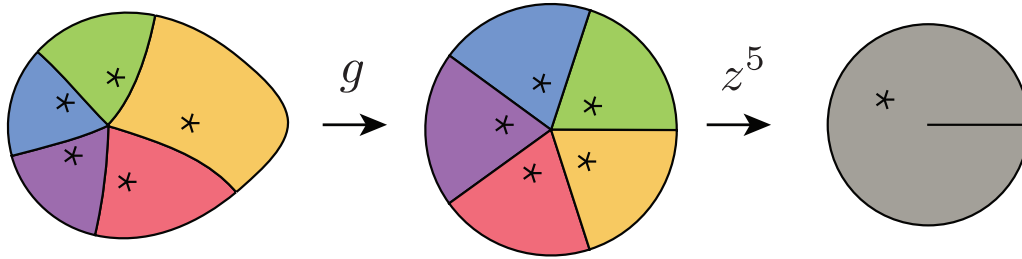


Figure 2.3 A zero of order 5

Proof. Using the power series expansion

$$f(z) = \sum_{n=k}^{\infty} a_n (z - z_0)^n$$

write

$$f(z) = (z - z_0)^k (a_k + h(z))$$

with

$$h(z) = \sum_{n=k+1}^{\infty} a_n z^{n-k}.$$

Note that h is holomorphic near z_0 with $h(z_0) = 0$ and that $a_k \neq 0$. Now define (using [Lemma 2.5.1](#))

$$g(z) = (z - z_0) \cdot (a_k + h(z))^{\frac{1}{k}}.$$

Corollary 2.5.4 *Let z_0 be a root of order k of the holomorphic function f . Then there is a neighborhood U of z_0 that is mapped by f to a disk $D_\epsilon(0)$ such that the equation $f(z) = w$ has precisely k solutions for $0 < |w| < \epsilon$.*

Proof. Write $f(z) = g(z)^k$ where g is a function with a simple zero at z_0 . By the inverse function theorem applied to g , there is a neighborhood U of z_0 that is mapped biholomorphically by g to a disk $D_\epsilon(0)$. This disk is mapped by $w \mapsto w^k$ to the disk $D_{\epsilon^k}(0)$ such that every point has precisely k distinct preimages except for 0, which has just one preimage. 

Theorem 2.5.5 (Identity Principle) *Let f be holomorphic in the connected domain U , and assume that $\{z : f(z) = 0\}$ has an accumulation point in U . Then $f \equiv 0$ in U .*

Proof. Let z_0 be such an accumulation point. Then $f(z_0) = 0$. We first show that the order of f at z_0 cannot be finite. If that is the case, then there is a neighborhood of z_0 where the only zero of f is at z_0 , contradicting that z_0 is an accumulation point of zeroes.

Now suppose that the order of the zero of f is not finite. Then all coefficients of its power series expansion are 0, and $f \equiv 0$ in a neighborhood of z_0 . This shows that the set of all zeroes of infinite order of f is nonempty, open, and trivially closed. As U is connected, we are done. 

Theorem 2.5.6 (Open Mapping Theorem) *If $f \in O(U)$ is not constant, then $f(U)$ is a domain.*

Proof. Clearly, if $f(U)$ is connected, as f is continuous. To show that $f(U)$ is open, let $w_0 = f(z_0) \in f(U)$. Then $f(z) - w_0$ has a zero of finite order at z_0 . Thus there is $\epsilon > 0$ such that a neighborhood $V \subset U$ of z_0 is mapped onto $D_\epsilon(0)$ by $f(z) - w_0$. Consequently, f maps V onto $D_\epsilon(w_0)$, so that $D_\epsilon(w_0) \subset f(U)$. 

Theorem 2.5.7 (Maximum Principle) *If $f \in O(U)$ is not constant, then $|f(z)|$ has no local maximum in U .*

Proof. Suppose that $|f(z)| \leq |f(z_0)|$ for all $z \in D_\epsilon(z_0) \subset U$. But this contradicts the **Open Mapping Theorem 2.5.6**, because $f(D_\epsilon(z_0))$ must also contain points with norm $> |f(z_0)|$. 

Here is an alternative proof for the maximum principle:

Proof. Suppose that $|f(z)| \leq |f(z_0)|$. By Cauchy's theorem,

$$\begin{aligned} f(z_0) &= \frac{1}{2\pi i} \int_{\partial D_r(z_0)} \frac{f(z)}{z - z_0} dz \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt \end{aligned}$$

so that

$$\begin{aligned} |f(z_0)| &\leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{it})| dt \\ &\leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0)| dt \\ &= |f(z_0)| \end{aligned}$$

Thus

$$\int_0^{2\pi} |f(z_0)| - |f(z_0 + re^{it})| dt = 0.$$

As the integrand is non-negative, we conclude

$$|f(z_0)| = |f(z_0 + re^{it})|$$

for all t so that $|f|$ is constant on $\partial D_r(z_0)$. Thus $|f|$ is constant in $D_r(z_0)$ (use argument for varying r), and therefore also f , by [Exercise 1.6.4](#). 

Note that there is no corresponding minimum principle, unless we make an additional assumption:

Theorem 2.5.8 (Minimum Principle) *If $f \in O(U)$ is not constant and non-zero, then $|f(z)|$ has no local minimum in U .*

Theorem 2.5.9 (Schwarz Lemma) *Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic with $f(0) = 0$. Then $|f(z)| \leq |z|$, and $|f'(0)| \leq 1$. In case of equality at some $z \neq 0$, we have $f(z) = e^{it}z$ for some $t \in \mathbb{R}$.*

Proof. Write $f(z) = z \cdot g(z)$ for a holomorphic function g . Note that $f'(0) = g(0)$. For any $0 < r < 1$, we have

$$\begin{aligned} |f(z)| = r|g(z)| \leq 1 &\Rightarrow |g(z)| \leq \frac{1}{r} && \text{on } \partial D_r(0) \\ \Rightarrow |g(z)| &\leq \frac{1}{r} && \text{on } D_r(0) \\ \Rightarrow |g(z)| &\leq 1 && \text{on } \mathbb{D} \\ \Rightarrow |f(z)| &\leq |z| && \text{on } \mathbb{D} \\ \Rightarrow |f'(0)| &\leq 1. \end{aligned}$$

If $|f(z_0)| = |z_0|$ for some $z_0 \neq 0$, then $|g(z_0)| = 1$. By the [Maximum Principle 2.5.7](#), g is constant in $\mathbb{D}$, and thus $g(z) = e^{it}$. In case that $|f'(0)| = 1$, we have $|g(0)| = 1$, so g is constant in $\mathbb{D}$ by the maximum principle. 

Definition 2.5.10 A biholomorphic map $f : U \rightarrow V$ is a bijective holomorphic function. Note that the inverse is automatically holomorphic. A biholomorphic map $f : U \rightarrow U$ is called an automorphism of U .

Corollary 2.5.11 Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be an automorphism. Then there are $\alpha \in \mathbb{R}$ and $w \in \mathbb{D}$ such that

$$f(z) = \mu_{w,\alpha}(z) = e^{i\alpha} \frac{z - w}{1 - \bar{w}z}.$$

Proof. Let $w = f(0) \in \mathbb{D}$. In [Lemma 1.3.2](#) we have seen that

$$\mu_w : z \mapsto \frac{z - w}{1 - \bar{w}z}$$

is an automorphism of $\mathbb{D}$. Thus $g = \mu_w \circ f$ is also an automorphism of $\mathbb{D}$ with $g(0) = 0$. As both g and g^{-1} satisfy the assumptions of [Schwarz' Lemma 2.5.9](#), we have $|z| = |g^{-1}(g(z))| \leq |g(z)| \leq |z|$, and hence equality. The equality

case of **Schwarz' Lemma 2.5.9** implies now that $g(z) = e^{i\alpha}z$, and f is a Möbius transformation of the form above. 

2.6 ISOLATED SINGULARITIES

Theorem 2.6.1 (Riemann Singularity Theorem) *Let $f \in O(U - \{z_0\})$, and assume that f is bounded near z_0 . Then there is a function $f \in O(U)$ that extends f .*

Proof. Assume that $z_0 = 0$. If f has a continuous extension to 0, then we claim we are done by **Morera's theorem 2.3.4**. To see this, consider a triangle $\Delta \subset U$. If z_0 lies outside of Δ , we are done. If z_0 lies on the boundary of Δ , we are also done by continuity. If z_0 lies inside Δ , cut Δ in two smaller triangles using a segment through a vertex and z_0 , and reduce the problem to the previous case.

Now consider the general case and let $g(z) = z \cdot f(z)$. This function has a continuous extension to 0. By the first part of the proof, g extends holomorphically to 0 with $g(0) = 0$. Near 0, write $g(z) = \sum_{n=1}^{\infty} a_n z^n$. Then $\sum_{n=1}^{\infty} a_n z^{n-1}$ is the desired holomorphic extension of f .

Here is a slightly more elegant alternative proof, avoiding Morera: Let $h(z) = z^2 f(z)$. Then h is even holomorphic at 0, and must have a zero of order at least 2 at 0. 

Definition 2.6.2 *Let $f \in O(U)$. An isolated point in $\mathbb{C} - U$ is called an isolated singularity of f .*

We classify isolated singularities into three cases:

1. *Removable singularity:* f can be extended holomorphically.
2. *Pole of order n :* $(z - z_0)^n \cdot f(z)$ has a removable singularity.
3. *Essential singularity:* otherwise.

Examples are given by $f(z) = z$, $f(z) = 1/z$, and $f(z) = e^{1/z}$ in $\mathbb{D} - 0$, respectively.

Definition 2.6.3 *A function $f \in O(U)$ is called meromorphic if all of its isolated singularities are poles.*

Recall that Cauchy's formula implies that for a holomorphic function

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

we can recover the coefficients by

$$a_n = \frac{1}{2\pi} \int_{|z|=r} \frac{f(z)}{z^{n+1}} dz.$$

What do these numbers mean if f is not holomorphic at 0? To answer this question, we introduce a new concept.

Definition 2.6.4 A Laurent series is an expression of the form

$$\sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

The expression

$$\sum_{n=-\infty}^{-1} a_n (z - z_0)^n$$

is called its principal part. A Laurent series converges if both

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n \quad \text{and} \quad \sum_{n=-\infty}^{-1} a_n (z - z_0)^n$$

converge, and the limit of the series is then the sum of these two expressions.

Lemma 2.6.5 Suppose that the power series

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n$$

converges for $|z - z_0| < R$, and the power series

$$\sum_{n=0}^{\infty} a_{-n} (z - z_0)^n$$

converges for $|z - z_0| < 1/r$. Then the Laurent series

$$\sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

converges for $r < |z - z_0| < R$ to a holomorphic function. Moreover, for any $r < \rho < R$,

$$a_n = \frac{1}{2\pi i} \int_{\partial D_\rho(z_0)} \frac{f(z)}{(z - z_0)^{n+1}} dz.$$

Proof. The convergence is clear by definition. For the last claim, evaluate the integral by integrating term by term and not that all powers of z except z^{-1} have an antiderivative and hence integrate to 0. 

Here is the converse:

Theorem 2.6.6 Let f be holomorphic in $r < |z| < R$. Then

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n.$$

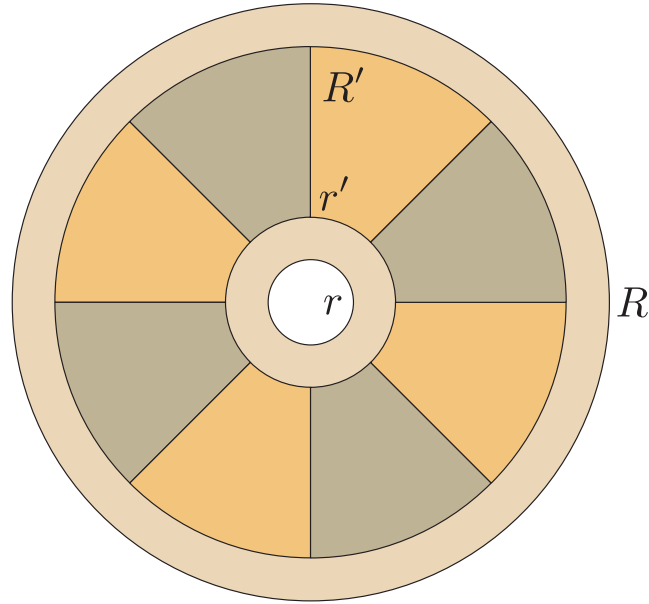


Figure 2.4 Laurent series expansion

Proof. Let $r < r' < \rho < R' < R$, such that for a fixed choice of z , $r' < |z| < R'$. Subdivide the annulus $r' < |z| < R'$ into wedges that lie in star-shaped subsets of $r < |z| < R$, and integrate over the boundaries of all these wedges, applying Cauchy's formula. Adding up these equations shows, using the trick from the proof of the **Power Expansion Theorem 2.2.3**:

$$\begin{aligned}
 f(z) &= \frac{1}{2\pi i} \int_{|w|=R'} \frac{f(w)}{w-z} dw + \frac{1}{2\pi i} \int_{|w|=r'} \frac{f(w)}{z-w} dw \\
 &= \frac{1}{2\pi i} \int_{|w|=R'} \frac{f(w)}{w(1-\frac{z}{w})} dw + \frac{1}{2\pi i} \int_{|w|=r'} \frac{f(w)}{z(1-\frac{w}{z})} dw \\
 &= \frac{1}{2\pi i} \sum_{n=0}^{\infty} \left(\int_{|w|=R'} \frac{f(w)}{w^{n+1}} dw \right) z^n + \frac{1}{2\pi i} \sum_{n=0}^{\infty} \left(\int_{|w|=r'} f(w) w^n dw \right) z^{-n-1}.
 \end{aligned}$$

We now replace the integrals over $|w| = r'$ and $|w| = R'$ by integrals over $|w| = \rho$ and are done. ◻

Corollary 2.6.7 If $|f(z)| \leq M$ on $|z| = \rho$, then $|c_n| \leq \frac{M}{\rho^n}$.

This gives us another proof of **Riemann's Removable Singularity Theorem 2.6.1**: Suppose f is bounded in $\mathbb{D} - 0$ and holomorphic. Then $|c_n| \leq \frac{M}{\rho^n}$ shows for $\rho \rightarrow 0$ that all $c_n = 0$ for $n < 0$, and f is holomorphic.

Corollary 2.6.8 (Casorati-Weierstrass) Let z_0 be an essential singularity of f . Then

$$f(0 < |z - z_0| < \epsilon)$$

is dense in $\mathbb{C}$ for any $\epsilon > 0$.

Proof. Suppose this is not true for some $\epsilon > 0$. Then $f(0 < |z - z_0| < \epsilon)$ does not contain some disk $D_\delta(w_0)$. Consider

$$g(z) = \frac{1}{f(z) - w_0}.$$

This function is bounded and holomorphic in $0 < |z - z_0| < \epsilon$ and hence has a removable singularity. Thus

$$f(z) = \frac{1}{g(z)} + w_0$$

has a pole or removable singularity at z_0 , contradicting the assumption.

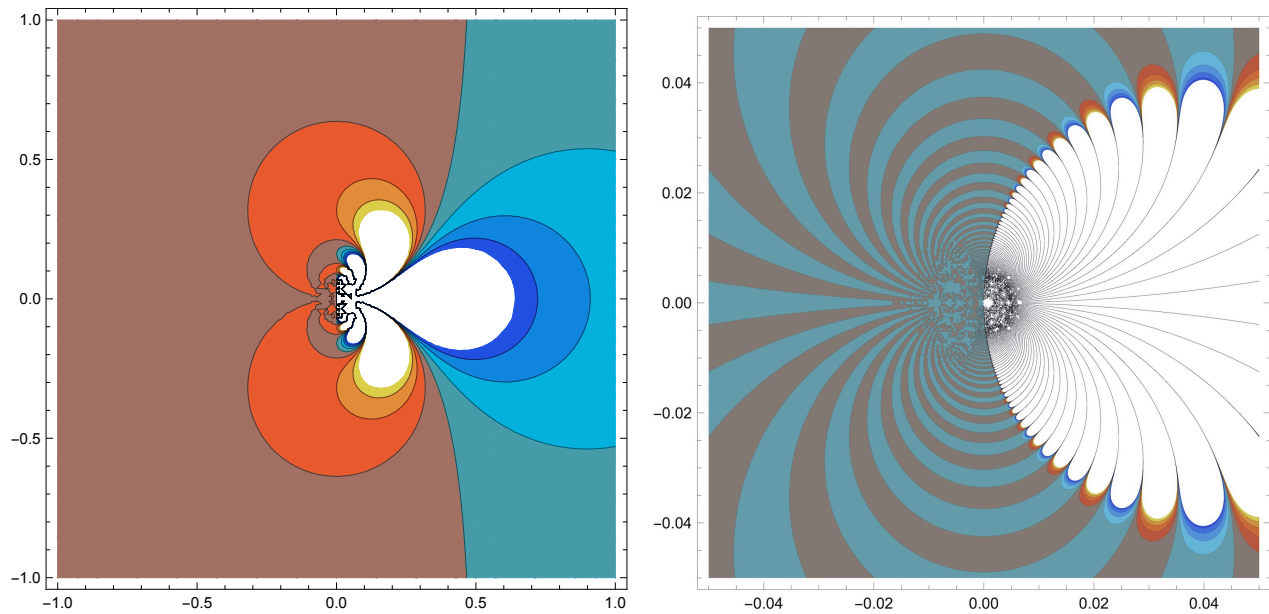


Figure 2.5 Contour plots of the real part of $e^{1/z}$ near 0

3 Products and Partial Fractions

This chapter discusses the construction of holomorphic and meromorphic functions with prescribed behavior at isolated points. We begin with an extensive example.

3.1 EXAMPLES

Theorem 3.1.1 *The infinite series*

$$f(z) = \sum_{n \in \mathbb{Z}} \frac{1}{(z - n)^2}$$

converges locally uniformly in $\mathbb{C} - \mathbb{Z}$

Proof. It suffices to show that

$$\sum_{n > r+1} \frac{1}{(z - n)^2}$$

converges uniformly in $|z| < r$. We estimate

$$\begin{aligned} \sum_{n > r+1} \frac{1}{|z - n|^2} &< \sum_{n > r+1} \frac{1}{(r - n)^2} \\ &\leq \sum_{n=1}^{\infty} \frac{1}{n^2} \\ &< \infty . \end{aligned}$$

Proposition 3.1.2 *The function*

$$g(z) = \left(\frac{\pi}{\sin(\pi z)} \right)^2$$

has the same principal parts at $\mathbb{Z}$ as $f(z)$.

Proof. Both f and g are $\mathbb{Z}$ -periodic. Thus, it suffices to compare the principal parts at 0. Clearly,

$$f(z) = \frac{1}{z^2} + \text{holo}$$

near $z = 0$. Then,

$$\begin{aligned} \frac{\sin(\pi z)}{\pi} &= z - \frac{\pi^2}{6} z^3 + \dots \\ &= z \left(1 - \frac{\pi^2}{6} z^2 + \dots \right) \end{aligned}$$

so that

$$\frac{\pi}{\sin(\pi z)} = \frac{1}{z} (1 + z^2 \cdot \text{holo}(z)) .$$

Squaring shows the claim. 

Lemma 3.1.3

$$\lim_{y \rightarrow \infty} g(x + iy) = 0$$

uniformly in x .

Proof.

$$\sin(z) = \frac{e^{iz} - e^{-iz}}{2i}$$

implies that

$$|\sin(x + iy)| = \frac{1}{2} |e^{ix}e^{-y} - e^{-ix}e^y| \rightarrow \infty$$

for $y \rightarrow \pm\infty$. 

For the next lemma, we will need the triangle inequality in the form

$$|x| + |y| \leq 2|x + iy|.$$

Lemma 3.1.4

$$\lim_{y \rightarrow \infty} f(x + iy) = 0$$

uniformly in x .

Proof. Because f is $\mathbb{Z}$ -periodic, we can assume that $0 \leq x \leq 1$. Then the above triangle inequality implies that

$$|z - n| \geq \frac{1}{2}(|y| + n - 1) \quad \text{for } n \geq 1$$

$$|z - m| \geq \frac{1}{2}(|y| + |m|) \quad \text{for } m \leq 0.$$

Thus

$$\begin{aligned} \sum_{n \in \mathbb{Z}} \frac{1}{|z - n|^2} &\leq 4 \sum_{n=1}^{\infty} \frac{1}{(|y| + n - 1)^2} + 4 \sum_{m=0}^{\infty} \frac{1}{(|y| + |m|)^2} \\ &= 8 \sum_{n=0}^{\infty} \frac{1}{(|y| + n)^2} \\ &\leq 8 \sum_{n=0}^{\infty} \frac{1}{n^2} \\ &\rightarrow 0 \end{aligned}$$

for $|y| \rightarrow \infty$. 

Theorem 3.1.5 (Partial Fraction Decomposition of $1/\sin^2(z)$)

$$\sum_{n \in \mathbb{Z}} \frac{1}{(z - n)^2} = \left(\frac{\pi}{\sin(\pi z)} \right)^2$$

for all $z \in \mathbb{C} - \mathbb{Z}$.

Proof. The function $f - g$ has removable singularities and therefore extends to an entire function. As $\lim_{|y| \rightarrow \infty} (f - g)(x + iy) = 0$ uniformly in x , $f - g$ is bounded and hence constant by **Liouville's Theorem 2.3.2**. By the behavior at ∞ , the constant is 0, and we have $f = g$. 

Corollary 3.1.6

$$\begin{aligned} \frac{\pi^2}{8} &= 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots \\ \frac{\pi^2}{6} &= 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots = \zeta(2) \end{aligned}$$

Proof. For the first identity, choose $z = \frac{1}{2}$. For the second, note that

$$\frac{1}{4}\zeta(2) = 1 + \frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{6^2} + \dots$$

so that

$$\zeta(2) = \frac{1}{4}\zeta(2) + \frac{\pi^2}{8}.$$

Theorem 3.1.7

$$k(z) = \frac{1}{z} + \sum_{n \in \mathbb{Z}^+} \frac{1}{z - n} + \frac{1}{n}$$

converges uniformly in $\mathbb{C} - \mathbb{Z}$

Proof. Use

$$\frac{1}{z - n} + \frac{1}{n} = \frac{z}{n(z - n)}$$

and proceed as before. 

Corollary 3.1.8

$$k'(z) = -f(z)$$

Proof. By **Weierstrass' Theorem 2.4.4**. 

Theorem 3.1.9 (Partial Fraction Decomposition of $\cot(z)$)

$$\pi \cdot \cot(\pi z) = \frac{1}{z} + \sum_{n \in \mathbb{Z}^*} \frac{1}{z - n} + \frac{1}{n}$$

Proof. With $h(z) = \pi \cdot \cot(\pi z)$ we have $h'(z) = -f(z)$, so $h - k$ is a constant. This constant is zero, because both h and k are odd. 

Theorem 3.1.10

$$\pi z \cot(\pi z) = 1 - 2 \sum_{n=1}^{\infty} \zeta(2n) z^{2n}$$

Proof. Using the geometric series

$$\sum_{n=1}^{\infty} \left(\frac{z}{k}\right)^{2n} = \frac{(z/k)^2}{1 - (z/k)^2} = \frac{z^2}{k^2 - z^2},$$

we compute

$$\begin{aligned} \pi z \cdot \cot(\pi z) &= 1 - 2 \sum_{k=1}^{\infty} \frac{z^2}{k^2 - z^2} \\ &= 1 - 2 \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \left(\frac{z}{k}\right)^{2n} \\ &= 1 - 2 \sum_{n=1}^{\infty} \zeta(2n) z^{2n}. \end{aligned}$$

We now use the definition of $\cot(z)$ to find a power series expansion.

$$\begin{aligned} z \cdot \cot(z) &= iz \frac{e^{iz} + e^{-iz}}{e^{iz} - e^{-iz}} \\ &= iz \cdot \frac{e^{2iz} + 1}{e^{2iz} - 1} \\ &= iz + \frac{2iz}{e^{2iz} - 1} \\ &= iz + r(z) \end{aligned}$$

where

$$r(z) = \begin{cases} \frac{2iz}{e^{2iz} - 1} & \text{if } z \neq 0 \\ 1 & \text{if } z = 0. \end{cases}$$

In particular, $r(z) + iz$ is even and we can expand

$$r(z) = 1 - iz + \sum_{n=1}^{\infty} \frac{B_{2n}}{(2n)!} z^{2n}$$

with the *Bernoulli numbers* B_{2n} as coefficients. For example,

$$B_2 = \frac{1}{6} \quad \text{and} \quad B_4 = -\frac{1}{30}.$$

It follows that

$$\pi z \cot(\pi z) = 1 + \sum_{n=1}^{\infty} (-1)^n \frac{2^{2n} B_{2n}}{(2n)!} \pi^{2n} z^{2n}$$

and thus

$$\zeta(2n) = (-1)^{n-1} 2^{2n-1} \frac{B_{2n}}{(2n)!} \pi^{2n}.$$

3.2 INFINITE PRODUCTS

Definition 3.2.1 $\prod_{k=1}^{\infty} a_k$ converges if there is an integer N such that

$$\lim_{n \rightarrow \infty} \prod_{k=N}^n a_k = \alpha$$

exists and is non-zero. We then write

$$\prod_{k=1}^{\infty} a_k = \alpha \cdot \prod_{k=1}^{N-1} a_k.$$

The definition is made so that even an infinite product is zero if and only if one of the factors is zero.

Example 3.2.1 The infinite product

$$\prod_{k=1}^{\infty} 1 - \frac{1}{k}$$

diverges.

Proof. Let $n \geq N > 1$. Then

$$\begin{aligned} \prod_{k=N}^n 1 - \frac{1}{k} &= \prod_{k=N}^n \frac{k-1}{k} \\ &= \frac{N-1}{N} \frac{N}{N+1} \cdots \frac{n-1}{n} \\ &= \frac{N-1}{n} \\ &\rightarrow 0 \end{aligned}$$

for $n \rightarrow \infty$, so the product diverges. Similarly, the infinite product

$$\prod_{k=1}^{\infty} 1 + \frac{1}{k}$$

diverges as

$$\begin{aligned}
\prod_{k=N}^n 1 + \frac{1}{k} &= \prod_{k=N}^n \frac{k+1}{k} \\
&= \frac{N+1}{N} \frac{N+2}{N+1} \cdots \frac{n+1}{n} \\
&= \frac{n+1}{N} \\
&\rightarrow \infty .
\end{aligned}$$

Lemma 3.2.2 Let $a_n \notin \mathbb{R}^{\leq 0}$. Then

$$\prod a_k < \infty \Leftrightarrow \sum \log a_k < \infty .$$

Proof. First suppose that $\sum \log a_k < \infty$. Then

$$e^{\sum \log a_k} = \prod a_k < \infty$$

as the exponential function is continuous. For the converse, it is tempting just to take the logarithm. However, the identity

$$\log(z \cdot z') = \log(z) + \log(z')$$

is only valid up to $2\pi i$. Fortunately, it is only necessary to show that

$$\sum_{k=N}^{\infty} \log a_k < \infty$$

for some N . Write

$$P_n = \prod_{k=1}^n a_k .$$

As

$$\lim P_n = P_{\infty} \neq 0$$

we can find $N > 0$ such that for all $m, n \geq N$

$$|P_m - P_n| < \frac{1}{4} |P_{\infty}| < \frac{1}{2} |P_m| .$$

Thus

$$\left| \frac{P_n}{P_m} - 1 \right| < \frac{1}{2}$$

for all $m, n \geq N$. In other words,

$$\prod_{k=m+1}^n a_k \in D_{1/2}(1) .$$

Now note that for $\operatorname{Re} z > 0$ and $\operatorname{Re} z' > 0$,

$$\log(z z') = \log(z) + \log(z')$$

so that

$$\log \prod_{k=m+1}^n a_k = \sum_{k=m+1}^n \log(a_k) .$$

as needed. 

Definition 3.2.3 $\prod_{k=1}^{\infty} a_k$ converges absolutely if and only if $\sum_{k=1}^{\infty} |\log(a_k)| < \infty$.

Lemma 3.2.4 $\prod_{k=1}^{\infty} 1 + a_k$ converges absolutely if and only if $\sum_{k=1}^{\infty} |a_k| < \infty$.

Proof.

$$\lim_{z \rightarrow 0} \frac{\log(1+z)}{z} = 1$$

implies that

$$\frac{1}{2} < \left| \frac{\log(1+z)}{z} \right| < \frac{3}{2}$$

for small z . Either convergence implies that $a_k \rightarrow 0$. Thus, for $n > N$,

$$\frac{1}{2}|a_k| < |\log(1+a_k)| < \frac{3}{2}|a_k| .$$

Example 3.2.2 The product

$$g(z) = \prod_{k=1}^{\infty} \left(1 - \frac{z^2}{k^2} \right)$$

converges locally uniformly, as

$$\sum_{k=1}^{\infty} \frac{|z|^2}{k^2} < \infty .$$

We compute

$$\begin{aligned} \frac{g'(z)}{g(z)} &= \sum_{k=1}^{\infty} \frac{-2z/k^2}{1 - z^2/k^2} \\ &= \sum_{k=1}^{\infty} \frac{2z}{z^2 - k^2} \\ &= \pi \cot(\pi z) - \frac{1}{z} . \end{aligned}$$

Thus

$$\log g(z) = \log \sin(\pi z) - \log(z) + \text{const}$$

and

$$g(z) = e^{\text{const} \frac{\sin(\pi z)}{z}}.$$

As $g(0) = 1$ we conclude that

$$\sin(\pi z) = \pi z \prod_{k=1}^{\infty} \left(1 - \frac{z^2}{k^2}\right).$$

3.3 PRODUCTS AND PARTIAL FRACTIONS IN $\mathbb{C}$

Now we have tools to construct examples with prescribed principal parts and zeroes.

Exercise 3.3.1 Let $A \subset \mathbb{C}$ be an infinite set without an accumulation point. Show that there is a sequence $a_n \in \mathbb{C}$ of pairwise distinct points with $a_n \rightarrow \infty$ such that $A = \{a_n\}$.

Theorem 3.3.2 (Mittag-Leffler) Let $a_n \in \mathbb{C}$ be pairwise distinct points without an accumulation point. Let $P_n(z)$ be a polynomial of positive degree without a constant term. Then there is a meromorphic function $f \in M(\mathbb{C})$ with poles just at the points a_n , and principal parts

$$h_n(z) = P_n\left(\frac{1}{z - a_n}\right).$$

Proof. As we cannot expect that the series

$$\sum_n P_n\left(\frac{1}{z - a_n}\right)$$

converges right away, we modify the summands by holomorphic terms that improve the convergence. We can assume that $a_n \rightarrow \infty$. For each n , find a polynomial Q_n such that

$$|h_n(z) - Q_n(z)| < \frac{1}{2^n} \quad \text{for } |z| < \frac{1}{2}|a_n|.$$

This is possible because h_n is holomorphic in the much larger disk $|z| < |a_n|$. By the **Power Expansion Theorem 2.2.3**, it has a power series expansion in that disk, which converges uniformly in any smaller disk.

Next we will show that $\sum_{n=1}^{\infty} h_n - Q_n$ converges locally uniformly in $\mathbb{C} - \{a_n\}$. Fix $r > 0$ and find an index N such that for $n > N$ we have $|a_n| > 2r$. This makes $h_n(z)$ holomorphic in $|z| < r$ for $n > N$.

Then $\sum_{n=N+1}^{\infty} h_n - Q_n$ converges absolutely and uniformly in $D_r(0)$ to a holomorphic function. This implies that $\sum_{n=1}^{\infty} h_n - Q_n$ converges locally uniformly away from the a_n to a holomorphic function and has poles at the a_n with the prescribed principal parts. 

Example 3.3.1

$$\frac{1}{z} + \sum_{n \in \mathbb{Z}} \frac{1}{z - n} - \frac{1}{n} = \pi \cdot \cot(\pi z).$$

Theorem 3.3.3 (Weierstrass Product Theorem) *Let $a_n \neq 0$ be a (possibly repetitive) sequence of complex numbers with $a_n \rightarrow \infty$. Then there is a sequence $m_n \in \mathbb{Z}^{\geq 0}$ such that*

$$f(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z}{a_n} \right) e^{\frac{z}{a_n} + \dots + \frac{1}{m_n} \left(\frac{z}{a_n} \right)^{m_n}}$$

converges locally uniformly on $\mathbb{C}$, and the holomorphic limit function f has a zero at $a \in \mathbb{C}$ of the order with which a appears in the sequence a_n .

Proof. The idea is to modify the product $\prod_{n=1}^{\infty} \left(1 - \frac{z}{a_n} \right)$ using factors that are entire functions and get the terms $1 - \frac{z}{a_n}$ closer to 1 to improve convergence. As

$$\left(1 - \frac{z}{a_n} \right) e^{-\log\left(1 - \frac{z}{a_n}\right)} = 1$$

we just replace $-\log\left(1 - \frac{z}{a_n}\right)$ by a Taylor polynomial $P_{m_n}(z)$ of sufficiently high order. Let's make that precise. We first determine the degrees m_n . Recall that

$$-\log(1 - z) = z + \frac{1}{2}z^2 + \frac{1}{3}z^3 + \dots$$

Now choose m_n such that the Taylor polynomial of $-\log(1 - z)$ of degree m_n approximates $-\log(1 - z)$ in the disk $D_{1/2}(0)$ up to $1/2^n$:

$$\left| \log(1 - z) + z + \frac{1}{2}z^2 + \dots + \frac{1}{m_n}z^{m_n} \right| < \frac{1}{2^n}.$$

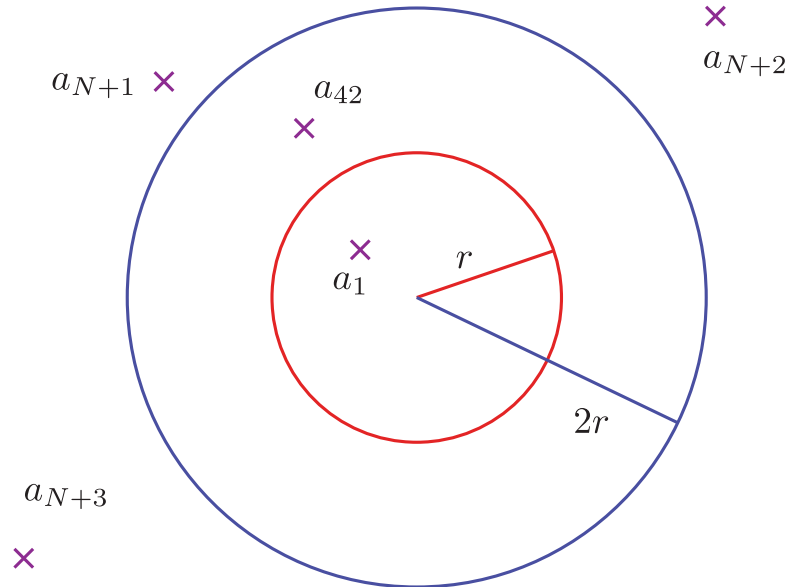


Figure 3.1 Convergence of Weierstrass products

For fixed $r > 0$ choose $N > 0$ such that $|a_n| > 2r$ for all $n > N$. Then

$$\sum_{n=N+1}^{\infty} \left(\log\left(1 - \frac{z}{a_n}\right) + \frac{z}{a_n} + \dots + \frac{1}{m_n} \left(\frac{z}{a_n} \right)^{m_n} \right)$$

converges uniformly on $D_r(0)$ as $\left|\frac{z}{a_n}\right| < \frac{1}{2}$, and hence represents a holomorphic function $g_0(z)$. Thus

$$e^{g_0(z)} = \prod_{n=N+1}^{\infty} \left(1 - \frac{z}{a_n}\right) e^{\frac{z}{a_n} + \dots + \frac{1}{m_n} \left(\frac{z}{a_n}\right)^{m_n}}$$

uniformly in $D_r(0)$, where it has no zeroes. Hence, the full product

$$f(z) = e^{g_0(z)} \prod_{n=1}^N \left(1 - \frac{z}{a_n}\right) e^{\frac{z}{a_n} + \dots + \frac{1}{m_n} \left(\frac{z}{a_n}\right)^{m_n}}$$

converges uniformly in D_r to a holomorphic function whose zeroes in D_r are precisely the $a_n \in D_r$. As this holds for all r , we are done. 

Note that the sequence m_n can be constructed universally for all choices of a_n . However, this construction will usually lead to a sequence $m_n \rightarrow \infty$, while in the example of the product expansion of

$$\sin(\pi z) = \pi z \prod_{k=1}^{\infty} \left(1 - \frac{z^2}{k^2}\right).$$

we are able to use $m_n = 0$.

Corollary 3.3.4 *The field of meromorphic functions $M(\mathbb{C})$ is the quotient field of the ring of entire functions $O(\mathbb{C})$.*

Proof. Clearly, the quotient f/g of two entire functions with $g \neq 0$ is a meromorphic function. On the other hand, given a meromorphic function $h \in M(\mathbb{C})$, form a sequence of points $a_n \rightarrow \infty$ enumerating the poles of h (with repetition to obtain the correct order if needed). By the product theorem, there is a holomorphic function $g \in O(\mathbb{C})$ with zeroes precisely at the a_k . This shows that $g \cdot h \in O(\mathbb{C})$. 

We improve on the statement of the product theorem by providing a better estimate for the magnitude of the m_n to obtain convergence. While not needed for the statement, we will make use of it when we prove the **Hadamard Product Theorem 3.6.1**. The key point is to provide a better estimate for the exponential factors, which we first formulate as

Lemma 3.3.5 *For $|z| < \frac{1}{2}$ and $m \geq 0$ we have*

$$\left| (1-z) e^{z + \frac{1}{2}z^2 + \dots + \frac{1}{m}z^m} - 1 \right| \leq 2e|z|^{m+1}.$$

Proof. We first estimate the tail of the power series expansion of $\log(1-z)$ for $|z| < \frac{1}{2}$:

$$\begin{aligned} \left| \frac{z^{m+1}}{m+1} + \frac{z^{m+2}}{m+2} + \dots \right| &\leq |z|^{m+1} |1 + |z| + |z|^2 + \dots| \\ &\leq |z|^{m+1} \frac{1}{1-|z|} \\ &\leq 2|z|^{m+1} \\ &\leq 1. \end{aligned}$$

Secondly we note that for all $w \in \mathbb{C}$ we have (using the power series)

$$|e^w - 1| \leq |w|e^{|w|}.$$

Thus

$$\left| (1-z)e^{z+\frac{1}{2}z^2+\dots+\frac{1}{m}z^m}-1 \right| = \left| e^{-\frac{z^{m+1}}{m+1}-\frac{z^{m+2}}{m+2}-\dots}-1 \right| \leq 2e|z|^{m+1}.$$

Proposition 3.3.6 *Let z_n be a sequence of complex numbers and m_n a sequence of positive integers so that*

$$\sum_{n=1}^{\infty} \left(\frac{r}{|z_n|} \right)^{m_n+1} < \infty \quad \text{for all } r > 0,$$

then the infinite product

$$h(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z}{z_n} \right) e^{\frac{z}{z_n} + \frac{1}{2} \left(\frac{z}{z_n} \right)^2 + \dots + \frac{1}{m_n} \left(\frac{z}{z_n} \right)^{m_n}}$$

converges locally uniformly in $\mathbb{C}$.

Proof. We proceed as in the proof of the **Weierstrass Product Theorem 3.3.3**. Let $r > 0$ and choose N so that for $n > N$ we have $|z_n| > 2r$, or $\left| \frac{z}{z_n} \right| < \frac{1}{2}$ for all $|z| \leq r$. By **Lemma 3.3.5**, applied to m_n and $\frac{z}{z_n}$ instead of m and z ,

$$\left| \left(1 - \frac{z}{z_n} \right) e^{\frac{z}{z_n} + \frac{1}{2} \left(\frac{z}{z_n} \right)^2 + \dots + \frac{1}{m_n} \left(\frac{z}{z_n} \right)^{m_n}} - 1 \right| \leq 2e \left| \frac{r}{z_n} \right|^{m_n+1}.$$

By assumption, the sum over the right-hand side converges. Hence, the sum over the left-hand side converges absolutely and uniformly in $|z| \leq r$. The claim now follows from **Lemma 3.2.4**. 

3.4 THE POISSON-JENSEN FORMULA

If we are given an entire function and would like to write it as a product, the **Weierstrass Product Theorem 3.3.3** does not provide any hint on how to choose the convergence generating factors. In the second next section, we will prove Hadamard's Product Theorem, which is a considerable improvement. In this section, we introduce entire functions of finite growth for which this theorem is valid. A key ingredient is the following identity, which allows us to control how quickly the zeroes of an entire function diverge to infinity.

Theorem 3.4.1 (Jensen's Formula) *Let f be holomorphic in a domain containing $\overline{D_r(0)}$, and let $z_1, \dots, z_n$ be the zeroes of f in $D_r(0)$, repeated if necessary to account for zeroes of higher order. We assume that f neither vanishes at 0 nor on $\partial D_r(0)$. Then*

$$\log |f(0)| = \sum_{k=1}^n \log \frac{|z_k|}{r} + \frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{it})| dt.$$

Proof. Note that Jensen's Formula is multiplicative, i.e., if it holds for f and g , it is also true for $f \cdot g$. Hence, it suffices to prove it for functions of the form $z - a$ and functions that do not vanish in $D_r(0)$. Let's first suppose that f does not vanish in $D_r(0)$. Then we can write $f(z) = e^{g(z)}$ for a holomorphic function g defined in $D_r(0)$. By the **Mean Value Theorem 2.2.1**, we have

$$g(0) = \frac{1}{2\pi} \int_0^{2\pi} g(re^{it}) dt.$$

Taking the real part and using $\operatorname{Re} \log(z) = \log|z|$ shows

$$\log|f(0)| = \frac{1}{2\pi} \int_0^{2\pi} \log|f(re^{it})| dt .$$

Now suppose that $f(z) = z - a$ with $|a| < r$. Then we have to show that

$$\log|a| = \log \frac{|a|}{r} + \frac{1}{2\pi} \int_0^{2\pi} \log|re^{it} - a| dt ,$$

which is equivalent to

$$\int_0^{2\pi} \log|e^{it} - a/r| dt = 0 .$$

Write $b = a/r$. Then we compute for any $|b| < 1$ that

$$\begin{aligned} \int_0^{2\pi} \log|e^{it} - b| dt &= \int_0^{2\pi} \log|e^{-it} - b| dt && \text{substitute } t \mapsto -t \\ &= \int_0^{2\pi} \log|1 - be^{it}| dt && \text{use } |e^{-it}| = 1 \\ &= \int_0^{2\pi} \log|\tilde{f}(e^{it})| dt && \text{with } \tilde{f}(z) = 1 - bz \\ &= 2\pi \log|\tilde{f}(0)| && \text{mean value theorem} \\ &= 0 \end{aligned}$$

If we want to evaluate a function somewhere other than at the center of the disk, we can move that point to the center of the disk by a Möbius transformation. We practice this first in a simple case:

Theorem 3.4.2 (Poisson Integral Formula) *Let f be holomorphic in a domain containing $\overline{D_r(0)}$. Then for $|z_0| < r$,*

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} \frac{r^2 - |z_0|^2}{|re^{it} - z_0|^2} f(re^{it}) dt$$

Proof. We consider

$$z = z(w) = r \frac{rw + z_0}{r + \overline{z_0}w} .$$

By **Corollary 1.3.2**, this is a biholomorphic map from $|w| \leq 1$ to $|z| \leq r$ and sends 0 to z_0 . By **Cauchy's Integral formula 2.1.1**,

$$f(z_0) = \frac{1}{2\pi i} \int_{|w|=1} f(z(w)) \frac{dw}{w} .$$

As

$$w = r \frac{z - z_0}{r^2 - \overline{z_0}z} ,$$

we compute

$$\begin{aligned}
\frac{dw}{w} &= \frac{dz}{z - z_0} + \frac{\bar{z}_0 dz}{r^2 - \bar{z}_0 z} \\
&\stackrel{|z|=r}{=} \left(\frac{z}{z - z_0} + \frac{\bar{z}_0}{\bar{z} - \bar{z}_0} \right) \frac{dz}{z} \\
&= \frac{r^2 - |z_0|^2}{|z - z_0|^2} \frac{dz}{z} .
\end{aligned}$$

This shows that

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} \frac{r^2 - |z_0|^2}{|re^{it} - z_0|^2} f(re^{it}) dt$$

as claimed. 

Definition 3.4.3 The Poisson kernel is defined as

$$P(z) = \frac{r^2 - |z|^2}{|re^{it} - z|^2} .$$

One advantage of using the Poisson kernel is that it is real-valued, making it easier to estimate than a complex-valued kernel. We collect two useful facts:

Lemma 3.4.4

$$\begin{aligned}
\frac{|w|^2 - |z|^2}{|w - z|^2} &= \frac{w}{w - z} + \frac{\bar{z}}{\bar{w} - \bar{z}} = \operatorname{Re} \frac{w + z}{w - z} \\
\frac{|w| - |z|}{|w| + |z|} &\leq \frac{|w|^2 - |z|^2}{|w - z|^2} \leq \frac{|w| + |z|}{|w| - |z|} .
\end{aligned}$$

Corollary 3.4.5 (Poisson-Jensen Formula) Let f be holomorphic in a domain containing $\overline{D_r(0)}$, and let $z_1, \dots, z_n$ be the zeroes of f in $D_r(0)$, repeated if necessary to account for zeroes of higher order. We assume that f does neither vanish on $\partial D_r(0)$. Then for any $z \in D_r(0)$ with $f(z) \neq 0$,

$$\log |f(z)| = \sum_{k=1}^n \log \left| \frac{r(z - z_k)}{r^2 - \bar{z}_k z} \right| + \frac{1}{2\pi} \int_0^{2\pi} \frac{r^2 - |z|^2}{|re^{it} - z|^2} \log |f(re^{it})| dt .$$

Proof. We can apply the same substitution as in the proof of the **Poisson Integral Formula 3.4.2**. Alternatively, we can argue directly as follows. Let

$$b(z) = \prod_{k=1}^m r \frac{z - z_k}{r^2 - \bar{z}_k z} ,$$

where the product is taken over all zeroes of f in $D_r(0)$. Note that $|b(z)| < 1$ for $|z| < r$ and $|b(z)| = 1$ for $|z| = r$, as this holds for each factor. Now consider

$$g(z) = \frac{f(z)}{b(z)} .$$

This function is holomorphic and non-vanishing in $D_r(0)$, thus we can define a holomorphic function

$$\log g(z) = \int^z \frac{g'(z)}{g(z)} dz.$$

We apply the **Poisson Integral Formula 3.4.2** to this function and take the real part to obtain

$$\log |g(z)| = \int_0^{2\pi} \frac{r^2 - |z|^2}{|re^{it} - z|^2} \log |g(re^{it})| dt.$$

As for $|z| = r$ we have $|b(z)| = 1$, the claim follows. 

Definition 3.4.6 Let f be holomorphic in the disk $D_r(0)$. The zero counting function of f is defined as

$$\begin{aligned} & : (0, r) \rightarrow \mathbb{R}^{\geq 0} \\ & \rho \mapsto \#\{z : f(z) = 0 \text{ and } 0 < |z| < \rho\}, \end{aligned}$$

where we count the zeroes with multiplicity.

We claim

Lemma 3.4.7 Let f be holomorphic in a domain containing $\overline{D_r(0)}$ with $f(0) \neq 0$. Let $z_1, \dots, z_n$ be the zeroes of f in $D_r(0)$, repeated if necessary. Then

$$\int_0^r (t) \frac{dt}{t} = \sum_{k=1}^n \log \frac{r}{|z_k|}.$$

Proof. Denote by

$$\chi_k(t) = \begin{cases} 1 & \text{if } r > |z_k| \\ 0 & \text{if } r < |z_k| \end{cases}$$

Now we compute

$$\begin{aligned} \sum_{k=1}^n \log \frac{r}{|z_k|} &= \sum_{k=1}^n \int_{|z_k|}^r \frac{1}{t} dt \\ &= \sum_{k=1}^n \int_0^r \frac{\chi_k(t)}{t} dt \\ &= \int_0^r \sum_{k=1}^n \frac{\chi_k(t)}{t} dt \\ &= \int_0^r \frac{(t)}{t} dt. \end{aligned}$$

Combining this with **Jensen's Formula 3.4.1**, we obtain

Corollary 3.4.8

$$\int_0^r (t) \frac{dt}{t} = \frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{it})| dt - \log |f(0)|.$$

3.5 FUNCTIONS OF FINITE ORDER

For a non-constant entire function f , we define its *growth function* by

$$M(r) = M_f(r) = \max_{|z|=r} |f(z)|.$$

By the **Maximum Principle 2.5.7**, this is strictly increasing, and by **Liouville's Theorem 2.3.2** we have $M(r) \rightarrow +\infty$ for $r \rightarrow +\infty$.

Definition 3.5.1 *The order of growth of f is*

$$\rho = \limsup_{r \rightarrow \infty} \frac{\log \log M(r)}{\log r} \leq \infty$$

Example 3.5.1

1. For $n \in \mathbb{Z}^{>0}$ and $a \neq 0$, the function $f(z) = e^{az^n}$ has order n . To see this, note that $\log M(r) = |a|r^n$, so that

$$\begin{aligned} \lim_{r \rightarrow \infty} \frac{\log \log M(r)}{\log r} &= \lim_{r \rightarrow \infty} \frac{n \log r + \log |a|}{\log r} \\ &= n. \end{aligned}$$

2. Let

$$f(z) = \cos(\sqrt{z}) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^n.$$

We estimate for $|z| = r$

$$\begin{aligned} |f(z)| &\leq \sum_{n=0}^{\infty} \frac{1}{(2n)!} r^n \\ &= \frac{1}{2} (e^{\sqrt{r}} + e^{-\sqrt{r}}) \\ &= \cosh \sqrt{r}, \end{aligned}$$

with equality for $z = -r$. Thus

$$M(r) = \frac{1}{2} (e^{\sqrt{r}} + e^{-\sqrt{r}})$$

and

$$\sqrt{r} - \log 2 < \log M(r) < \sqrt{r}$$

so that

$$\frac{1}{2} + \frac{\log \left(1 - \frac{\log 2}{\log r}\right)}{\log r} < \frac{\log \log M(r)}{\log r} < \frac{1}{2}.$$

This shows that f has order $\frac{1}{2}$.

3. Let $f(z) = e^{e^z}$. Then $M(r) = e^{e^r}$ and thus

$$\frac{\log \log M(r)}{\log r} = \frac{r}{\log r} \rightarrow \infty .$$

Therefore, the order of f is ∞ .

We now relate the order of an entire function to the number of its zeroes.

Theorem 3.5.2 *Let f be an entire function of order ρ . Then*

$$\limsup_{r \rightarrow \infty} \frac{\log(r)}{\log r} \leq \rho .$$

Proof. We can assume that $f(0) \neq 0$; otherwise, we consider $f(z)/z^n$, where n is the order of the zero at 0. This does not change the order of f but decreases the number of its zeroes. As f has order ρ , we have that

$$M(r) \leq ae^{br^\rho}$$

for positive constants a, b and r sufficiently large.

By **Corollary 3.4.8**,

$$\int_0^r (t) \frac{dt}{t} = \frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{it})| dt - \log |f(0)| .$$

Replacing r by $2r$ and choosing r sufficiently large implies

$$\begin{aligned} \int_r^{2r} (t) \frac{dt}{t} &\leq \frac{1}{2\pi} \int_0^{2\pi} \log |f(2re^{it})| dt - \log |f(0)| \\ &\leq \log M(2r) - \log |f(0)| \\ &\leq cr^\rho \end{aligned}$$

for some positive number c . On the other hand, since (r) is increasing,

$$\begin{aligned} \int_r^{2r} (t) \frac{dt}{t} &\geq (r) \int_r^{2r} \frac{dt}{t} \\ &= (r) \log(2) . \end{aligned}$$

Combining these two estimates proves the claim. 

As another application, we prove

Theorem 3.5.3 *If f is an entire function of order ρ and zeroes $z_n \neq 0$ ordered so that*

$$0 < |z_1| \leq |z_2| \leq \dots \leq |z_n| \leq \dots ,$$

then

$$\sum_{n=1}^{\infty} \frac{1}{|z_n|^s} < \infty \quad \text{for } s > \rho .$$

Proof. As $s > \rho$, we have

$$\log(r) < as \log r$$

for any $a < 1$ and r sufficiently large. Therefore

$$n \leq (|z_n|)^s < |z_n|^{as}$$

so that for all n sufficiently large

$$\frac{1}{|z_n|^s} < \frac{1}{n^{1/a}} .$$

The claim follows. 

3.6 HADAMARD'S PRODUCT THEOREM

The goal of this section is to prove

Theorem 3.6.1 (Hadamard's Product Theorem) *Let f be an entire function of finite order ρ . Let $m = \lfloor \rho \rfloor$. Then*

$$f(z) = z^k e^{g(z)} \prod_{n=1}^{\infty} \left(1 - \frac{z}{z_n}\right) e^{\frac{z}{z_n} + \dots + \frac{1}{m} \left(\frac{z}{z_n}\right)^m},$$

where g is a polynomial of degree at most m .

This theorem considerably strengthens the **Weierstrass Product Theorem 3.3.3**.

Proof. By **Corollary 3.5.3**, we have

$$\sum_{n=1}^{\infty} \left(\frac{r}{|z_n|}\right)^{m+1} < \infty .$$

Thus, by **Proposition 3.3.6**, the product converges to an entire function with the desired zeroes. It remains to show that the function $g(z)$ is a polynomial of degree at most m .

We will show that $g^{m+1}(z) \equiv 0$. Without changing the order of f we can assume that $f(0) \neq 0$. From the **Poisson-Jensen Formula 3.4.5**, we obtain for any r with $f(z) \neq 0$ for $|z| = r$.

$$\log |f(z)| = \sum_{k=1}^{(r)} \log \left| \frac{r(z - z_k)}{r^2 - \bar{z}_k z} \right| + \frac{1}{2\pi} \int_0^{2\pi} \operatorname{Re} \frac{re^{it} + z}{re^{it} - z} \log |f(re^{it})| dt .$$

By the **Cauchy-Riemann Equations 1.6.1**, for any holomorphic function $h(z)$,

$$\left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \operatorname{Re} h(z) = h'(z) .$$

Therefore

$$\frac{f'(z)}{f(z)} = \sum_{k=1}^{(r)} \left(\frac{1}{z - z_k} + \frac{\bar{z}_k}{r^2 - \bar{z}_k z} \right) + \frac{1}{2\pi} \int_0^{2\pi} \frac{2re^{it}}{(re^{it} - z)^2} \log |f(re^{it})| dt .$$

Differentiating this m times gives

$$\begin{aligned} \left(\frac{d}{dz}\right)^m \frac{f'(z)}{f(z)} &= \sum_{k=1}^{(r)} \frac{(-1)^m m!}{(z - z_k)^{m+1}} + \sum_{k=1}^{(r)} \frac{\bar{z}_k^{m+1} m!}{(r^2 - \bar{z}_k z)^{m+1}} \\ &\quad + \frac{(m+1)!}{\pi} \int_0^{2\pi} \frac{re^{it}}{(re^{it} - z)^{m+2}} \log |f(re^{it})| dt. \end{aligned}$$

Our next goal is to estimate for fixed z and r large

$$J(r) = \frac{1}{2\pi} \int_0^{2\pi} \frac{re^{it}}{(re^{it} - z)^{m+2}} \log |f(re^{it})| dt.$$

As

$$\int_0^{2\pi} \frac{re^{it}}{(re^{it} - z)^{m+2}} dt = \frac{1}{i} \int_{|w|=1} \frac{1}{(w - z)^{m+2}} dw = 0,$$

we can modify $J(r)$ like so:

$$J(r) = -\frac{1}{2\pi} \int_0^{2\pi} \frac{re^{it}}{(re^{it} - z)^{m+2}} (\log M(r) - \log |f(re^{it})|) dt.$$

Here $M(r) = \max_{|z|=r} |f(z)|$ so that the second factor of the integrand becomes non-negative. Thus

$$|J(r)| \leq \frac{r}{2\pi(r - |z|)^{m+2}} \int_0^{2\pi} (\log M(r) - \log |f(re^{it})|) dt.$$

Combining this with [Jensen's Formula 3.4.1](#)

$$\log |f(0)| \leq \frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{it})| dt$$

we obtain for $r > 2|z|$ very generously

$$|J(r)| \leq \frac{2^{m+2}}{r^{m+1}} (\log M(r) - \log |f(0)|).$$

Now let

$$\epsilon = \frac{1}{2}(m + 1 - \rho) > 0$$

so that for r sufficiently large

$$\log M(r) < r^{\rho+\epsilon}$$

and thus

$$|J(r)| < C_1 \frac{r^{\rho+\epsilon}}{r^{m+1}} = \frac{C_1}{r^\epsilon}.$$

Next we estimate

$$\begin{aligned} \sum_{k=1}^{(r)} \frac{\bar{z}_k^{m+1} m!}{(r^2 - \bar{z}_k z)^{m+1}} &\leq (r) \frac{r^{m+1}}{(r^2 - r|z|)^{m+1}} \\ &\leq \frac{2^{m+1}(r)}{r^{m+1}} \\ &\leq \frac{C_2}{r^\epsilon}, \end{aligned}$$

where we used in the last step that

$$(r) < r^{\rho+\epsilon}$$

by **Theorem 3.5.2**.

Combining these two estimates shows for fixed z , r sufficiently large (dependent on z) and $\epsilon > 0$ independent of z

$$\left| \left(\frac{d}{dz} \right)^m \frac{f'(z)}{f(z)} - \sum_{k=1}^{(r)} \frac{(-1)^m m!}{(z - z_k)^{m+1}} \right| < \frac{C_3}{r^\epsilon} \xrightarrow{r \rightarrow \infty} 0$$

On the other hand,

$$\frac{f'(z)}{f(z)} = g'(z) + \sum_{k=1}^{\infty} \left(\frac{1}{z - z_k} + \frac{1}{z_k} + \cdots + \frac{z^{m-1}}{z_k^m} \right)$$

where the right hand side converges locally uniformly in $\mathbb{C} - \{z_k\}$. Differentiating this m times gives

$$\left(\frac{d}{dz} \right)^m \frac{f'(z)}{f(z)} = g^{(m+1)}(z) + \sum_{k=1}^{\infty} \frac{(-1)^m m!}{(z - z_k)^{m+1}}.$$

3.7 THE Γ -FUNCTION

Recall from Analysis that

$$\gamma = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} - \log n \right) \approx 0.57721$$

converges. This number is called the *Euler-Mascheroni number*, and shows up in many places. It is expected, though not known, that this number is irrational.

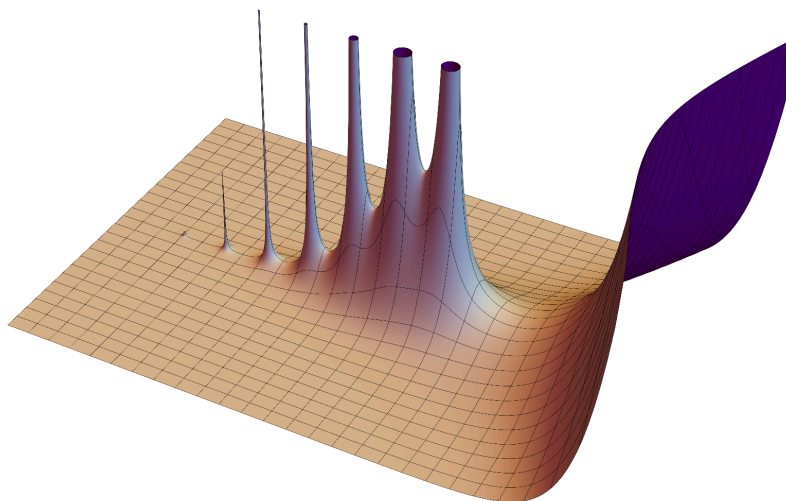


Figure 3.2 Absolute value of the Γ -function

The next theorem introduces the Γ -function.

Theorem 3.7.1 *The limit*

$$\frac{1}{\Gamma(z)} = \lim_{n \rightarrow \infty} \frac{z(z+1) \cdot \dots \cdot (z+n)}{n! n^z} .$$

converges locally uniformly in $\mathbb{C}$. We have

$$\frac{1}{\Gamma(z)} = ze^{\gamma z} \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right) e^{-\frac{z}{n}}$$

so that $\Gamma(z)$ is a meromorphic function in $\mathbb{C}$ with simple poles in $\mathbb{Z}^{\leq 0}$ and without zeroes.

Proof. As $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, the infinite product

$$g(z) = z \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right) e^{-\frac{z}{n}}$$

converges by **Proposition 3.3.6** locally uniformly in $\mathbb{C}$ to an entire function with simple zeroes at $\mathbb{Z}^{\leq 0}$. To see the connection with the first formula, we rewrite

$$\begin{aligned} \frac{1}{n^z} &= e^{-z \log n} \\ &= e^{z(1 + \frac{1}{2} + \dots + \frac{1}{n} - \log n)} e^{-(z + \frac{z}{2} + \dots + \frac{z}{n})} \end{aligned}$$

so that

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{z(z+1) \cdot \dots \cdot (z+n)}{n! n^z} &= \lim_{n \rightarrow \infty} \frac{z(1+z)(1+\frac{z}{2}) \cdot \dots \cdot (1+\frac{z}{n})}{n^z} \\ &= \lim_{n \rightarrow \infty} e^{z(1 + \frac{1}{2} + \dots + \frac{1}{n} - \log n)} e^{-(z + \frac{z}{2} + \dots + \frac{z}{n})} z(1+z) \cdot \dots \cdot (1+\frac{z}{n}) \\ &= ze^{\gamma z} \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right) e^{-\frac{z}{n}} \end{aligned}$$

as claimed. 

Theorem 3.7.2 *The Γ -function satisfies the functional equation*

$$\Gamma(z+1) = z\Gamma(z) .$$

Proof. We compute

$$\begin{aligned} \Gamma(z+1) &= \lim_{n \rightarrow \infty} \frac{n! n^{z+1}}{(z+1) \cdot \dots \cdot (z+n+1)} \\ &= \lim_{n \rightarrow \infty} \frac{n}{z+n+1} z \frac{n! n^z}{z(z+1) \cdot \dots \cdot (z+n)} \\ &= z\Gamma(z) . \end{aligned}$$

Corollary 3.7.3 *For $n \in \mathbb{Z}^{\geq 0}$ we have*

$$\Gamma(n+1) = n! .$$

Proof. $\Gamma(1) = 1$ follows from the definition, and the rest follows by induction from the functional equation. 

Theorem 3.7.4 *We have*

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}.$$

Proof. We compute

$$\begin{aligned}\Gamma(z)\Gamma(1-z) &= \lim_{n \rightarrow \infty} \frac{n! n^z}{z(z+1) \cdots (z+n)} \frac{n! n^{1-z}}{(1-z)(2-z) \cdots (n+1-z)} \\ &= \lim_{n \rightarrow \infty} \frac{1}{z(1+z) \cdots (1+\frac{z}{n})} \frac{1}{(1-z)(1-z) \cdots (1-\frac{z}{n})} \frac{n}{n+1-z} \\ &= \lim_{n \rightarrow \infty} \frac{1}{z(1-z^2) \cdots (1-\frac{z^2}{n^2})} \\ &= \frac{\pi}{\sin(\pi z)}\end{aligned}$$

by **Example 3.2.2**. 

In particular, we have $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

Lemma 3.7.5 *For $n \in \mathbb{Z}^{\geq 0}$,*

$$\operatorname{res}_{z=-n} \Gamma(z) = \frac{(-1)^n}{n!}.$$

Proof.

$$\begin{aligned}\operatorname{res}_{z=-n} \Gamma(z) &= \lim_{z \rightarrow -n} (z+n) \Gamma(z) \\ &= \lim_{z \rightarrow -n} (z+n) \frac{\Gamma(z+1)}{z} \\ &= \lim_{z \rightarrow -n} (z+n) \frac{\Gamma(z+n+1)}{z \cdots (z+n)} \\ &= \lim_{z \rightarrow -n} \frac{\Gamma(z+n+1)}{z \cdots (z+n-1)} \\ &= \frac{(-1)^n \Gamma(1)}{n!}.\end{aligned}$$

Lemma 3.7.6 *If $\operatorname{Re} z > 0$, then $|\Gamma(z)| \leq |\Gamma(\operatorname{Re} z)|$.*

Proof. Write $z = x + iy$. Then

$$|n^z| = |e^{(x+iy) \log n}| = e^{x \log n} = n^x$$

and

$$|z+k|^2 = |x+k|^2 + |y|^2 \geq |x+k|^2.$$

The claim now follows from the definition of $\Gamma(z)$.



We now use some of these properties to characterize the Γ -function.

Theorem 3.7.7 *Let $g(z)$ be holomorphic in a domain U containing the vertical strip $1 \leq \operatorname{Re} z \leq 2$. Suppose that*

1. $g(z+1) = zg(z)$ whenever defined;
2. $g(z)$ is bounded in $1 \leq \operatorname{Re} z \leq 2$;
3. $g(1) = 1$.

Then $g(z) = \Gamma(z)$.

Proof. Denote by U_n the domain U translated by $n \in \mathbb{Z}$. These domains overlap and cover all of $\mathbb{C}$. Let $n \in \mathbb{Z}^{\geq 0}$. For $z \in U_n$, define

$$g(z) = (z-1) \cdot \dots \cdot (z-n) \cdot g(z-n),$$

and for $z \in U_{-n}$, define

$$g(z) = \frac{g(z+n)}{z(z+1) \cdot \dots \cdot (z+n-1)}.$$

By property (1), this extension is well defined and holomorphic in $\mathbb{C} - \mathbb{Z}^{\leq 0}$.

In $\mathbb{Z}^{\leq 0}$, g has simple poles with the same residues as Γ , as the proof of the formula for the residues only used the functional equation and $\Gamma(1) = 1$. Thus $f(z) = \Gamma(z) - g(z)$ is an entire function with $f(1) = 0$.

Consider $h(z) = f(z)f(1-z)$. This function has period 2, because

$$\begin{aligned} h(z+1) &= f(z+1)f(-z) \\ &= zf(z) \cdot \frac{f(1-z)}{-z} \\ &= -h(z). \end{aligned}$$

By assumption, $|f|$ is bounded by a constant C in $1 \leq \operatorname{Re} z \leq 2$. This implies that f is bounded in $0 \leq \operatorname{Re} z \leq 1$, $|\operatorname{Im} z| \geq 1$ as well, as

$$|f(z)| = \left| \frac{1}{z} f(z+1) \right| \leq C.$$

As f is holomorphic, it is also bounded in the compact set $0 \leq \operatorname{Re} z \leq 1, |\operatorname{Im} z| \leq 1$. Thus f is bounded in $0 \leq \operatorname{Re} z \leq 2$. This implies that h is bounded in $0 \leq \operatorname{Re} z \leq 2$ as well, and by periodicity, in all of $\mathbb{C}$. By [Liouville's Theorem 2.3.2](#), h is constant. As $f(1) = 0$, $h(z) = f(z)f(1-z)$ is identically 0. If f were not identically zero, it wouldn't vanish in some open set. In this set, $f(1-z)$ must vanish. Hence $f(1-z)$ vanishes identically in all of $\mathbb{C}$, and so does f . This proves $g = \Gamma$.



As an application, we show:

Theorem 3.7.8 *For $\operatorname{Re} z > 0$,*

$$\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt.$$

Proof. Let

$$f(z) = \int_0^\infty e^{-t} t^z \frac{dt}{t}.$$

To show the existence and holomorphicity of f , consider $0 < a < \operatorname{Re} z < b < \infty$. Then

$$|t^{z-1}e^{-t}| = t^{\operatorname{Re} z - 1}e^{-t} \leq \begin{cases} t^{a-1} & \text{if } 0 < t \leq 1 \text{ and } \operatorname{Re} z < 1 \\ t^{b-1}e^{-t} & 1 \leq t. \end{cases}$$

Thus both integrals

$$\int_0^1 t^{a-1} dt \text{ and } \int_1^\infty t^{b-1}e^{-t} dt$$

exist, and the improper integral $\int_0^\infty e^{-t} t^z \frac{dt}{t}$ exists as a locally uniform limit as well. For a fixed $n > 0$ and $\frac{1}{n} \leq t \leq n$, the integrand $t^{z-1}e^{-t}$ is holomorphic in z , continuous in t , and locally uniformly bounded. By [Theorem 5.1.8](#), the functions

$$z \mapsto \int_{\frac{1}{n}}^n e^{-t} t^z \frac{dt}{t}$$

are holomorphic. As a locally uniform limit of holomorphic functions, f is therefore holomorphic. We estimate for $z = x + iy$

$$|f(z)| \leq \int_0^\infty e^{-t} t^x \frac{dt}{t}$$

so that in particular f is bounded in $1 \leq \operatorname{Re} z \leq 2$. Evidently,

$$f(1) = \int_0^\infty e^{-t} dt = 1.$$

To show the functional equation, we use integration by parts:

$$\begin{aligned} f(z+1) &= \int_0^\infty e^{-t} t^{z+1} \frac{dt}{t} \\ &= -e^{-t} t^z \Big|_{t=0}^{t=\infty} + z \int_0^\infty e^{-t} t^z \frac{dt}{t} \\ &= z \cdot f(z). \end{aligned}$$

As another application, we evaluate the *Beta-Integral*:

Theorem 3.7.9 For $\operatorname{Re}(a) > 0$ and $\operatorname{Re}(b) > 0$ we have

$$\int_0^1 t^{a-1} (1-t)^{b-1} dt = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

Proof. Note that this integral can be improper at $t = 0$ and $t = 1$. One shows convergence and holomorphic dependence as above for the Γ -function. Now we fix b with $\operatorname{Re}(b) > 0$ and consider the function

$$f(z) = \frac{\Gamma(z+b)}{\Gamma(b)} \int_0^1 t^{z-1} (1-t)^{b-1} dt.$$

To show that $f(z) = \Gamma(z)$, we verify the assumptions of **Theorem 3.7.7**.

Fix $0 < \delta < 1$. The integral is holomorphic in $1 - \delta < \operatorname{Re} z < 2 + \delta$ and bounded because of

$$\left| \int_0^1 t^{z-1} (1-t)^{b-1} dt \right| \leq \int_0^1 t^{x-1} (1-t)^{\operatorname{Re} b-1} dt.$$

Likewise $\Gamma(z+b)$ is holomorphic and bounded in $1 - \delta < \operatorname{Re} z < 2 + \delta$ as $\operatorname{Re} b > 0$. Thus, the same holds for $f(z)$. We compute

$$\begin{aligned} f(1) &= \frac{\Gamma(1+b)}{\Gamma(b)} \int_0^1 (1-t)^{b-1} dt \\ &= \frac{\Gamma(1+b)}{b\Gamma(b)} \\ &= 1. \end{aligned}$$

Finally, we verify the functional equation. Integration by parts shows for $\operatorname{Re} z > 0$ and $\operatorname{Re} b > 0$

$$\begin{aligned} \int_0^1 t^z (1-t)^{b-1} dt &= -\frac{t^z}{b} (1-t)^b \Big|_{t \rightarrow 0+}^{t \rightarrow 1-} + \frac{z}{b} \int_0^1 t^{z-1} (1-t)^b dt \\ &= \frac{z}{b} \int_0^1 t^{z-1} (1-t)^{b-1} dt - \frac{z}{b} \int_0^1 t^z (1-t)^{b-1} dt \end{aligned}$$

so that

$$\int_0^1 t^z (1-t)^{b-1} dt = \frac{z}{z+b} \int_0^1 t^{z-1} (1-t)^{b-1} dt$$

and therefore $f(z+1) = zf(z)$ as needed.



4 Global Complex Analysis

4.1 CAUCHY'S THEOREM

In this section, we address the following question: Let U be a domain. For which closed paths γ in U do we have

$$\int_{\gamma} f(z) dz = 0$$

for all holomorphic functions $f \in O(U)$?

So far, we know that if U is star-shaped, this holds for all closed paths. But we have also seen that for U a domain containing the unit circle but not 0,

$$\int_{\partial \mathbb{D}} \frac{1}{z} dz = 2\pi$$

In other words, integrals over $1/z$ detect potentials for Cauchy's integral theorem failing. We turn this into a definition.

Definition 4.1.1 Let γ be a closed path, and $a \notin \gamma$. Then

$$n(\gamma, a) = \frac{1}{2\pi i} \int_{\gamma} \frac{dw}{w - a}$$

is called the winding number of γ with respect to a .

As a geometric motivation of the name, assume that $a = 0$ without loss. For two rays footed at 0 ending at z_1 and z_2 that are not opposites there is a unique angle $\phi \in (-\pi, \pi)$ such that

$$e^{i\phi} \frac{z_1}{|z_1|} = \frac{z_2}{|z_2|}.$$

We denote this angle by $\angle(z_1, z_2)$. Given a closed curve $\gamma : [a, b] \rightarrow \mathbb{C}$ with $0 \notin \gamma$, choose a partition

$$a = t_0 < \dots < t_n = b$$

such that each sub-arc $\gamma|_{[t_k, t_{k+1}]}$ lies in an open half plane. Define

$$\angle(\gamma) = \sum_{k=1}^n \angle(\gamma(t_{k-1}), \gamma(t_k))$$

Note that refining the partition does not change $\angle(\gamma)$. As any two partitions have a common subdivision, $\angle(\gamma)$ is well defined and can be computed using any sufficiently fine partition.

Lemma 4.1.2 $\angle(\gamma)$ is an integral multiple of 2π .

Proof. For a sufficiently fine subdivision of $[a, b]$, let

$$\phi_k = \angle(\gamma(t_{k-1}), \gamma(t_k)).$$

Then

$$e^{i(\phi_1 + \dots + \phi_n)} \frac{\gamma(a)}{|\gamma(a)|} = \frac{\gamma(b)}{|\gamma(b)|}.$$

As γ is closed,

$$e^{i(\phi_1 + \dots + \phi_n)} = 1.$$

Lemma 4.1.3 Let $\gamma_t : [a, b] \rightarrow \mathbb{C}$ be a homotopy of closed curves such that $0 \notin \gamma_t$ for all t . Then $\angle(\gamma_t)$ is constant.

Proof. Using a sufficiently fine partition for all t near a fixed t_0 , we see that $\angle(\gamma_t)$ is continuous in t_0 . As it is an integer, it must be constant. 

Proposition 4.1.4

$$\angle(\gamma, a) = \frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z - a} = n(\gamma, a).$$

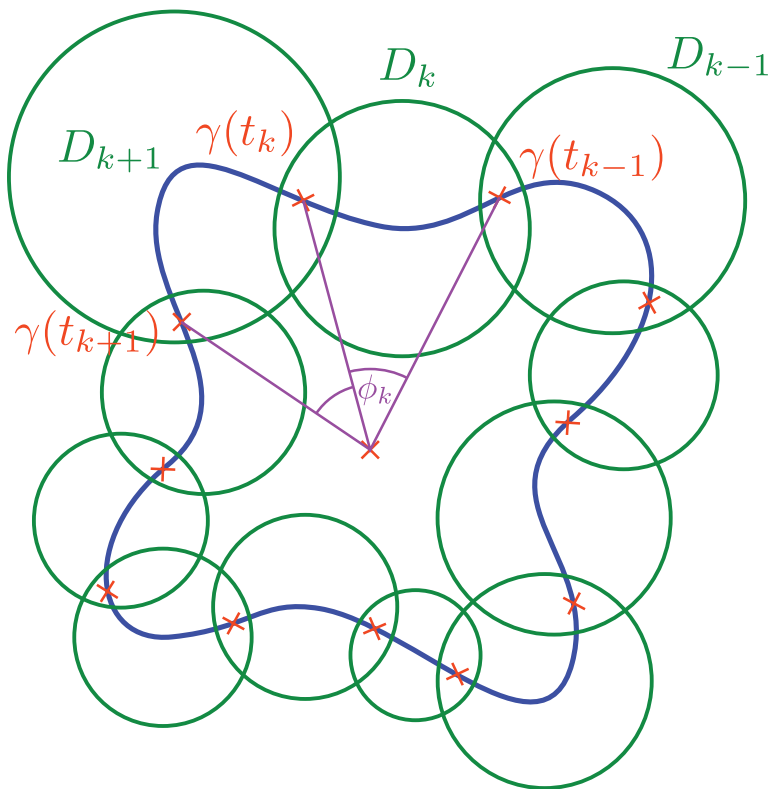


Figure 4.1 The winding number integral

Proof. Assume $a = 0$. Cover γ with finitely many disks D_k ($k = 1, \dots, n$) that do not contain 0 and such $D_{k-1} \cap D_k \neq \emptyset$. Let $\log_k$ be an antiderivative of $\frac{1}{z}$ in D_k such that $\log_{k-1} = \log_k$ in $D_{k-1} \cap D_k$. This is possible as any two antiderivatives differ by a constant, so that we can proceed by induction, using that $D_{k-1} \cap D_k$ is connected and non-empty. Note that we cannot and do not require that $\log_1 = \log_n$. Choose a partition t_k so that $\gamma(t_{k-1})$ and $\gamma(t_k)$ lie both in D_k for each k , i.e. ensure that $\gamma(t_k) \in D_k \cap D_{k+1}$. Then

$$\begin{aligned} \operatorname{Im} \int_{\gamma|_{[t_{k-1}, t_k]}} \frac{dz}{z} &= \operatorname{Im} (\log_k(\gamma(t_k)) - \log_{k-1}(\gamma(t_{k-1}))) \\ &= \langle (\gamma(t_{k-1}), \gamma(t_k)) \rangle . \end{aligned}$$

Summing over k proves the claim for the imaginary part. The contributions to the real part add up to 0, because $\operatorname{Re} \log_k(z) = \log |z|$ can be chosen independent of k . 

Definition 4.1.5 A chain in a domain U is a formal finite linear combination $\sigma = \sum_k n_k \gamma_k$ with $n_k \in \mathbb{Z}$ and γ_k a curve in U . A cycle is a chain where all γ_k are closed curves.

We extend the definition of the integral and the winding number to chains by linearity.

Definition 4.1.6 A cycle is called null homologous in a domain U if for all $a \notin U$ we have

$$n(\sigma, a) = 0 .$$

Two chains σ_1 and σ_2 are called homologous if their difference $\sigma_1 - \sigma_2$ is a null homologous cycle.

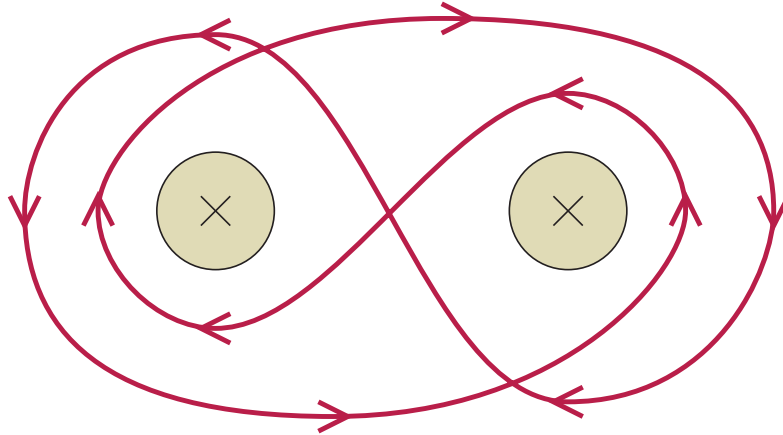


Figure 4.2 A null homologous cycle that is not null homotopic

Theorem 4.1.7 (Cauchy's Integral Theorem) Let γ be null homologous in a domain U , and $f \in O(U)$. Then

$$\int_{\gamma} f(z) dz = 0 .$$

If $z \notin \gamma$, then

$$n(\gamma, z) f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{w - z} dw$$

Proof. We claim that for any $z \in U - \gamma$,

$$\int_{\gamma} \frac{f(w) - f(z)}{w - z} dw = 0 .$$

We will consider the left-hand side as a function of z , extend it to an entire function, and show that it is identically 0. We begin by extending the integrand continuously to points on the diagonal $\Delta = \{(z, z) : z \in U\}$. Let

$$g(w, z) = \begin{cases} \frac{f(w) - f(z)}{w - z} & \text{if } w \neq z \\ f'(w) & \text{if } w = z \end{cases}$$

This function is continuous in both arguments. This is clear for $w \neq z$. For a point (z_0, z_0) on the diagonal of $U \times U$, consider $g(w, z) - g(z_0, z_0)$. If $w = z$, then

$$g(w, z) - g(z_0, z_0) = f'(z) - f'(z_0) .$$

If $w \neq z$, then

$$\begin{aligned} g(w, z) - g(z_0, z_0) &= \frac{f(w) - f(z)}{w - z} - f'(z_0) \\ &= \frac{1}{w - z} \int_z^w f'(t) - f'(z_0) dt \end{aligned}$$

For a given $\epsilon > 0$, choose δ such that for all $t \in B_\delta(z_0)$ we have

$$|f'(t) - f'(z_0)| < \epsilon .$$

Then in both cases we have for $w, z \in B_\delta(z_0)$

$$|g(w, z) - g(z_0, z_0)| < \epsilon$$

so that g is continuous at points on the diagonal as well.

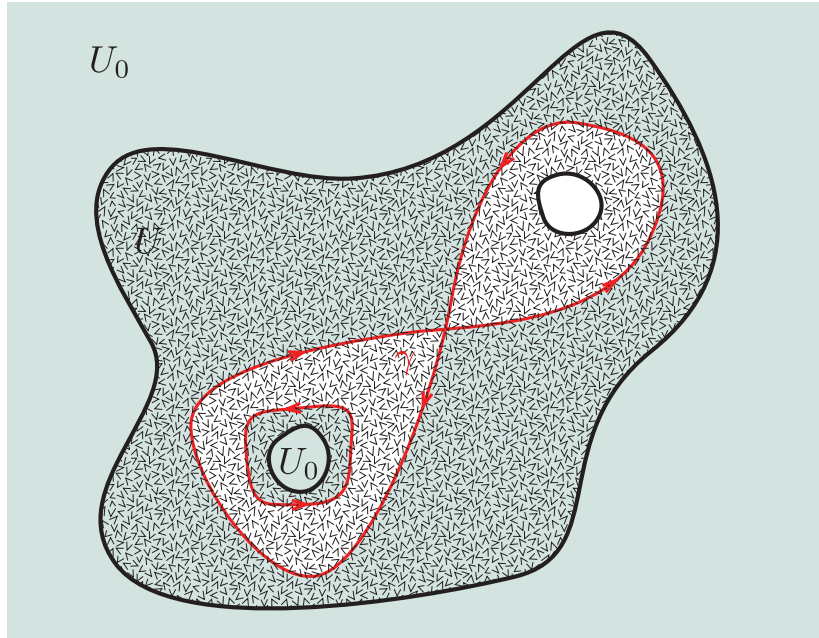
Now let

$$h_0(z) = \int_\gamma g(w, z) dw .$$

Our goal is to show that $h_0 \equiv 0$. Clearly h_0 is continuous in U , even for $z \in \gamma$. We next show that h_0 is holomorphic in U , using [Morera's Theorem 2.3.4](#). Let $\Delta \subset U$ be a triangle. Then

$$\begin{aligned} \int_\Delta h_0(z) dz &= \int_\Delta \int_\gamma g(w, z) dw dz \\ &= \int_\gamma \int_\Delta g(w, z) dz dw \\ &= 0 . \end{aligned}$$

To see the last equality, note that for fixed w the function $z \mapsto g(w, z)$ is holomorphic in $U - w$ and continuous at w , hence holomorphic in all of U by [Riemann's Removable Singularity Theorem 2.6.1](#).

Figure 4.3 Constructing h_0 and h_1

Next, we extend h_0 beyond U . Let

$$U_0 = \{z \in \mathbb{C} : n(\gamma, z) = 0\}.$$

This is an open set containing $\mathbb{C} - U$. On U_0 , define a holomorphic function

$$h_1(z) = \int_{\gamma} \frac{f(w)}{w - z} dw.$$

Then, in $U \cap U_0$,

$$h_0(z) = h_1(z)$$

so that we can define without conflict

$$h(z) = \begin{cases} h_0(z) & \text{if } z \in U \\ h_1(z) & \text{if } z \in U_0. \end{cases}$$

This extends h_0 to $U \cup U_0$. By assumption, γ is null homologous. This means that

$$n(\gamma, z_0) = 0$$

for all $z_0 \notin U$. Thus $U \cup U_0 = \mathbb{C}$, and h is an entire function.

For $|z|$ sufficiently large, we have $h(z) = h_1(z)$ and thus

$$|h(z)| \leq \frac{\max_{\gamma} |f|}{d(z, \gamma)} \text{length}(\gamma) \rightarrow 0$$

for $z \rightarrow \infty$. Thus $h \equiv 0$ by **Liouville's theorem 2.3.2**. This shows the second claim. For the first claim, note that for any $p \in U$,

$$F(z) = (z - p) f(z)$$

is holomorphic in U with $F(p) = 0$. Applying the first claim shows

$$\begin{aligned} 0 &= n(\gamma, p) F(p) \\ &= \frac{1}{2\pi i} \int_{\gamma} \frac{F(w)}{w-p} dw \\ &= \frac{1}{2\pi i} \int_{\gamma} f(w) dw . \end{aligned}$$

Proof. We now give an alternative proof. We first show the second claim. For this, we can assume that γ is a closed curve by connecting the components with paths to and fro.

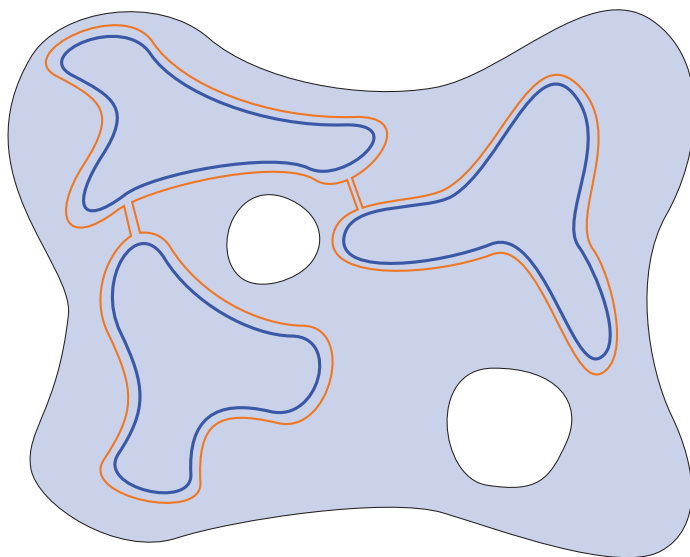


Figure 4.4 Making a cycle a single curve

Secondly, we can assume that γ is an ϵ -path:

Choose $0 < \epsilon < \frac{1}{3}d(\gamma, \partial U)$, and overlay U with horizontal and vertical lines distance ϵ apart. We are now going to deform γ so that it lies on these lines. To do so, let $\delta > 0$ such that

$$|\gamma(t) - \gamma(t')| < \epsilon$$

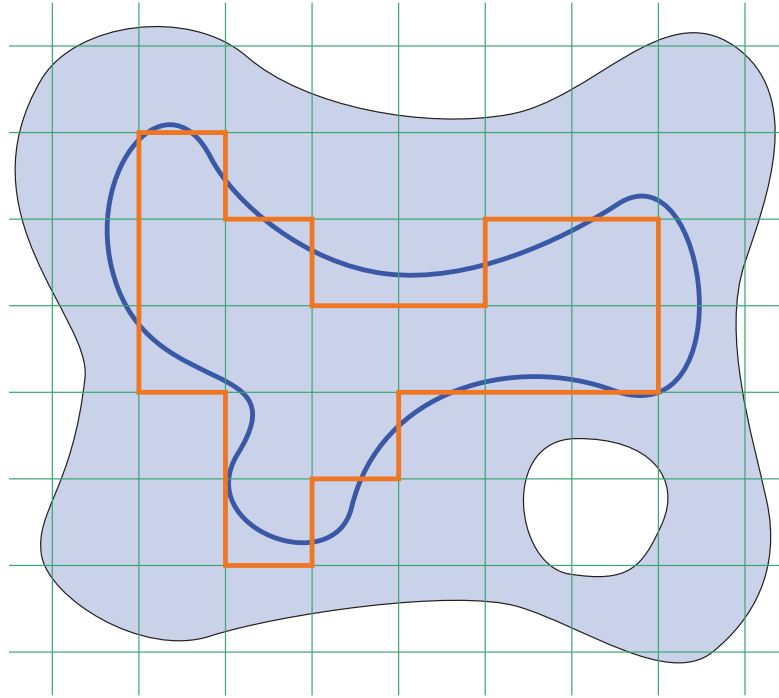
for $|t - t'| < \delta$, using uniform continuity. Choose n so that $\frac{1}{n} < \delta$.

Now find lattice points v_k such that

$$\left| \gamma\left(\frac{k}{n}\right) - v_k \right| < \sqrt{2} \cdot \epsilon .$$

This means that $\gamma\left(\frac{k}{n}\right)$ lies in a lattice square that has v_k as a corner. Then connect v_k to v_{k+1} by a shortest path following the lattice lines. Denote by γ_ϵ the juxtaposition of all these paths. We claim that

$$\int_{\gamma} f(z) dz = \int_{\gamma_\epsilon} f(z) dz$$

Figure 4.5 Making an ϵ -path

This follows from the basic version of **Cauchy's theorem 2.1.1** as the difference $\gamma - \gamma_\epsilon$ can be written as a product of closed paths that lie in star-shaped regions (namely squares).

Next, we claim that we can assume that γ is an ϵ -path that does not enclose any point. To see this, let a_k be the centers of the squares such that $n(\gamma, a_k) \neq 0$, meaning that a_k lies 'within' the path. Let b_k be the lower left corner of that square. Because γ stays away from the boundary of U , any of these squares must be either entirely inside of U or outside of U . The latter case is impossible, as we have assumed that γ is nullhomologous in U , so its winding number with respect to points outside of U must be 0.

Now pick any vertex in the lattice, and consider a lattice path from that vertex to that square, around it, and back the same way. By **Cauchy's theorem 2.1.1** again, this loop has $\int f(z) dz = 0$, and its winding number with respect to a_k is 1. By adding suitable multiples of such loops to γ , we can eliminate all squares with non-zero winding number without affecting $\int_\gamma f(z) dz$.

Finally, any closed ϵ -path that does not enclose any square uses all of its edges in both directions equally often. To see this, consider a segment that is used a times in one direction, and b time in the other direction. Modify the path along this segment so that it uses the two squares A and B adjacent to the right of the segment, respectively. Denote the centers of these squares by v_A and v_B . Then

$$a = n(\gamma, v_A) = n(\gamma, v_B) = b,$$

because there is no edge separating v_A and v_B . This proves the second claim.

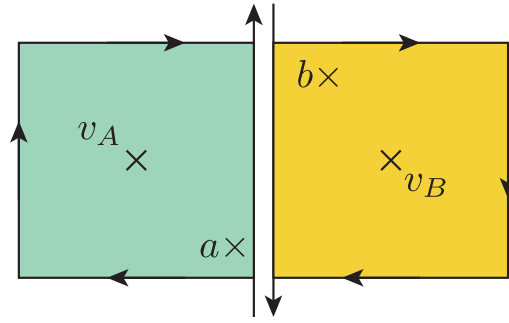


Figure 4.6 Back and Forth

To show the first claim (the general Cauchy integral formula)

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z-a} dz = n(\gamma, a) \cdot f(a),$$

modify γ by adding to γ a small loop about a , and apply what we just proved.



Definition 4.1.8 Let s be an isolated singularity of a function f that is holomorphic in a disk $D_{\epsilon}(s)$. Let

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z-s)^n$$

be the Laurent series of f at s . Then

$$\text{res}_s f(z) = a_{-1}$$

is called the residue of f at s .

Corollary 4.1.9 (Residue Theorem) Let f be holomorphic in U except for a set S of isolated singularities. Let γ be null homotopic in U , not passing through any singularity. Then

$$\frac{1}{2\pi i} \int_{\gamma} f(z) dz = \sum_{s \in S} n(\gamma, s) \cdot \text{res}_s f$$

Proof. First note that the sum is finite. If not, then there are infinitely many $s \in S$ with $n(\gamma, s) \neq 0$. These points will accumulate at a point $s_0 \in \hat{\mathbb{C}} - U$. But $n(\gamma, s_0) = 0$. As the winding number is locally constant, $n(\gamma, s) = 0$ for s near s_0 , contradicting that points with non-zero winding number accumulate at s_0 .

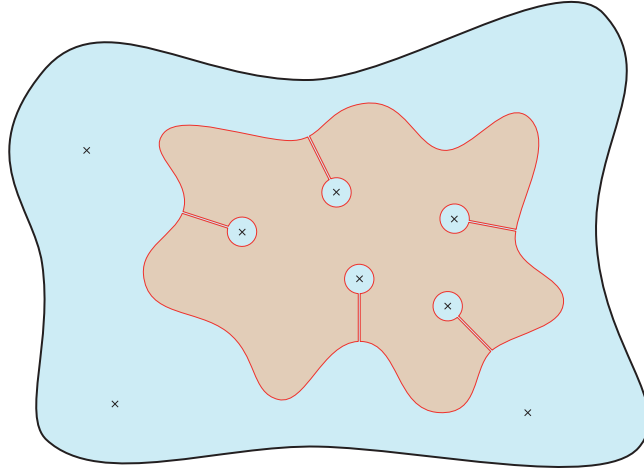


Figure 4.7 Avoiding singularities

Now let $s_1, \dots, s_n$ be the points in S enclosed by γ . Then, as usual, modifying γ with paths β_k ,

$$\int_{\gamma - \sum_{k=1}^n \beta_k} f(z) dz = 0$$

and the claim follows from Cauchy's formula.



4.2 EVALUATION OF INTEGRALS

As an application of the **Residue Theorem 4.1.9**, we evaluate some tricky infinite sums.

Proposition 4.2.1 *Let f be a rational function with poles at $z_k \notin \mathbb{Z}$. Assume that the degree of the denominator exceeds the degree of the numerator by at least 2. This means that*

$$|f(z)| \leq \frac{M}{|z|^2}$$

for $|z|$ sufficiently large. Then

$$\sum_{n=-\infty}^{\infty} f(n) = - \sum_k \text{res}_{z_k} (f(z) \pi \cot(\pi z)) .$$

Proof. Notice that

$$\begin{aligned} \text{res}_0 \pi \cot(\pi z) &= \lim_{z \rightarrow 0} \pi z \frac{\cos(\pi z)}{\sin(\pi z)} \\ &= \lim_{z \rightarrow 0} \frac{\pi}{\cos(\pi z)} \\ &= 1 , \end{aligned}$$

and the same is true for all integers by periodicity. This allows us to rewrite

$$f(n) = \text{res}_n (\pi \cot(\pi z) \cdot f(z)) .$$

Now consider

$$R_m = \{z : |\operatorname{Re} z| < m + \frac{1}{2}, |\operatorname{Im} z| < m\},$$

where we choose m so big that R_m contains all poles of f . By the residue theorem,

$$\frac{1}{2\pi i} \int_{\partial R_m} f(z) \cdot \pi \cot(\pi z) dz = \sum_{n=-m}^m f(n) + \sum_k \operatorname{res}_{z_k}(f(z) \pi \cot(\pi z)).$$

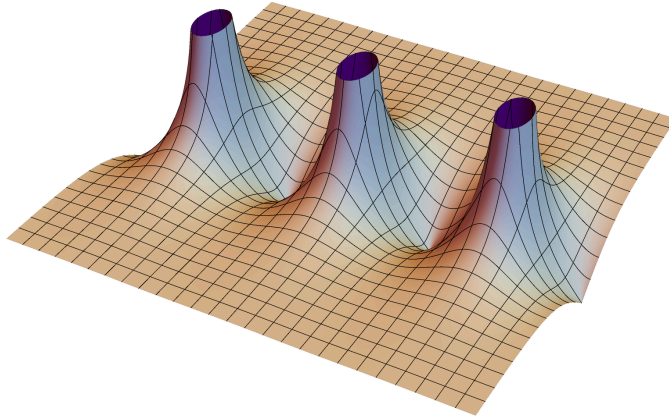


Figure 4.8 Navigating the Cotangent Landscape

Next, we estimate the integrand on the boundary of R_m . To this end, compute

$$\begin{aligned} i \cot(\pi z) &= i \frac{\cos(\pi z)}{\sin(\pi z)} \\ &= -i \frac{e^{2\pi i z} + 1}{e^{2\pi i z} - 1} \\ &= \frac{e^{2\pi i x} e^{-2\pi y} + 1}{e^{2\pi i x} e^{-2\pi y} - 1}. \end{aligned}$$

On the vertical edges $x = \pm(m + \frac{1}{2})$ we have $e^{2\pi i x} = -1$ so that

$$|\cot(\pi z)| = \frac{|-e^{-2\pi y} + 1|}{-e^{-2\pi y} + 1} \leq 1$$

and thus

$$\begin{aligned} \left| \frac{1}{2\pi i} \int_{\uparrow} f(z) \cdot \pi \cot(\pi z) dz \right| &\leq \frac{1}{2} \int_{\uparrow} |f(z)| \cdot |\cot(\pi z)| |dz| \\ &\leq \frac{1}{2} \frac{M}{m^2} 2m \\ &= \frac{M}{m}. \end{aligned}$$

On the horizontal segments $y = \pm m$ we use

$$|\cot(\pi z)| \leq \frac{e^{2m\pi} + 1}{e^{2m\pi} - 1} \leq 2$$

so that

$$\left| \frac{1}{2\pi i} \int_{\rightarrow} f(z) \cdot \pi \cot(\pi z) dz \right| \leq \frac{1}{2} (2m+1) \frac{2M}{m^2} \\ \leq \frac{3M}{m} .$$

This shows that for $m \rightarrow \infty$,

$$\int_{\partial R_m} f(z) \cdot \pi \cot(\pi z) dz \rightarrow 0 ,$$

and the claim follows. 

Example 4.2.1 For $w \notin \mathbb{Z}$, let

$$f(z) = \frac{1}{z^2 - w^2} .$$

Then

$$\operatorname{res}_{z \pm w} f(z) \pi \cot(\pi z) = \lim_{z \rightarrow \pm w} \frac{z \mp w}{z^2 - w^2} \\ = \frac{\pi}{2w} \cot(\pi w) .$$

Thus

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2 - w^2} = -\pi \frac{\cot(\pi w)}{w}$$

which is again the partial fraction decomposition of the cotangent function.

4.3 COUNTING ZEROS

The moral of the **Residue Theorem 4.1.9** is that everything about a holomorphic function f that can be computed as a residue can also be computed as an integral. The question arises: which features of f can be expressed as residues? As a first example, we use residues to count the number of zeroes.

Let

$$f(z) = \sum_{n=N}^{\infty} a_n (z-p)^n$$

with $a_N \neq 0$. Then

$$\operatorname{ord}_p f = N$$

is the order of f at p . If $N < 0$, p is a pole of order $-N$, and if $N > 0$, p is a zero of order N . Note that if $\operatorname{ord}_p f = N$, then

$$f(z) = (z-p)^N \cdot g(z)$$

with a holomorphic function $g(z)$ that does not vanish at p . Then

$$\frac{f'(z)}{f(z)} = \frac{N(z-p)^{N-1} g(z) + (z-p)^N g'(z)}{(z-p)^N g(z)} \\ = \frac{N}{z-p} + \frac{g'(z)}{g(z)}$$

where $\frac{g'(z)}{g(z)}$ is holomorphic at p . This shows that $\frac{f'(z)}{f(z)}$ has a simple pole at p , and

$$\operatorname{res}_p \frac{f'(z)}{f(z)} = \operatorname{ord}_p f$$

tells us the order of the zero of f at p . This proves

Proposition 4.3.1 (Zero-Counting Integral) *Let f be meromorphic in U , and let γ be a null homologous cycle in U that does not meet any of the zeroes or poles of f . Then*

$$\int_{\gamma} \frac{f'(z)}{f(z)} dz = \sum_p n(\gamma, p) \cdot \operatorname{ord}_p f$$

Example 4.3.1 Let f be a polynomial of degree N . Then

$$\left| \frac{f'(z)}{f(z)} \right| \approx \frac{N}{|z|}$$

so that f has precisely N zeroes in any sufficiently large disk.

We also give a geometric interpretation. Suppose that γ encloses the singularities p_k of f just once, i.e. that $n(\gamma, p_k) = 1$. By the chain rule

$$\begin{aligned} n(f \circ \gamma, 0) &= \int_{f \circ \gamma} \frac{dz}{z} \\ &= \int_{\gamma} \frac{f'(z)}{f(z)} dz \\ &= \sum_p n(\gamma, p) \cdot \operatorname{ord}_p f. \end{aligned}$$

This means that we can compute the number of zeroes of a meromorphic function f inside (say) a circle by determining the winding number of the image of the circle under f with respect to 0.

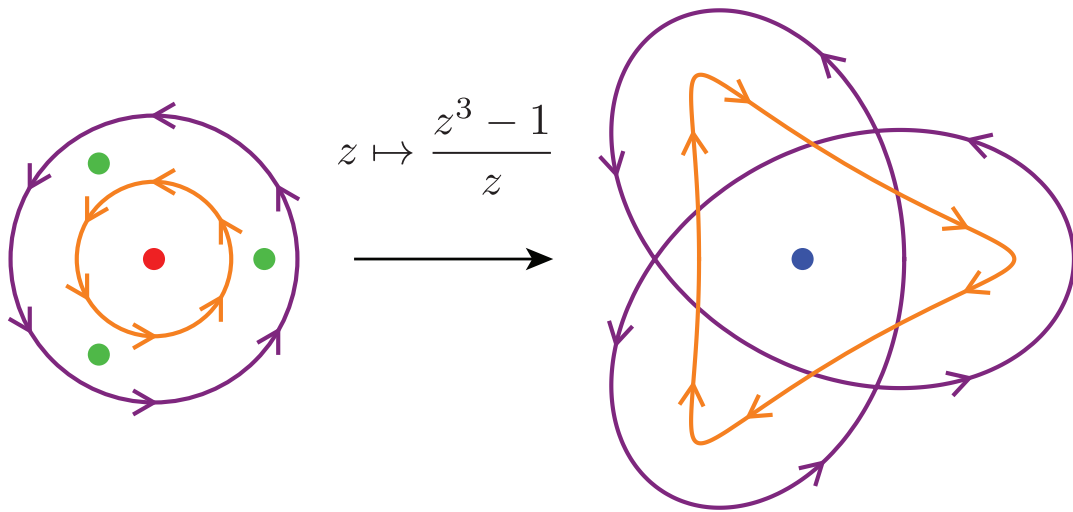


Figure 4.9 Zero counting and winding numbers

That we can express the number of zeroes of a function (which is an integer) as an integral (which depends notoriously continuously on parameters) suggests that we look at how small distortions of holomorphic functions affect their zeroes. This is the content of the following theorem.

Theorem 4.3.2 (Rouché) *Let $\overline{D} \subset U$ be a disk. Let f and g be holomorphic in U such that*

$$|g(z)| < |f(z)|$$

for all $z \in \partial D$. Then f and $f + g$ have the same number of zeroes in D .

Proof. For $0 \leq t \leq 1$, consider

$$f_t(z) = f(z) + tg(z) .$$

For $z \in \partial D$ we have

$$\begin{aligned} |f_t(z)| &= |f(z) + tg(z)| \\ &\geq |f(z)| - t|g(z)| \\ &\geq |f(z)| - |g(z)| \\ &> 0 . \end{aligned}$$

Thus, the number of zeroes of $f + tg$ in D is given by the integral

$$\frac{1}{2\pi i} \int_{\partial D} \frac{f' + tg'}{f + tg} dz$$

which is continuous in t and an integer, thus constant. 

The following theorem shows that not having zeroes survives limits, unless catastrophe strikes.

Theorem 4.3.3 (Hurwitz) *Let f_n be holomorphic and assume that $f_n \rightarrow f$ locally uniformly in U . Assume also that $f_n(z) \neq 0$ in U . Then either $f \equiv 0$, or $f(z) \neq 0$ for all $z \in U$ as well.*

Proof. Let's assume by contradiction that $f \not\equiv 0$ in U and that $f(p) = 0$ for some $p \in U$. As $f \not\equiv 0$, we can find a disk $\overline{D} \subset U$ containing p such that $f(z) \neq 0$ for all $z \in \partial D$. Then $|f(z)| \geq \epsilon$ for some $\epsilon > 0$ and all $z \in \partial D$. As $f_n \rightarrow f$ locally uniformly, there is an index n such that

$$|f_n(z) - f(z)| < \epsilon < |f(z)|$$

for all $z \in \partial D$. By **Rouché's Theorem 4.3.2**, f and f_n have the same number of zeroes in D . But this means that f_n also has a zero in D , contradicting the assumption. 

Corollary 4.3.4 *Let f_n be holomorphic and assume that $f_n \rightarrow f$ locally uniformly in U . Assume also that $f_n(z)$ is injective in U . Then either $f \equiv 0$, or $f(z)$ is injective as well.*

Proof. Assume $f(z_1) = w = f(z_2)$, and let $D_\epsilon(z_1)$ and $D_\epsilon(z_2)$ be disjoint disks in which f assumes the value w only once each at z_1 and z_2 , respectively. Then

$$\frac{1}{2\pi i} \int_{\partial D_\epsilon(z_k)} \frac{f'(z)}{f(z) - w} dz = 1$$

But this contradicts that for n sufficiently large,

$$\frac{1}{2\pi i} \int_{\partial D_\epsilon(z_1) \cup \partial D_\epsilon(z_2)} \frac{f'_n(z)}{f_n(z) - w} dz \leq 1.$$

This proof didn't use **Rouché's Theorem 4.3.2**. Here is a more general version of the corollary that does:

Corollary 4.3.5 *Let f_n be holomorphic and assume that $f_n \rightarrow f$ locally uniformly in U . Assume also that $f_n(z)$ assumes the value w at most N times in U , counted with multiplicity. Then either $f \equiv 0$, or $f(z)$ assumes the value w at most N times in U as well, counted with multiplicity.*

Proof. Without loss, let $w = 0$, and assume by contradiction that $f \not\equiv 0$ has at least $N + 1$ zeroes z_k in U . Let $\overline{D}_\epsilon(z_k) \subset U$ be disjoint disks. Then there is $\delta > 0$ such that $|f(z)| > \delta$ for all $z \in \partial D_\epsilon(z_k)$. Thus, for sufficiently large n ,

$$|f_n(z) - f(z)| \leq \delta \leq |f(z)|$$

so that by **Rouché's Theorem 4.3.2**, f_n and f have the same number of zeroes in $\bigcup D_k$. 

This also reproves the injectivity statement above.

4.4 RUNGE THEORY

The goal of this section is to approximate holomorphic functions on compact sets by rational functions and polynomials. The proof of Runge's first theorem has two key ingredients: We represent the holomorphic function on K by an integral over a path enclosing the compact set, and we approximate this integral by Riemann sums.

Proposition 4.4.1 *Let K be a compact set and f be continuous in a set U containing K . Let γ be a path in $U - K$. For every $\epsilon > 0$ there is a rational function R_γ with poles only on γ such that for all $z \in K$,*

$$\left| R_\gamma(z) - \int_\gamma \frac{f(w)}{w - z} dw \right| < \epsilon.$$

Proof. For a parametrization $\gamma : [0, 1] \rightarrow U - K$ of γ , we have

$$\int_\gamma \frac{f(w)}{w - z} dw = \int_0^1 \frac{f(\gamma(t))}{\gamma(t) - z} \gamma'(t) dt.$$

For given $\epsilon > 0$, we approximate the integral by its Riemann sum: Choose $n > 0$ large enough so that

$$\left| \int_0^1 \frac{f(\gamma(t))}{\gamma(t) - z} \gamma'(t) dt - \frac{1}{n} \sum_{k=1}^n \frac{f(\gamma(k/n))}{\gamma(k/n) - z} \gamma'(k/n) \right| < \epsilon.$$

This can be done uniformly for $z \in K$ as K is compact and the integrand is continuous in z . As the Riemann sum side is a rational function in z with simple poles on γ , we are done. 

We next show that any compact set $K \subset U$ can be 'enclosed' by a curve γ . More precisely:

Lemma 4.4.2 *Let $K \subset U$. Then there is a polygonal curve $\gamma \subset U - K$ such that for all $z \in K$ we have $n(z, \gamma) = 1$.*

Proof. The proof is similar to the beginning of our second proof of the Cauchy integral theorem. Let $\epsilon < d(K, \mathbb{C} - U)$, and tile $\mathbb{C}$ with squares of diameter ϵ . Let Σ be the finite set of squares that intersect K , and let γ be the boundary cycle of all squares in Σ . This cycle consists of those edges of squares in Σ that do not have another square of Σ adjacent. In particular, these edges lie outside of K but inside of U by the definition of ϵ . For $z \in K$ and not on the boundary of any of the squares in Σ , we have for all squares $S \in \Sigma$

$$n(z, \partial S) = \begin{cases} 1 & \text{for } z \in S \\ 0 & \text{for } z \notin S \end{cases}.$$

Thus $n(z, \gamma) = 1$. By continuity of the winding number integral, this extends to all of K .

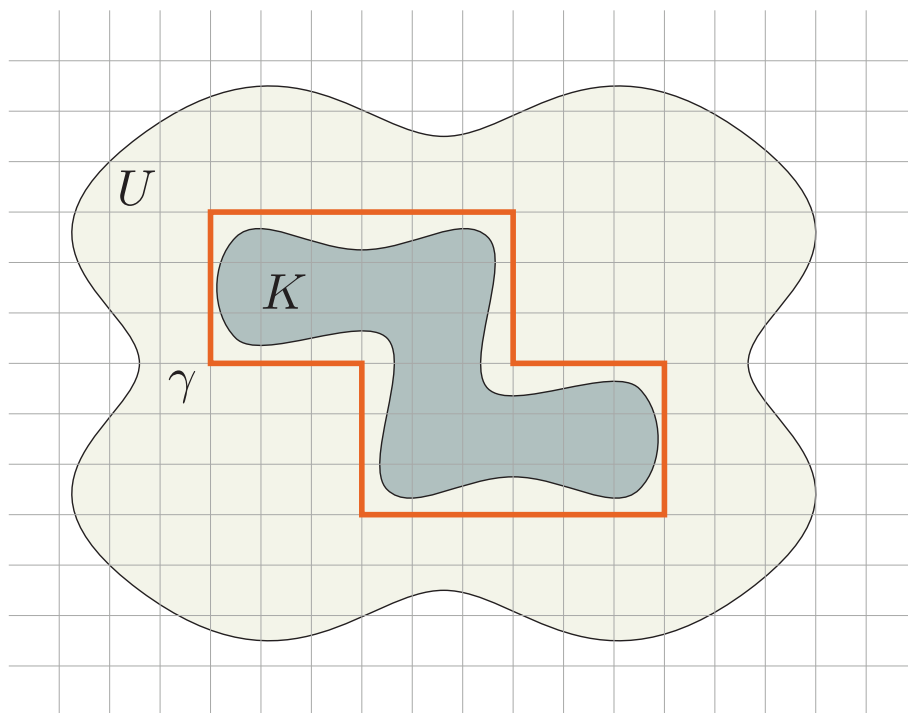


Figure 4.10 Grid path

Combining proves

Theorem 4.4.3 (Runge) *Let f be a holomorphic function defined in an open set U containing a compact set K . Then f can be approximated uniformly on K by rational functions with simple poles in $U - K$.*

Proof. Choose a curve γ as in **Lemma 4.4.2**. By Cauchy's integral theorem, we have for all $z \in K$

$$f(z) = \frac{1}{2\pi} \int_{\gamma} \frac{f(w)}{w - z} dw.$$

Given $\epsilon > 0$, **Proposition 4.4.1** provides a rational function R with simple poles on γ such that

$$\left| R_\gamma(z) - \frac{1}{2\pi} \int_\gamma \frac{f(w)}{w-z} dw \right| < \epsilon.$$

The precise control we have over the placement of the poles of the approximating rational function allows us to push them to infinity. We first show that we can push the poles locally.

Lemma 4.4.4 *Let K be compact, and let $z_0, z_1 \in \mathbb{C} - K$ such that $|z_1 - z_0| < d(z_1, K)$. Then $\frac{1}{z-z_0}$ can be approximated uniformly on K by a polynomial in $\frac{1}{z-z_1}$.*

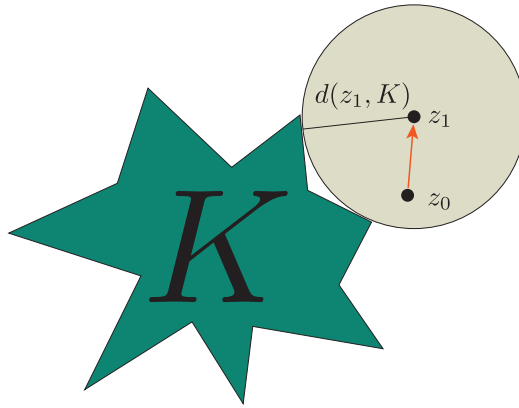


Figure 4.11 Pole pushing

Proof. We write

$$\begin{aligned} \frac{1}{z-z_0} &= \frac{1}{z-z_1} \frac{1}{1 - \frac{z_0-z_1}{z-z_1}} \\ &= \sum_{k=0}^{\infty} \frac{(z_0-z_1)^k}{(z-z_1)^{k+1}} \end{aligned}$$

where the geometric series converges because

$$\left| \frac{z_0-z_1}{z-z_1} \right| \leq \frac{|z_0-z_1|}{d(z_1, K)} < 1$$

Note that if we can approximate $\frac{1}{z-z_0}$ uniformly by a polynomial in $\frac{1}{z-z_1}$, then we can also approximate a polynomial in $\frac{1}{z-z_0}$ uniformly by a polynomial in $\frac{1}{z-z_1}$.

Corollary 4.4.5 *Let K be compact such that $\mathbb{C} - K$ is connected, and let $z_0 \in \mathbb{C} - K$. Then for all $z_1 \in \mathbb{C} - K$, $\frac{1}{z-z_0}$ can be approximated uniformly on K by a polynomial in $\frac{1}{z-z_1}$.*

Proof. The set U of all $z_1 \in \mathbb{C} - K$ such that $\frac{1}{z-z_0}$ can be approximated uniformly on K by a polynomial in $\frac{1}{z-z_1}$ is non-empty as it contains z_0 , and open by **Lemma 4.4.4**. As $\mathbb{C} - K$ is connected, $U = \mathbb{C} - K$. 

We can now prove:

Theorem 4.4.6 (Runge's Second Theorem) *Let f be a holomorphic function defined in an open set U containing a compact set K . Assume that $\mathbb{C} - K$ is connected. Then f can be approximated uniformly on K by polynomials.*

Proof. By **Runge's First Theorem 4.4.3**, we can approximate f uniformly on K by a rational function with simple poles outside of K . As $\mathbb{C} - K$ is connected **Corollary 4.4.5** says that we can approximate this rational function by a polynomial in $\frac{1}{z-z_0}$ for any $z_0 \in \mathbb{C} - K$.

Thus, it suffices to show that we can approximate for *one* $z_0 \in \mathbb{C} - K$ the rational function $\frac{1}{z-z_0}$ uniformly on K by polynomials.

Let R be so large that $K \subset D_R(0)$. For z_0 with $|z_0| > R$, the partial sums of the geometric series

$$\frac{1}{z-z_0} = -\frac{1}{z_0} \frac{1}{1-\frac{z}{z_0}} = -\sum_{k=0}^{\infty} \frac{z^k}{z_0^{k+1}}$$

give us a uniform approximation of $\frac{1}{z-z_0}$ by polynomials on K . ◻

As an application, we construct now a sequence of polynomials that has a pointwise limit in $\mathbb{C}$ to a discontinuous function. More precisely:

Theorem 4.4.7 *There is a sequence of polynomials p_n such that*

1. $p_n(0) \rightarrow 1$;
2. $p_n(z) \rightarrow 0$ pointwise for $z \neq 0$;
3. p_n converges locally uniformly in $\mathbb{C} - \mathbb{R}^{\geq 0}$ to 0.

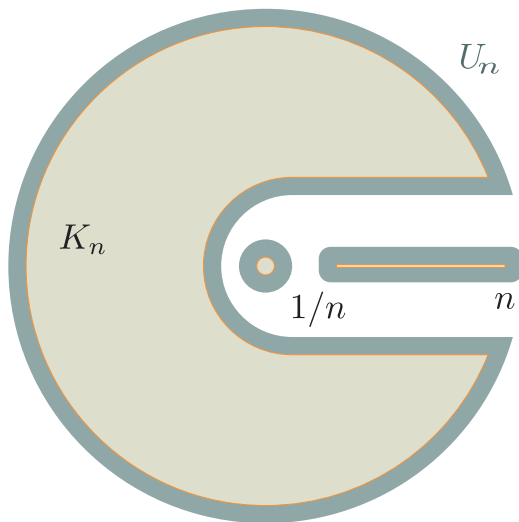


Figure 4.12 Pointwise, not uniform

Proof. The polynomials are constructed using **Runge's Theorem 4.4.6**. Fix $n > 0$, and let

$$K_n = \{0\} \cup \left[\frac{1}{n}, n \right] \cup \left\{ z : \frac{1}{n} \leq |z| \leq n, d(z, \mathbb{R}^{\geq 0}) \geq \frac{1}{n} \right\}.$$

Note that the K_n are an increasing sequence of compact sets whose union is all of $\mathbb{C}$, and whose complements are connected. Let $U_n = \{z : d(z, K_n) < \frac{1}{4n}\}$. This set still has three disjoint components. We define a holomorphic

function f_n on U_n by setting it 1 on $D_{1/4n}(0)$ and 0 on the two other components of U_n . By **Runge's Theorem 4.4.6**, there is a polynomial p_n that approximates f_n uniformly on K_n so that

$$|f(z) - p_n(z)| \leq \frac{1}{n} \quad \text{for all } z \in K_n.$$

We immediately have that $p_n(0) \rightarrow 1$. For any $z_0 \notin \mathbb{R}^{\geq 0}$, let $D_\epsilon(0)$ be a neighborhood of z_0 not meeting $\mathbb{R}^{\geq 0}$. The, for n sufficiently large, $D_\epsilon(0) \subset K_n$, and f_n will converge uniformly on $D_\epsilon(0)$ to 0. Finally, for $z_0 \in \mathbb{R}^{>0}$, z will lie for n large enough in the component $[\frac{1}{n}, n]$ of K_n on which f is zero, so $p_n(z_0) \rightarrow 0$ pointwise. $\square$

As another application, we show that the unit disk is a so-called *Stein manifold*. This is done by proving

Theorem 4.4.8 *There are three holomorphic functions $f_1, f_2, f_3 : \mathbb{D} \rightarrow \mathbb{C}$ such that the map $f : z \mapsto (f_1(z), f_2(z), f_3(z))$ is injective, has non-vanishing derivative $(f'_1(z), f'_2(z), f'_3(z))$ for all $z \in \mathbb{D}$, and that the image $f(\mathbb{D})$ is a closed subset of $\mathbb{C}^3$.*

This implies that $f(\mathbb{D})$ is a closed submanifold of a complex vector space, which is one way of saying that $\mathbb{D}$ is a Stein manifold.

Proof. We will construct two holomorphic functions $f_1, f_2 \in O(\mathbb{D})$ such that for every sequence $z_n \in \mathbb{D}$ with $z_n \rightarrow z_0 \in \partial\mathbb{D} - 0$, $|f_1(z_n)|^2 + |f_2(z_n)|^2 \rightarrow \infty$. Then these two functions together with $f_3(z) = \frac{1}{1-z}$ will satisfy our needs.

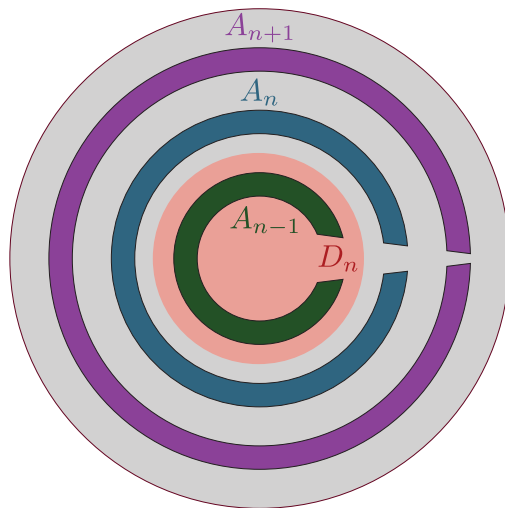


Figure 4.13 Constructing f_1

For the first function f_1 , consider a sequence A_n of disjoint horseshoes obtained from annuli by removing a wedge centered at the point 1 and opening towards 0. We don't care about the width of these horseshoes, but do want them to approach the unit circle. Our goal is to construct the function $f_1 \in O(\mathbb{D})$ so that $|f_1| > 2^n$ on A_n . Take compact nested disks $D_n \subset \mathbb{D}$ such that ∂D_n separates A_{n-1} from A_n , or more precisely such that

1. $\bigcup_n D_n = \mathbb{D}$;
2. $A_1 \cup \dots \cup A_{n-1} \subset D_n$;
3. $D_n \cap A_n = \emptyset$.

Then the compact set $A_n \cup D_n$ has a connected complement. We use **Runge's Theorem 4.4.6** to inductively define polynomials p_n . We begin with $p_0 = 3$. Now suppose that $p_0, \dots, p_{n-1}$ are constructed. Let

$$h_n(z) = \begin{cases} 0 & \text{for } z \in D_n \\ 2^n + 2 - p_0(z) - \cdots - p_{n-1}(z) & \text{for } z \in A_n \end{cases},$$

and let p_n approximate f uniformly on $A_n \cup D_n$ so that

$$|p_n(z)| < \frac{1}{2^n} \quad \text{for } z \in D_n$$

and

$$|p_0(z) + \cdots + p_n(z)| > 2^n + 1 \quad \text{for } z \in A_n.$$

By the first inequality, the series $\sum_{n=1}^{\infty} p_n(z)$ converges locally uniformly in $\mathbb{D}$ to a function f_1 . On the other hand the second inequality tells us that for $z \in A_n$,

$$\begin{aligned} |f_1(z)| &\geq |p_0(z) + \cdots + p_n(z)| - \sum_{k=n+1}^{\infty} |p_k(z)| \\ &\geq 2^n + 1 - \sum_{k=n+1}^{\infty} 2^{-k} \\ &\geq 2^n. \end{aligned}$$

This is all we need to know about f_1 .

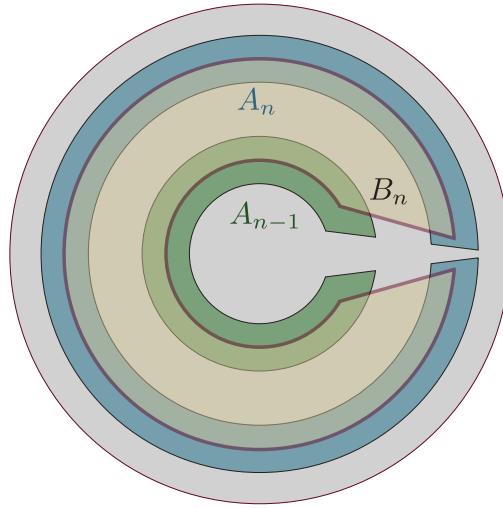


Figure 4.14 Constructing f_2

Next, we construct f_2 . To this end, let B_n be another sequence of horseshoe domains so that B_n covers the annulus between A_{n-1} and A_n except for a trapezoid that has the real axis as symmetry axis. As above, we can find a function $f_2 \in O(\mathbb{D})$ such that $|f_2(z)| \geq 2^n$ on B_n .

If we have a sequence $z_k \rightarrow z_0 \in \partial\mathbb{D} - 1$, z_k will lie for k large enough in the sets A_n or B_n with increasing n , so that either way $|f_1(z_n)|^2 + |f_2(z_n)|^2 \rightarrow \infty$ as desired.

5 Conformal Mappings

In this chapter, we study *Geometric Function Theory*, which is concerned with the behavior of holomorphic functions as maps between domains in the plane. We know already that such maps are conformal and map open sets to open sets.

5.1 MONTEL'S THEOREM

Definition 5.1.1 A sequence $f_n \in O(U)$ is locally uniformly bounded if for every point $p \in U$ there is a disk $D_r(p)$ and a constant C such that

$$|f_n(z)| \leq C$$

for all $z \in D_r(p)$.

Note that this definition allows that each individual f_n is unbounded in U .

Definition 5.1.2 A sequence $f_n \in O(U)$ is locally equicontinuous if for every point $p \in U$ there is a disk $D_r(p)$ such that the sequence f_n is equicontinuous in $D_r(p)$, i.e. such that for every $\epsilon > 0$ there is a $\delta > 0$ such that for all n and all $z, z' \in D_r(p)$ with $|z - z'| < \delta$ we have $|f_n(z) - f_n(z')| < \epsilon$.

Lemma 5.1.3 Let $Q \subset U$ be a countable and dense subset, and let f_n be a pointwise bounded sequence of functions in U . Then f_n has a subsequence that converges on Q .

Proof. Let $\{q_n\}$ be an enumeration of Q . Choose subsequence f_n^1 of f that converges at q_1 . Continue inductively to choose subsequences f_n^{k+1} of f_n^k that converge at $q_1, \dots, q_{k+1}$. Then the diagonal sequence f_n^n converges on Q . 

Theorem 5.1.4 (Arzela-Ascoli) Let f_n be pointwise bounded and locally equicontinuous in U . Then f_n has a locally uniform convergent subsequence.

Proof. By the lemma, we can assume that f_n converges on $Q = \mathbb{Q}^2 \cap U$. Let $p \in U$, and $D = \overline{D}_r(p) \subset U$ be a disk on which f_n are equicontinuous. Let $\epsilon > 0$, and find $\delta > 0$ such that for all $z, z' \in D$ with $|z - z'| < \delta$ we have for all n that $|f_n(z) - f_n(z')| < \epsilon$.

As Q is dense in U , the disks $D_\delta(q)$ cover D . As D is compact, there are $q_1, \dots, q_n$ such that already

$$\bigcup_{k=1}^n D_\delta(q_k) \supset D.$$

As the f_n converge pointwise on Q , there is an index N such that for all $m, n > N$ and all $k = 1, \dots, n$ we have

$$|f_m(q_k) - f_n(q_k)| < \epsilon.$$

Now we verify that f_n converges uniformly in D . For $z \in D$, find k such that $|z - q_k| < \delta$. Then

$$\begin{aligned} |f_m(z) - f_n(z)| &< |f_m(z) - f_m(q_k)| + |f_m(q_k) - f_n(q_k)| + |f_n(q_k) - f_n(z)| \\ &< 3\epsilon. \end{aligned}$$

The unpleasant part of this theorem is verifying the assumption of local equicontinuity. For holomorphic functions, we can avoid this:

Proposition 5.1.5 *Let $f_n \in O(U)$ be locally uniformly bounded. Then the f_n are locally equicontinuous.*

Proof. For $p \in U$ find a neighborhood $D = D_{2r}(p)$ such that $|f_n(z)| \leq C$ for all n and all $z \in D$. Then for all $z, z' \in B_r(p)$ we estimate

$$\begin{aligned} |f_n(z) - f_n(z')| &= \left| \frac{1}{2\pi i} \int_{\partial D_{2r}(p)} \frac{f_n(w)}{w - z} - \frac{f_n(w)}{w - z'} dw \right| \\ &\leq \frac{1}{2\pi} 2\pi r C \frac{|z - z'|}{r^2} \\ &= 2C \frac{|z - z'|}{r} \\ &< \epsilon \end{aligned}$$

if we let $\delta = \frac{\epsilon r}{2C}$ and choose $|z - z'| < \delta$.



Corollary 5.1.6 (Montel) *A locally uniformly bounded sequence of holomorphic functions has a locally uniformly convergent subsequence.*

Under the mild additional assumption that the sequence of functions converges on a set with an accumulation point, we can avoid taking subsequences:

Corollary 5.1.7 (Vitali) *Let f_n be a locally uniformly bounded sequence of holomorphic functions in a domain U . Let $z_k \in U$ converge to $z_0 \in U$, where $z_0 \neq z_k$. Assume that for all k the sequence $f_n(z_k)$ converges. Then f_n converges locally uniformly in U .*

Proof. By Montel's theorem, there is a subsequence of the f_n that converges uniformly. Thus, it suffices to show that any two convergent subsequences have the same limit. But these two limits agree at the points z_k . By the **Identity Principle 2.5.5**, the limits agree in all of U .



As an application, we show a general result about integrals of holomorphic functions that depend continuously on a parameter.

Theorem 5.1.8 *Let U be a domain and $t \in [0, 1]$. and $f_t \in O(U)$ for $t \in [0, 1]$. Assume that*

1. *For each $t \in [0, 1]$, $z \mapsto f_t(z)$ is holomorphic in U ;*
2. *For each $z \in U$, $t \mapsto f_t(z)$ is continuous in t ;*
3. *For every compact set $A \subset U$ there is a constant C such that $|f_t(z)| \leq C$ for all $t \in [0, 1]$ and all $z \in A$.*

Then

1. *$f_t(z)$ and $f'_t(z)$ are continuous in $[0, 1] \times U$;*
2. *$g(z) = \int_0^1 f_t(z) dt$ is holomorphic;*
3. *$g'(z) = \int_0^1 f'_t(z) dt$.*

Proof. Let $t_n \in [0, 1]$ be a sequence with $t_n \rightarrow t_0$, and abbreviate $f_n = f_{t_n}$. Let $z_k \rightarrow z_0$ in U . Then the functions f_n satisfy the assumptions of **Vitali's Theorem 5.1.7**, hence converge locally uniformly to $f_0(z)$. This implies $f_n(z_n) \rightarrow f_0(z_0)$, so that $(t, z) \mapsto f_t(z)$ is continuous. By **Weierstrass' Theorem 2.4.4**, $f'_n(z) \rightarrow f'_0(z)$ locally uniformly as well, and the above argument shows that $(t, z) \mapsto f'_t(z)$ is also continuous.

To prove the second statement, let

$$g_n(z) = \frac{1}{n} \sum_{k=1}^n f_{k/n}(z) .$$

Then

$$g_n(z) \rightarrow g(z) = \int_0^1 f_t(z) dt$$

for each fixed $z \in U$. As averages the g_n are locally uniformly bounded and holomorphic in z , so satisfy the assumptions of **Vitali's Theorem 5.1.7**. Hence, the convergence of the g_n is locally uniform. This shows that g is holomorphic, and that $g'_n \rightarrow g'$, i.e.

$$\frac{1}{n} \sum_{k=1}^n f'_{k/n}(z) \rightarrow g'(z) .$$

As $f'_t(z)$ is continuous in t , the left hand side also converges to $\int_0^1 f'_t(z) dt$.



5.2 THE RIEMANN MAPPING THEOREM

In this section, we will prove the Riemann mapping theorem that says that all domains “without holes” are either the complex plane $\mathbb{C}$ or biholomorphic to the unit disk. The intuitive notion “without holes” is made precise by the concept of a simply connected domain. In topology, one defines:

Definition 5.2.1 A connected domain U is simply connected if every closed curve in U is null homologous.

Here are two equivalent notions. We claim

Theorem 5.2.2 For a connected domain $U \subset \mathbb{C}$ the following are equivalent:

1. Every closed curve in U is null homotopic;
2. U is simply connected;
3. $\hat{\mathbb{C}} - U$ is connected.

Proof. We first prove $1 \Rightarrow 2$. If U is simply connected, then trivially every closed curve in U has winding number zero with respect to any point outside U , as the winding number does not change under homotopies that don't cross the reference point. This shows $1 \Rightarrow 2$.

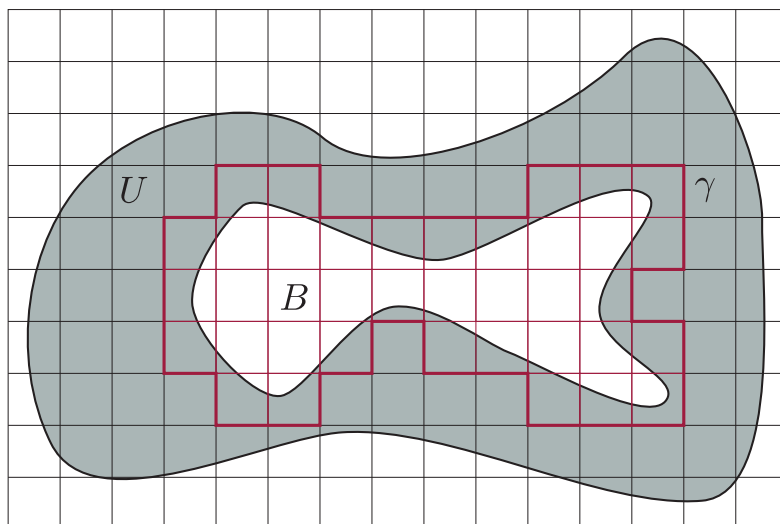


Figure 5.1 Constructing a curve with positive winding number

For $2 \Rightarrow 3$, let's assume that $\hat{\mathbb{C}} - U$ is not connected. Let B be a component of $\hat{\mathbb{C}} - U$ that does not contain ∞ . Then B is bounded and closed, and hence compact. It therefore has a positive distance 3ϵ to its complement in $\hat{\mathbb{C}} - U$. Overlay $\mathbb{C}$ with an ϵ -grid of horizontal and vertical lines, and let S_i be the (finitely many) grid $\epsilon \times \epsilon$ squares that cover B . The boundary γ of their union lies in U . Pick a point $b \in B$ that is on no grid line (which can be achieved by shifting the grid if necessary). Then $b \in S_i$ for precisely one index i , so that the winding number of the cycle $c = \sum_i \partial S_i$ with respect to b is one. That cycle c , however, has the same winding number with respect to b as γ . Hence γ is not null homologous in U .

For $3 \Rightarrow 2$, we can assume $\hat{\mathbb{C}} - U$ contains not just ∞ . Then note that $n(\gamma, p) = 0$ for any cycle γ in U and $p \in \mathbb{C} - U$ sufficiently large. As the winding number is continuous in p as long as p stays away from γ , $n(\gamma, p)$ is constant in $\hat{\mathbb{C}} - U$, and hence identically 0, so that γ is null homologous.

To prove $2 \Rightarrow 1$, we will use the Riemann mapping theorem. The claim follows, because both $\mathbb{C}$ and $\mathbb{D}$ are simply connected. 

Corollary 5.2.3 A domain U is simply connected if and only if

$$\int_{\gamma} f(z) dz = 0$$

for all closed curves γ in U and all $f \in O(U)$.

Proof. One direction follows from **Cauchy's Theorem 4.1.7**. Assume now that U is *not* simply connected. Then there is a cycle γ in U that is not null homologous, i.e., there is $p \in \mathbb{C} - U$ with $n(\gamma, p) \neq 0$. Then the function

$$f(z) = \frac{1}{z - p}$$

is holomorphic in U but

$$\int_{\gamma} f(z) dz \neq 0.$$

Corollary 5.2.4 *Let U be simply connected and $f : U \rightarrow V$ be biholomorphic. Then $V = f(U)$ is also simply connected.*

Proof. Chain rule and corollary above. 

Lemma 5.2.5 *Let $U \neq \mathbb{C}$ be simply connected. Then there is an injective holomorphic map $f : U \rightarrow \mathbb{D}$.*

Proof. Without loss we can assume that $0 \notin U$ (using that $U \neq \mathbb{C}$), and that $z_0 = 1 \in U$. Define

$$\begin{aligned} \log : U &\rightarrow \mathbb{C} \\ z &\mapsto \int_1^z \frac{1}{w} dw . \end{aligned}$$

This is well-defined by **Cauchy's Theorem 4.1.7**, using that U is simply connected. Now we can also define

$$\begin{aligned} \sqrt{\cdot} : U &\rightarrow \mathbb{C} \\ z &\mapsto e^{\frac{1}{2}\log(z)} . \end{aligned}$$

Note that $\sqrt{\cdot}$ is trivially injective. Moreover, not both w and $-w$ can be in $\sqrt{U}$. Otherwise, there are $z_1, z_2 \in U$ such that $w = \sqrt{z_1}$ and $-w = \sqrt{z_2}$. Then $z_1 = z_2$ and thus $w = -w = 0$ so that $z_1 = z_2 = 0$, contradicting that $0 \notin U$. We now use this to show that $\sqrt{U}$ misses an open disk.

By the **Open Mapping Theorem 2.5.6**, $\sqrt{U}$ is open. Let $D = D_\epsilon(1) \subset \sqrt{U}$ be an open disk. Then $-D \cap \sqrt{U} = \emptyset$, i.e.

$$\{z : |z + 1| < \epsilon\} \subset \mathbb{C} - \sqrt{U} .$$

Thus

$$z \mapsto \frac{\epsilon}{z + 1}$$

maps $\sqrt{U}$ into $\mathbb{D}$. 

Now we are ready to prove

Theorem 5.2.6 (Riemann Mapping Theorem) *Let $U \neq \mathbb{C}$ be simply connected, and let $z_0 \in U$. Then there is a unique biholomorphic map $f : U \rightarrow \mathbb{D}$ with $f(z_0) = 0$ and $f'(z_0) > 0$.*

Proof. Without loss, we can assume that $z_0 = 0$. Let

$$\mathcal{R} = \{f : U \rightarrow \mathbb{D} : f \text{ is injective, holomorphic, } f(0) = 0, \text{ and } f'(0) > 0\} .$$

By **Lemma 5.2.5**, there is an injective holomorphic map $U \rightarrow \mathbb{D}$. We compose it with a Möbius transformation so that 0 is mapped to 0 and the derivative at 0 is a positive real number. This shows that $\mathcal{R} \neq \emptyset$.

Let $f_n \in \mathcal{R}$ be a sequence such that

$$f'_n(0) \rightarrow \sup\{f'(0) : f \in \mathcal{R}\} .$$

Note that the supremum is finite. To see this, let $\overline{D}_\epsilon(0) \subset U$ be a disk. Then, as in the **Cauchy Estimate 2.4.2**, we have (using that $|f| < 1$)

$$|f'(0)| = \left| \frac{1}{2\pi i} \int_{\partial D_\epsilon(0)} \frac{f(z)}{z^2} dz \right| \leq \frac{1}{\epsilon}.$$

By **Montel's Theorem 5.1.6**, the sequence f_n has a locally uniform convergent subsequence $f_{n_k} \rightarrow f$. By **Weierstrass' Theorem 2.4.4**, f is holomorphic, and $f'_{n_k} \rightarrow f'$. In particular, $f'(0) \neq 0$, and f cannot be identically zero. By **Hurwitz' Theorem 4.3.3**, f is injective. We claim that f is also surjective.

Suppose not. We will reach a contradiction by explicitly increasing $f'(0)$. By assumption, there is $w \in \mathbb{D}$ such that $w \notin f(U)$. Postcompose f with a Möbius transformation μ so that $\mu(w) = 0$ and $\mu(0) > 0$. Then $0 \notin V = \mu \circ f(U) \subset \mathbb{D}$. In V , we can again define $\log$ and $\sqrt{\cdot}$, as V is simply connected by **Corollary 5.2.4**. We normalize $\sqrt{\cdot}$ so it is a positive real number at $\mu(0)$. Let

$$g = \nu \circ \sqrt{\mu \circ f}$$

where ν is an automorphism of $\mathbb{D}$ so that 0 is mapped to 0 by g and such that $g'(0) > 0$. Then $g \in \mathcal{R}$. We show that $g'(0) > f'(0)$. Note that $\rho(z) = \mu^{-1}(\nu^{-1}(z)^2)$ maps $\mathbb{D}$ into $\mathbb{D}$ and $0 \mapsto 0$. By **Schwarz' Lemma 2.5.9**, $|\rho'(0)| < 1$. (We could also have estimated this directly). Strict inequality holds because ρ is not an automorphism. As $f = \rho \circ g$, we have $|f'(0)| < |g'(0)|$, contradicting the definition of f .

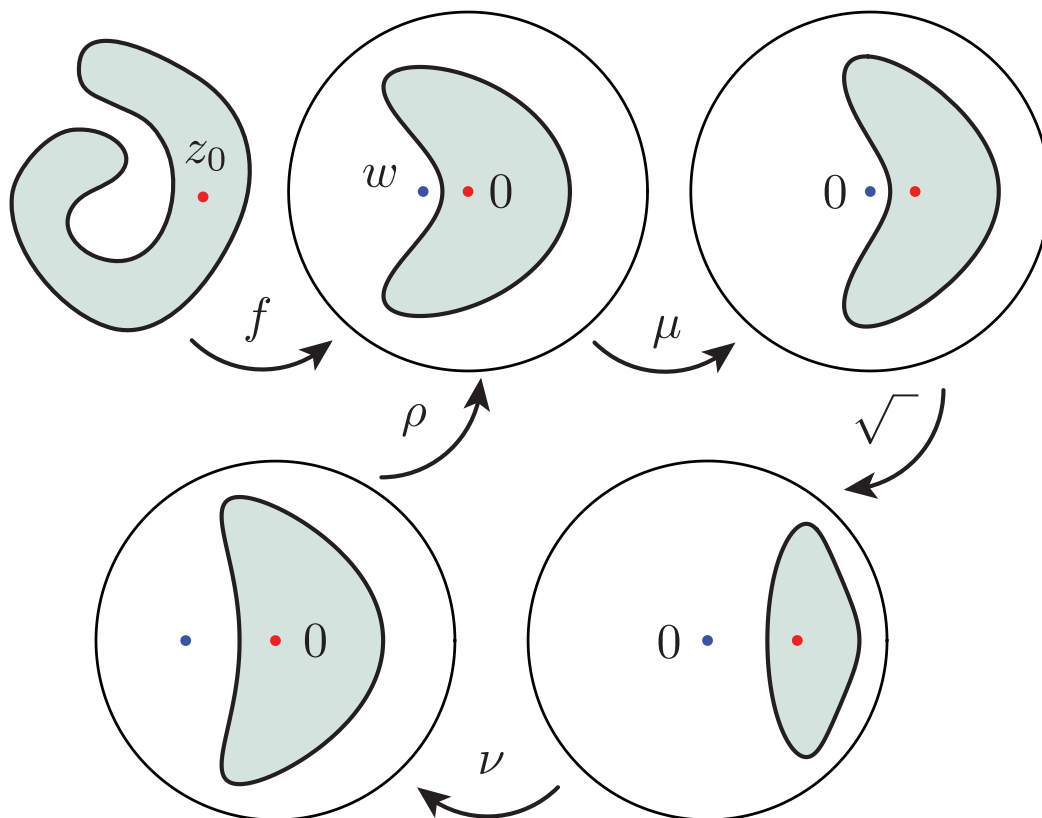


Figure 5.2 Improving Surjectivity

5.3 EXTENSION TO THE BOUNDARY

The Riemann mapping function provides us with a biholomorphic map between the interiors of two domains, but nothing is said about a possible extension to the boundaries. That this can be problematic can be seen in simple

domains with slits.

There is an extensive theory about the boundary behavior of holomorphic maps. For bounded domains, the best result is due to Carathéodory.

Definition 5.3.1 A Jordan curve is the continuous and injective image of a circle. A Jordan domain is a connected domain bounded by a Jordan curve.

By the Jordan Curve Theorem, any Jordan domain is simply connected.

Theorem 5.3.2 (Carathéodory) Let U and V be Jordan domains, and $f : U \rightarrow V$ be a biholomorphic map. Then f has a unique extension to a homeomorphism $\bar{f} : \bar{U} \rightarrow \bar{V}$.

The proof requires specialized tools from planar topology and is technical. We will prove the following special case, which is good enough for all applications we have in mind, and shows the basic idea.

Theorem 5.3.3 Let U and V be domains with piecewise smooth boundary, and $f : U \rightarrow V$ a biholomorphic map. Then f has a unique continuous extension to the boundary.

Proof. We will show that f is uniformly continuous in U . Then the existence of the extension follows from point-set topology: a uniformly continuous map from a subset U of a metric space into a complete metric space has a unique continuous extension to the closure of U .

Assume by contradiction that f is not uniformly continuous in U . Then there exists $\epsilon > 0$ such that for all $n > 0$ there are $z_n, z'_n \in U$ with $|z_n - z'_n| < \frac{1}{n}$ and $|f(z_n) - f(z'_n)| > 3\epsilon$. By compactness of $\bar{U}$, we can assume that z_n and z'_n converge to a common limit $z_0 \in \partial U$.

By compactness of $\bar{V}$ we can also assume that the sequences $w_n = f(z_n)$ and $w'_n = f(z'_n)$ converge to different points w_0, w'_0 with $|w_0 - w'_0| \geq 3\epsilon$. As $f : U \rightarrow V$ is in particular a homeomorphism, we necessarily have that $w_0, w'_0 \in \partial V$. Denote by $D = D_\epsilon(w_0)$ and $D' = D_\epsilon(w'_0)$ two disks centered at w_0 and w'_0 distance ϵ apart.

Because the boundary of V is piecewise smooth, we can find continuous curves $\eta, \eta' : [0, 1) \rightarrow V$ and a monotone sequence $t_n \in [0, 1)$ with $t_n \rightarrow 1$ such that $w_n = \eta(t_n)$ and $w'_n = \eta'(t_n)$. Let $\gamma = f^{-1} \circ \eta$ and $\gamma' = f^{-1} \circ \eta'$ the preimages in U . By construction, $\gamma(t_n) = z_n$ and $\gamma'(t_n) = z'_n$. In particular, both γ and γ' have points that are arbitrarily close to z_0 . Fix $R > 0$ so that $\gamma(0)$ and $\gamma'(0)$ lie outside $D_R = D_R(z_0) \cap U$. For $0 < r < R$, denote by z_r (resp. z'_r) an intersection of γ (resp. γ') with $\partial D_r(z_0)$.

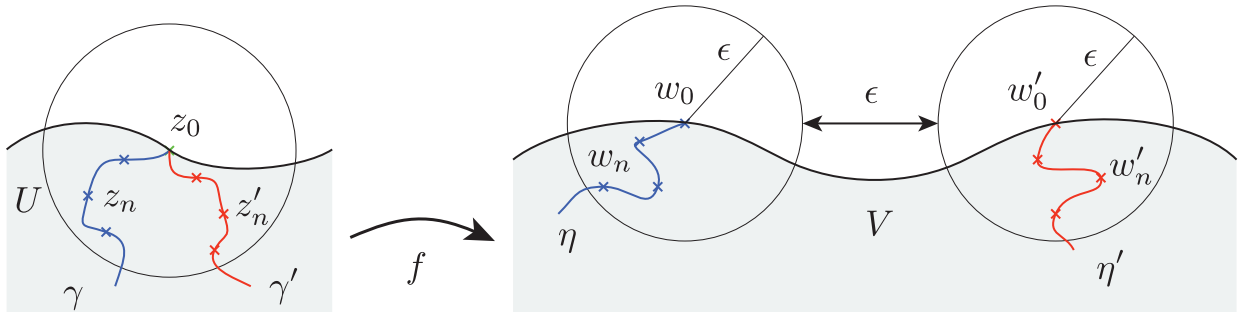


Figure 5.3 Uniform continuity

Using polar coordinates, write $z_r = re^{i\theta}$ and $z'_r = re^{i\theta'}$. Then we estimate, using a path from z_r to z'_r on $\partial D_r(z_0)$,

$$\begin{aligned}
\epsilon^2 &\leq |f(z_r) - f(z'_r)|^2 \\
&\leq \left| \int_{z_r}^{z'_r} f'(z) dz \right|^2 \\
&= \left| \int_{\theta}^{\theta'} r f'(z_0 + re^{it}) dt \right|^2 \\
&\leq \int_0^{2\pi} |f'(z_0 + re^{it})|^2 dt \cdot \int_0^{2\pi} r^2 dt \quad \text{Cauchy-Schwarz} \\
&\leq 2\pi r \int_0^{2\pi} r \cdot |f'(z_0 + re^{it})|^2 dt
\end{aligned}$$

Integrating

$$\frac{\epsilon^2}{r} \leq 2\pi \int_0^{2\pi} r \cdot |f'(z_0 + re^{it})|^2 dt$$

from ρ to R gives

$$\epsilon^2(\log R - \log \rho) \leq 2\pi \text{area } f(D_R(z_0)) .$$

While the right-hand side is bounded, the left diverges for $\rho \rightarrow 0$, and we have reached a contradiction. As the same argument applies to f^{-1} , we are done. 

5.4 THE SCHWARZ-CHRISTOFFEL FORMULA

The proof of the **Riemann Mapping Theorem 5.2.6** does not indicate how to construct the mapping function explicitly. There is an important special case where this is possible, namely when the domain is a Euclidean polygon.

We begin with a preparation.

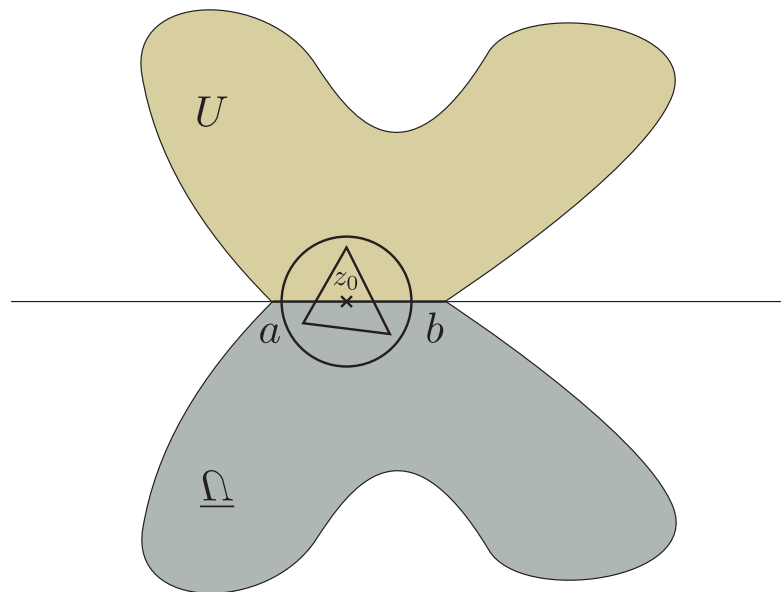


Figure 5.4 The Schwarz reflection principle

Theorem 5.4.1 (Schwarz Reflection Principle) *Let $U \subset \mathbb{H}$ be a domain and $(a, b) \subset \mathbb{R} \cap \partial U$ a piece of the boundary of U that lies on the real axis. Assume that f is holomorphic in U , continuous in $U \cup (a, b)$, and real valued on (a, b) . Then*

$$g(z) = \begin{cases} f(z) & \text{if } z \in U \cup (a, b) \\ \overline{f(\bar{z})} & \text{if } z \in \bar{U} \end{cases}$$

extends f to the domain $V = U \cup (a, b) \cup \bar{U}$.

Proof. We only have to show that f is holomorphic at points in (a, b) . For $z_0 \in (a, b)$, let $D_\epsilon(z_0) \subset V$ be a disk contained in V . We show that g is holomorphic in $D_\epsilon(z_0)$. Let $\Delta \subset D_\epsilon(z_0)$ be a triangle. To see that $\int_{\partial \Delta} g(z) dz = 0$, note that at worst (a, b) cuts Δ into two polygons Δ_1 and Δ_2 . The function g is holomorphic in the interior of Δ_j and continuous in $\bar{\Delta}_j$, so that $\int_{\partial \Delta_j} g(z) dz = 0$. Thus g is holomorphic in $D_\epsilon(z_0)$ by **Morera's Theorem 2.3.4**. 

Definition 5.4.2 *A Euclidean polygon is a bounded simply connected domain in the plane such that all boundary components consist of straight segments, meeting in a finite set of vertices.*

Theorem 5.4.3 (Schwarz-Christoffel Formula) *Let P be a Euclidean polygon with vertices at points v_k , $k = 1, \dots, n$, and interior angle α_k at v_k , labeled counterclockwise. Let*

$$a_k = \frac{\alpha_k}{2\pi} - 1.$$

Then there are real numbers $t_1 < \dots < t_n$ such that the map

$$f(z) = \lambda \int^z (w - t_1)^{a_1} \cdots (w - t_n)^{a_n} w + \mu$$

maps the upper half plane $\mathbb{H}$ biholomorphically onto P .

Proof. By the **Riemann mapping theorem 5.2.6**, there is a biholomorphic function $f : \mathbb{H} \rightarrow P$. By **Theorem 5.3.3**, this function extends as a continuous and injective function to $\mathbb{R}$. Denote by $t_k \in \mathbb{R}$ the preimages of the polygon vertices v_k . We can assume that none of the t_k is at ∞ . Using the **Reflection Principle 5.4.1**, we extend f to the lower half plane by reflecting across a segment $t_{k-1}t_k \subset \mathbb{R}$ and the corresponding edge $v_{k-1}v_k$. We then extend f back to the upper half plane, reflecting across another segment $t_{j-1}t_j$. Let's denote this second extension by $\tilde{f}$. Because $\tilde{f}$ is obtained from f by two Euclidean reflections, they differ by an orientation-preserving Euclidean isometry. This implies that $\tilde{f}(z) = af(z) + b$ for $|a| = 1$ and $b \in \mathbb{C}$.

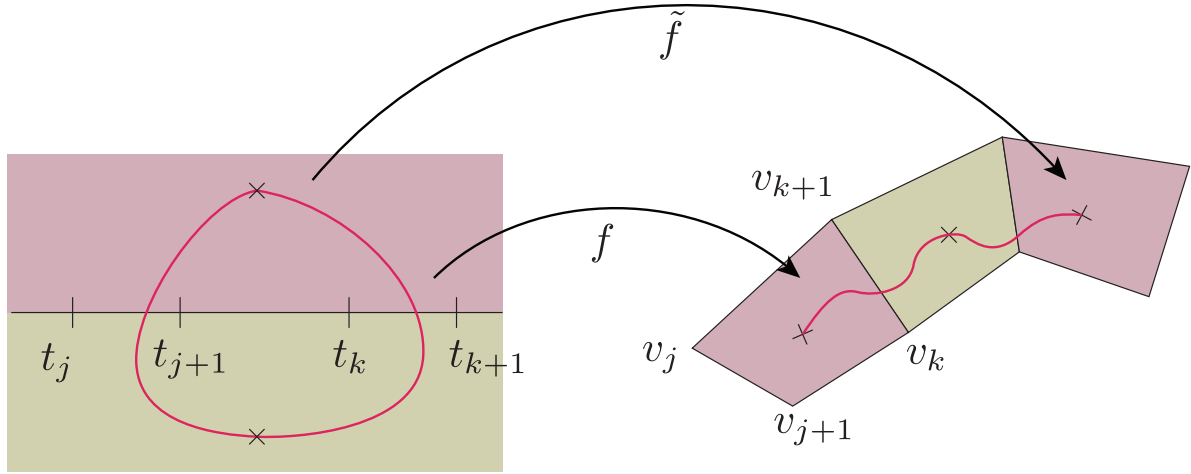


Figure 5.5 The Schwarz reflection formula

As a consequence, the function

$$g(z) = \frac{f''(z)}{f'(z)}$$

will be single valued in $\mathbb{C} - \{t_1, \dots, t_n\}$

Our next goal is to derive an explicit expression for $g(z)$. We begin by deriving an expression for g near t_k . To this end, we need to analyze the behavior of $f(z)$ near t_k .

Introduce a local coordinate $w = z - t_k$ around a fixed t_k . As g is not affected by translations and rotations of the polygon, we can assume that $f(t_k) = 0$ and that the segment $f(t_k)f(t_{k+1})$ lies on the positive real axes. Now consider the function $h(w) = f(w)^{\pi/\alpha_j}$. By the Schwarz reflection principle, this function is single-valued and bounded around $w = 0$ and hence extends holomorphically to 0 with a simple zero. Thus, we can write $h(w) = w \cdot h_0(w)$ with h_0 holomorphic and non-zero at 0. Hence

$$f(w) = h(w)^{\frac{\alpha_j}{\pi}}$$

with a regular function h and thus

$$\frac{f''}{f'}(w) = \frac{a_j}{w} + \dots$$

where we have denoted

$$a_j = \frac{\alpha_j}{\pi} - 1$$

Observe that necessarily

$$\sum a_j = -2$$

as the *exterior* angles in a Euclidean polygon add up to 2π .

Now consider the function

$$g_1(z) = g(z) - \sum_j \frac{a_j}{z - t_j}.$$

By construction, this is an entire holomorphic function. We would like to show that it is actually 0. To this end, it suffices to show that it is zero at ∞ .

As we assumed that ∞ is not mapped to a vertex, f is regular at ∞ , meaning that

$$f(1/z) = c_0 + c_1 z + \dots$$

with $c_{-1} \neq 0$, so that

$$f'(1/z) = -z^2 \cdot (c_1 + 2c_2 z + \dots)$$

$$f''(1/z) = 2z^2 \cdot (c_1 + 3c_2 z + \dots)$$

Hence g has a single order zero at ∞ . The same holds for g_1 , which hence must be identically zero. Therefore, we obtain

$$g(z) = \sum_j \frac{a_j}{z - t_j}$$

In the case that ∞ is a vertex, we first make a translation $z \mapsto z + c$ to ensure that 0 is not a vertex. Under this translation, the formula for g remains valid (with new t_j), as ∞ remains a vertex. Then we consider $\tilde{f}(z) = f(1/z)$. This function maps the upper half plane to the same polygon, but the vertices are now all finite at $1/t_j$ and at 0. Hence, we can write

$$\frac{\tilde{f}''(z)}{\tilde{f}'(z)} = \sum_j \frac{a_j}{z - 1/t_j} + \frac{a_\infty}{z}$$

On the other hand,

$$\frac{\tilde{f}''(z)}{\tilde{f}'(z)} = -\frac{2}{z} - \frac{1}{z^2} \frac{f''(1/z)}{f'(1/z)}$$

Replacing z by $1/w$ gives

$$\begin{aligned} -2w - w^2 \frac{f''(w)}{f'(w)} &= \sum_j \frac{a_j}{1/w - 1/t_j} + a_\infty w \\ &= \sum_j \left(\frac{a_j w^2}{t_j - w} + a_j w \right) + a_\infty w \end{aligned}$$

so that

$$\frac{f''(w)}{f'(w)} = \sum_j \frac{a_j}{w - t_j}$$

as before. The general rule for the behavior at infinity is: If it occurs as a vertex, just drop the term in your formula, and everything remains valid.

Hence f satisfies the differential equation

$$\frac{f''}{f'}(z) = \sum_j \frac{a_j}{z - t_j}$$

which can be integrated to

$$f(w) = C_1 + C_2 \int^w \prod_j (z - t_j)^{a_j} dz.$$

If the polygon has more than three vertices, its shape is not determined by its angles. If some of the edge lengths are given, the t_i in the Schwarz-Christoffel formula need to be determined numerically.

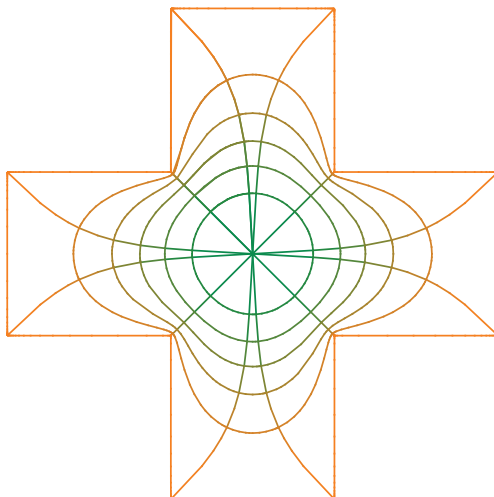


Figure 5.6 Mapping polar coordinates in the unit disk to the Swiss Cross

Exercise 5.4.4 Find a biholomorphic map from the upper half plane to the domain U obtained from the upper half plane by removing the segment $[0, i]$ of the imaginary axis.

6 Elliptic Functions

6.1 ELLIPTIC FUNCTIONS

Definition 6.1.1 A lattice Λ is given by

$$\Lambda = \{n_1\omega_1 + n_2\omega_2 : n_1, n_2 \in \mathbb{Z}\}$$

with generators $\omega_1, \omega_2 \in \mathbb{C}$ linearly independent over $\mathbb{R}$. The two generators form a basis of the lattice. The parallelogram Π spanned by ω_1 and ω_2 is called a fundamental parallelogram of Λ .

Note that neither the basis nor the fundamental parallelogram is uniquely determined by a lattice.

Example 6.1.1 The square lattice can be generated by 1 and i . The hexagonal lattice can be generated by 1 and $\frac{1}{2}(1 + \sqrt{3}i)$.

Definition 6.1.2 A function f is called elliptic with respect to the lattice Λ if it is meromorphic in $\mathbb{C}$ and satisfies $f(z + \omega) = f(z)$ for all $\omega \in \Lambda$. We denote the field of elliptic functions with respect to Λ by $\mathcal{E}(\Lambda)$.

Note that derivatives of elliptic functions are again elliptic. An elliptic function has only finitely many zeroes and poles within a fundamental parallelogram.

Definition 6.1.3 Let $f \in \mathcal{E}(\Lambda)$ be elliptic, and Π a translated fundamental parallelogram so that no poles of f lie on $\partial\Pi$. Then the order of f is the number of its poles in Π , counted with multiplicity.

Lemma 6.1.4 Elliptic functions of order 0 are constant.

Proof. An elliptic function of order 0 is holomorphic, hence entire. As $\overline{\Pi}$ is compact, it is bounded. **Liouville's Theorem 2.3.2** shows it is constant. 

Theorem 6.1.5 Let $f \in \mathcal{E}(\Lambda)$ with poles $w_1, \dots, w_n \in \Pi$. Then

$$\sum_{k=1}^n \text{res}_{w_k} f = 0.$$

Proof. By the residue theorem,

$$\begin{aligned} \sum_{k=1}^n \text{res}_{w_k} f &= \frac{1}{2\pi i} \int_{\partial\Pi} f(z) dz \\ &= 0 \end{aligned}$$

by periodicity, as integrals over opposite edges cancel. 

Corollary 6.1.6 There are no elliptic functions of order 1.

Proof. Such a function would have a single pole of order 1. Such a pole has a non-zero residue. ✎

Corollary 6.1.7 *An elliptic function of order n assumes every value precisely n times in $\mathbb{C}/\Lambda$.*

Proof. Translate Π so that f does not assume w on $\partial\Pi$. Then

$$\frac{1}{2\pi i} \int_{\partial\Pi} \frac{f'(z)}{f(z) - w} dz = 0$$

because the integrand is elliptic. On the other hand, the left-hand side counts the number of zeroes and poles of $f(z) - w$ in Π . ✎

6.2 WEIERSTRASS THEORY

So far, we haven't seen a single non-trivial elliptic function. Here is the first:

Example 6.2.1 For a lattice Λ ,

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \in \Lambda - 0} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right)$$

converges locally uniformly and absolutely in $\mathbb{C} - \Lambda$ and defines an elliptic function of order 2. This function is called the *Weierstrass $\wp$ -function*.

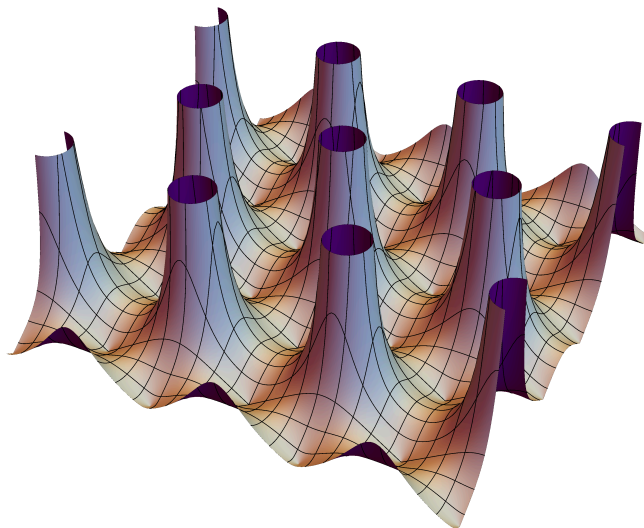


Figure 6.1 Graph of $|\wp(z)|$ for the hexagonal lattice

Proof. We first show convergence. Fix $r > 0$. Let $\omega \in \Lambda$ with $|\omega| > 2r$, and $z \in \mathbb{C}$ with $|z| < r$. Then $|z/\omega| < 1/2$, and thus

$$\begin{aligned} \left| \frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right| &= \left| \frac{z(2\omega - z)}{(z - \omega)^2 \omega^2} \right| \\ &= \left| \frac{z}{\omega^3} \right| \cdot \left| \frac{2 - z/\omega}{(1 - z/\omega)^2} \right| \\ &\leq 10 \frac{r}{|\omega|^3}. \end{aligned}$$

Thus, it suffices to show that

$$\sum_{\omega \in \Lambda - 0} \frac{1}{|\omega|^3} < \infty .$$

Consider nested parallelogram perimeters P_n through lattice points as in **Figure 6.2** such that on each P_n there are $8n$ lattice points. Let $\epsilon > 0$ so that $|\omega| > \epsilon$ for all $\omega \in P_1$. Then and $|\omega| \geq n\epsilon$ for all $\omega \in P_n$. Thus

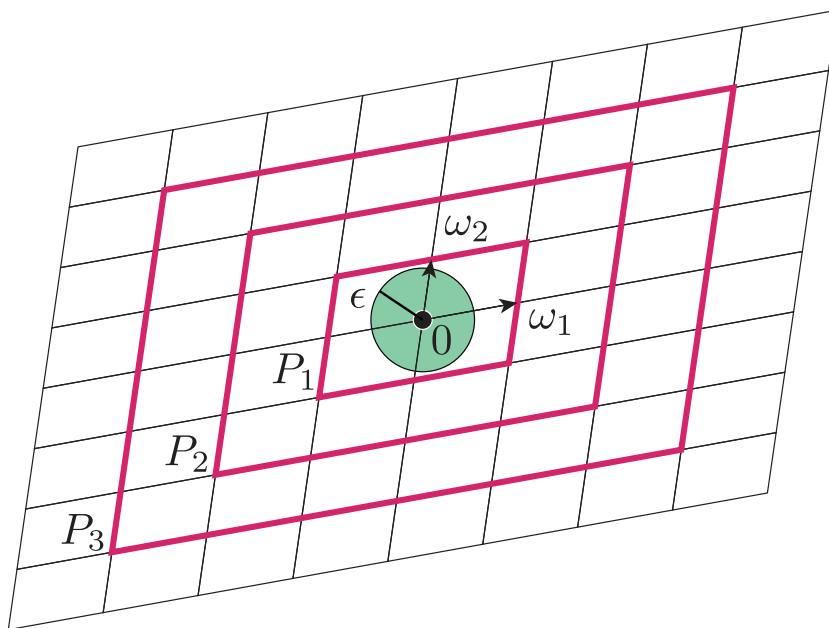


Figure 6.2 Estimating sums over lattice points

$$\begin{aligned} \sum_{\omega \in \Lambda - 0} \frac{1}{|\omega|^3} &= \sum_{n=1}^{\infty} \sum_{\omega \in P_n} \frac{1}{|\omega|^3} \\ &\leq \sum_{n=1}^{\infty} \frac{8n}{\epsilon^3 n^3} \\ &< \infty \end{aligned}$$

Thus $\wp(z)$ is meromorphic in $\mathbb{C}$ with double order poles at all lattice points. We now show that it is elliptic. Note that

$$\wp'(z) = -2 \sum_{\omega \in \Lambda} \frac{1}{(z - \omega)^3} .$$

This $\wp'(z)$ is evidently elliptic of order 3. For a lattice point ω , both $\wp(z)$ and $\wp(z + \omega)$ are antiderivatives, so that for all z

$$\wp(z + \omega) = \wp(z) + c_{\omega} .$$

Replacing z by $z - \omega$ and using that $\wp$ is even shows

$$\begin{aligned}
\wp(z) &= \wp(z - \omega) + c_\omega \\
&= \wp(-z + \omega) + c_\omega \\
&= \wp(-z) + 2c_\omega \\
&= \wp(z) + 2c_\omega
\end{aligned}$$

so that $c_\omega = 0$, and $\wp$ is elliptic as claimed. 

By definition

$$\lim_{z \rightarrow 0} \wp(z) - \frac{1}{z^2} = 0$$

so that we can write (traditionally)

$$\wp(z) = \frac{1}{z^2} - \frac{g_2}{20}z^2 + \frac{g_3}{28}z^4 + \dots$$

with numbers g_2 and g_3 determined (as *Eisenstein series*, as we will see) by the lattice.

Lemma 6.2.1 *The Weierstrass $\wp$ -function satisfies the ordinary differential equation*

$$\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3.$$

Proof. We expand

$$\wp'(z)^2 = 4z^{-6} - \frac{2}{5}g_2z^{-2} - \frac{4}{7}g_3 + \dots$$

$$4\wp(z)^3 = 4z^{-6} + \frac{3}{5}g_2z^{-2} + \frac{3}{7}g_3 + \dots$$

$$g_2\wp(z) = g_2z^{-2} + \dots$$

Thus

$$f(z) = \wp'(z)^2 - 4\wp(z)^3 + g_2\wp(z) + g_3$$

is an elliptic function without poles and value 0 at 0, hence identically 0. 

Proposition 6.2.2 *With*

$$h(z) = \wp(z) - \frac{1}{z^2} = \sum_{k=1}^{\infty} c_{2k}z^{2k}$$

we have

$$c_{2k} = (2k+1) \cdot \sum_{\omega \in \Lambda - 0} \frac{1}{\omega^{2k+2}}.$$

Proof. We have

$$h^{(n)}(z) = (-1)^n (n+1)! \sum_{\omega \in \Lambda - 0} \frac{1}{(z - \omega)^{n+2}}.$$

Thus

$$\begin{aligned} c_{2k} &= \frac{1}{(2k)!} h^{(2k)}(0) \\ &= (2k+1) \sum_{\omega \in \Lambda - 0} \frac{1}{\omega^{2k+2}} . \end{aligned}$$

In particular,

$$\begin{aligned} g_2 &= 60 \sum_{\omega \in \Lambda - 0} \frac{1}{\omega^4} \\ g_3 &= 140 \sum_{\omega \in \Lambda - 0} \frac{1}{\omega^6} \end{aligned}$$

These *Eisenstein series* generalize the Riemann ζ -function and form one of many connections between Complex Analysis and Number Theory.

Corollary 6.2.3

$$\begin{aligned} \wp''(z) &= 6\wp(z)^2 - \frac{1}{2}g_2 \\ \wp'''(z) &= 12\wp(z)\wp'(z) \end{aligned}$$

Note that g_2 and g_3 determine Λ , as they determine $\wp$. One can show that if $g_2^3 - 27g_3^2 \neq 0$, there exists a lattice with these values.

So far, all our elliptic functions have at least poles of order 2. To construct elliptic functions with simple poles (at more than one point), we introduce a non-elliptic function.

Definition 6.2.4 Let $\zeta(z)$ be the unique odd function with

$$\zeta'(z) = -\wp(z) .$$

This function is called the Weierstrass ζ -function. It is not elliptic and has as its Laurent expansion

$$\zeta(z) = \frac{1}{z} - \frac{1}{60}g_2z^3 - \frac{1}{40}g_3z^5 + \dots$$

Definition 6.2.5 We define the quasi periods of ζ by

$$\zeta(z + \omega_k) = \zeta(z) + \eta_k .$$

Theorem 6.2.6 (Mittag-Leffler for tori) *Let $f \in M(\mathbb{C})$ be elliptic with poles $z_k \in \Pi$, $k = 1, \dots, m$, and principal parts*

$$h_k(z) = \sum_{j=1}^{n_k} a_{k,j} (z - z_k)^{-j} .$$

Then, up to a constant,

$$f(z) = \sum_{k=1}^m \left(\sum_{j=2}^{n_k} \frac{(-1)^j}{(j-1)!} a_{k,j} \wp^{(j-2)}(z - z_k) + a_{k,1} \zeta(z - z_k) \right) .$$

Vice versa, given principal parts as above with

$$\sum_{k=1}^m a_{k,1} = 0 ,$$

then $f(z)$ is an elliptic function with these principal parts in Π .

Proof. The first part is clear. The second part follows, because the condition on the $a_{k,1}$ guarantees that f is elliptic. 

Example 6.2.2 Let $b \in \mathbb{C}$ with $2b \notin \Lambda$. Consider the elliptic function

$$f(z) = \frac{\wp'(z) - \wp'(b)}{\wp(z) - \wp(b)}$$

We determine the principal part at 0 as

$$\begin{aligned} f(z) &= \frac{-2z^{-3} - \wp'(b) + \dots}{z^{-2} - \wp(b)} \\ &= -\frac{2}{z} - 2\wp(b) \cdot z + \dots \end{aligned}$$

Further principal parts can occur at the zeroes of the denominator $\wp(z) - \wp(b)$. As $\wp$ is even, there are zeroes at $z = \pm b$. Note that b and $-b$ are distinct in $\mathbb{C}/\Lambda$, as $2b \notin \Lambda$. As $\wp$ has order 2, there can't be any further zeroes. Next, we claim that the zeroes of $\wp(z) - \wp(b)$ are simple. As $\wp'(z)$ is odd it has three simple zeroes at $\omega_1/2$, $\omega_2/2$, and $\frac{1}{2}(\omega_1 + \omega_2)$. None of these points can be $\pm b$ so that $\wp'(z) \neq 0$ at $\pm b$.

Consequently, f does not have a pole at b , because the numerator is vanishing there as well. Therefore, f must have a simple pole at $-b$ with residue +2. By **Theorem 6.2.6** we can write

$$f(z) = 2\zeta(z + b) - 2\zeta(z) + c .$$

To determine c , we evaluate

$$\begin{aligned} c &= \lim_{z \rightarrow 0} \left(f(z) + \frac{2}{z} \right) - 2\zeta(z + b) + 2 \left(\zeta(z) - \frac{1}{z} \right) \\ &= -2\zeta(b) . \end{aligned}$$

This shows

$$\frac{\wp'(z) - \wp'(b)}{\wp(z) - \wp(b)} = 2\zeta(z + b) - 2\zeta(z) - 2\zeta(b) ,$$

which is the addition theorem for the Weierstrass ζ -function. Differentiating this with respect to z and b , adding, and using the differential equation for $\wp''(z)$ shows the addition theorem for the $\wp$ -function:

$$\wp(z+b) = -\wp(z) - \wp(b) + \frac{1}{4} \left(\frac{\wp'(z) - \wp'(b)}{\wp(z) - \wp(b)} \right)^2.$$

Historical motivation: Computation of the arc length of an ellipse leads to integrals like

$$\int \frac{dx}{\sqrt{4x^3 - g_2x - g_3}}$$

which are therefore called *elliptical integrals*. The substitution $x = \wp(z)$ reduces this to

$$\begin{aligned} \int \frac{dx}{\sqrt{4x^3 - g_2x - g_3}} &= \int \frac{\wp'(z)}{\sqrt{\wp'(z)^2}} dz \\ &= \int dz \\ &= z \\ &= \wp^{-1}(x) \end{aligned}$$

so that elliptical integrals can be solved by the inverses of the Weierstrass $\wp$ -function. This explains the name *elliptical functions*.

Theorem 6.2.7 *Let $f \in \mathcal{E}(\Lambda)$ be an elliptical function. Then there exist rational functions ϕ and ψ such that*

$$f(z) = \phi(\wp(z)) + \psi(\wp(z)) \cdot \wp'(z).$$

Proof. First assume that f is even. Choose a fundamental parallelogram centered at 0. Pick c, c^* such that $f(z) \neq c, c^*$ for $z \in \partial\Pi \cup \{0\}$. Then there are $z_k, z_k^* \in \Pi$ such that

$$f(\pm z_k) = c \quad \text{and} \quad f(\pm z_k^*) = c^*$$

for $k = 1, \dots, m$, where m is the order of f , and the points $\pm z_k, \pm z_k^*$ lie in Π . If necessary, we repeat the points to accommodate the order of zeroes and poles. Note that at these points $\wp$ is finite. Therefore, the elliptic function

$$g(z) = \frac{f(z) - c}{f(z) - c^*} \cdot \prod_{k=1}^m \frac{\wp(z) - \wp(z_k^*)}{\wp(z) - \wp(z_k)}$$

has no zeroes or poles whatsoever and is hence constant. Solving for f proves the claim $f(z) = \phi(\wp(z))$ for a rational function ϕ .

In the general case, note that the two functions

$$\frac{1}{2}(f(z) + f(-z)) \quad \text{and} \quad \frac{f(z) - f(-z)}{2\wp'(z)}$$

are both even so that we can apply the first part and write both as rational functions of $\wp(z)$. This proves the claim in general. 

6.3 THE MODULAR FUNCTION

We will now study the dependence of the Weierstrass $\wp$ -function on the lattice Λ . We first note that

$$\wp(z, \Lambda) = a^2 \wp(az, a\Lambda)$$

so that we don't seriously have to distinguish lattices that are isomorphic in the sense that $\Lambda' = a\Lambda$. We therefore normalize a basis for a lattice as

$$\omega_1 = 1 \text{ and } \omega_2 = \tau \in \mathbb{H} = \{\text{Im } z > 0\}.$$

and denote the resulting lattice by Λ_τ . Thus we use points in $\mathbb{H}$ to describe a lattice. There are still different points in $\mathbb{H}$ that describe the same lattice: If

$$\tau' = \frac{a\tau + b}{c\tau + d} \quad \text{for } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}),$$

then Λ_τ and $\Lambda_{\tau'}$ are isomorphic, as A maps the basis $1, \tau$ of Λ_τ just to a different basis.

We now define the *modular function*

$$\lambda(\tau) = \frac{\wp(\frac{1}{2} + \frac{\tau}{2}, \tau) - \wp(\frac{\tau}{2}, \tau)}{\wp(\frac{1}{2}, \tau) - \wp(\frac{\tau}{2}, \tau)}.$$

This is the cross ratio of the value of the $\wp$ function at the 2-division points of the lattice. While we cannot expect this modular function to be invariant under $SL_2(\mathbb{Z})$, it is invariant under the *congruence subgroup*

$$\Gamma(2) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{2} \right\}.$$

Proof. It suffices to show that elements of $\Gamma(2)$ fix 2-division points. Let's show this for the point $\tau/2$. This simply because

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 \\ 1/2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} a \\ c \end{pmatrix} \equiv \begin{pmatrix} 0 \\ 1/2 \end{pmatrix} \pmod{1}.$$

Next, we show:

Theorem 6.3.1 *The modular function omits 0 and 1.*

Proof. As $\wp$ is an elliptic function of degree 2 and $\wp'(z) = 0$ at $\frac{1}{2}, \frac{\tau}{2}$ and $\frac{1+\tau}{2}$, the $\wp$ -function does not take the values $\wp(\frac{1}{2}), \wp(\frac{\tau}{2})$ and $\wp(\frac{1+\tau}{2})$ anywhere else. Thus $\lambda \neq 0, 1$. 

We will maybe show later that λ maps the region

$$\{\tau : \text{Im } \tau > 0, |\tau - \frac{1}{2}| > \frac{1}{2}, 0 < \text{Re } \tau < 1\}$$

biholomorphically onto the upper half plane.

7 Geometric Function Theory

7.1 HERMITIAN METRICS

Definition 7.1.1 A Hermitian metric on a domain U is a C^2 function $\rho : U \rightarrow \mathbb{R}^{>0}$ which vanishes only at isolated points. We denote a domain with a Hermitian metric by (U, ρ) .

Example 7.1.1

1. The Euclidean metric on any domain $U \subset \mathbb{C}$ is given by $\rho = 1$.
2. The hyperbolic metric on $\mathbb{D}$ is given by

$$\lambda_{\mathbb{D}}(z) = \frac{2}{1 - |z|^2}.$$

3. The hyperbolic metric on the upper half plane $\mathbb{H}$ is given by

$$\lambda_{\mathbb{H}}(z) = \frac{1}{\operatorname{Im} z}.$$

4. The spherical metric on any domain $U \subset \mathbb{C}$ is given by

$$\sigma(z) = \frac{2}{1 + |z|^2}.$$

We will use Hermitian metrics to measure the lengths of curves:

Definition 7.1.2 Let $\gamma : [a, b] \rightarrow (U, \rho)$ be a piecewise C^1 -curve. We define its length with respect to the Hermitian metric ρ by

$$\operatorname{length}_{\rho}(\gamma) = \int_a^b \rho(\gamma(t)) |\gamma'(t)| dt.$$

Please check that this definition is independent of the parametrization of the curve. This, in turn, can be used to define a distance between points by

$$d(z_1, z_2) = \inf \{ \operatorname{length}_{\rho}(\gamma) : \gamma \in \Gamma \},$$

where Γ is the set of all C^1 curves connecting z_1 with z_2 .

An important method to construct new Hermitian metrics is by *pull-back*:

Definition 7.1.3 Let $f : U \rightarrow (V, \rho)$ be a non-zero holomorphic map. Then we define a Hermitian metric on U by

$$(f^* \rho)(z) = \rho(f(z)) |f'(z)|.$$

$f^* \rho$ is called the pull-back of ρ by f . As f has only isolated zeroes, this is again a Hermitian metric.

The definition is made so that it is compatible with the length of curves:

Lemma 7.1.4 For a holomorphic map $f : (U, f^*\rho) \rightarrow (V, \rho)$ and a curve γ in U we have

$$L_{f^*\rho}(\gamma) = L_\rho(f(\gamma)) .$$

Proof. By definition,

$$\begin{aligned} L_{f^*\rho}(\gamma) &= \int_a^b (f^*\rho)(\gamma(t)) |\gamma'(t)| dt \\ &= \int_a^b (\rho)(f(\gamma(t))) |f'(\gamma(t))| |\gamma'(t)| dt \\ &= \int_a^b (\rho)(f(\gamma(t))) \left| \frac{d}{dt} f(\gamma(t)) \right| dt \\ &= L_\rho(f(\gamma)) . \end{aligned}$$

Example 7.1.2 For $|z_0| < 1$, let $\mu(z) = \frac{z-z_0}{1-\bar{z}_0 z}$ be an automorphism of $\mathbb{D}$. Then

$$\begin{aligned} (\mu^* \lambda_{\mathbb{D}})(z) &= \lambda_{\mathbb{D}}(\mu(z)) |f'(z)| \\ &= \frac{2}{1 - |\mu(z)|^2} |\mu'(z)| \\ &= \frac{2}{1 - \frac{|z-z_0|^2}{|1-\bar{z}_0 z|^2}} \frac{1 - |z_0|^2}{|1 - \bar{z}_0 z|^2} \\ &= 2 \frac{1 - |z_0|^2}{|1 - \bar{z}_0 z|^2 - |z - z_0|^2} \\ &= \frac{2}{1 - |z|^2} . \end{aligned}$$

Exercise 7.1.5 Let $f : \mathbb{H} \rightarrow \mathbb{D}$ be given by $f(z) = \frac{i-z}{i+z}$. Show that $f^* \lambda_{\mathbb{D}} = \lambda_{\mathbb{H}}$.

Exercise 7.1.6 Let $f : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ be given by $f(z) = \frac{az+b}{-bz+\bar{a}}$ with $|a|^2 + |b|^2 = 1$. Show that $f^* \sigma = \sigma$ at all points where this is defined.

We will need to relax the conditions on a Hermitian metric slightly, for instance, in order to be able to consider pull-backs also when $f'(z) = 0$.

Definition 7.1.7 A pseudometric on a domain U is a continuous non-negative function ρ that is C^2 when positive.

7.2 CURVATURE OF HERMITIAN METRICS

To introduce the notion of curvature for Hermitian metrics, we need the Laplace operator

Definition 7.2.1 Let $h : U \rightarrow \mathbb{R}$ be a real-valued C^2 function. Then

$$\Delta u = u_{xx} + u_{yy} = 0 .$$

is called the Laplace operator.

It is convenient to have the Laplace operator expressed in terms of complex derivatives.

Definition 7.2.2 The differential operators

$$\frac{\partial}{\partial z} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right)$$

are called Wirtinger derivatives.

Note that function $f : U \rightarrow \mathbb{C}$ satisfies the **Cauchy-Riemann equations 1.6.1** if and only if $\frac{\partial}{\partial \bar{z}} f = 0$.

Lemma 7.2.3 The Wirtinger derivatives satisfy the following chain rules:

$$\begin{aligned} \frac{\partial}{\partial z} (f \circ g) &= \left(\frac{\partial f}{\partial z} \circ g \right) \frac{\partial g}{\partial z} + \left(\frac{\partial f}{\partial \bar{z}} \circ g \right) \frac{\partial \bar{g}}{\partial z} \\ \frac{\partial}{\partial \bar{z}} (f \circ g) &= \left(\frac{\partial f}{\partial z} \circ g \right) \frac{\partial g}{\partial \bar{z}} + \left(\frac{\partial f}{\partial \bar{z}} \circ g \right) \frac{\partial \bar{g}}{\partial \bar{z}} . \end{aligned}$$

Exercise 7.2.4 Using Wirtinger derivatives, we have

$$\Delta = 4 \frac{\partial^2}{\partial z \partial \bar{z}} .$$

Using polar coordinates $z = re^{it}$, we have

$$\Delta = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial t^2} .$$

For our purposes, all we really need to know about the Laplace operator is that if a C^2 function $u(z)$ has a local maximum at a point p , then $\Delta u(p) \leq 0$. This is true because at a local maximum, the Hessian of u is negative semidefinite, and the Laplace operator is the trace of the Hessian.

Finally, the Laplace operator satisfies a simple chain rule for the composition with holomorphic functions.

Exercise 7.2.5 Let $f : U \rightarrow V$ be holomorphic and $u : V \rightarrow \mathbb{R}$ be C^2 . Then

$$\Delta(u \circ f)(z) = |f'(z)|^2 (\Delta u)(f(z)) .$$

This has the remarkable consequence that if a function u is *harmonic*, i.e. $\Delta u = 0$, then its postcomposition with a holomorphic function is also harmonic. The notion of curvature improves this by defining a quantity that is invariant under holomorphic maps.

Definition 7.2.6 Let ρ be a Hermitian metric. Then the function

$$K = -\frac{1}{\rho^2} \Delta \log \rho$$

is called the curvature of ρ .

The curvature of a Hermitian metric is the same as the Gauß curvature of a Riemannian surface introduced in Differential Geometry. We will not need this fact.

Example 7.2.1 Clearly, the Euclidean metric has curvature 0. We compute the curvature of the hyperbolic metric in polar coordinates

$$\begin{aligned} K_{\lambda_{\mathbb{D}}} &= -\frac{1}{\rho^2} \Delta \log \lambda_{\mathbb{D}} \\ &= -\frac{(1-r^2)^2}{2^2} \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial t^2} \right) \log \frac{2}{1-r^2} \\ &= \frac{1}{4} (1-r^2)^2 \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} \right) \log (1-r^2) \\ &= -\frac{1}{4} (1-r^2)^2 \left(\frac{2(1-r^2) + 4r^2}{(1-r^2)^2} + \frac{2}{1-r^2} \right) \\ &= -\frac{1}{4} (1-r^2)^2 \frac{2(1-r^2) + 4r^2 + 2(1-r^2)}{(1-r^2)^2} \\ &= -1 \end{aligned}$$

For our purposes, it is important that the Gauß curvature is a *conformal invariant* in the following sense:

Lemma 7.2.7 Let ρ be a Hermitian metric in a domain V , and let $f : U \rightarrow V$ be holomorphic. Then we have

$$K_{f^*\rho}(z) = K_{\rho}(f(z))$$

for all $z \in U$ with $f'(z) \neq 0$.

Proof. We use that for holomorphic functions,

$$\Delta(u \circ f)(z) = |f'(z)|^2 (\Delta u)(f(z)) .$$

and compute

$$\begin{aligned} K_{f^*\rho}(z) &= -\frac{1}{(f^*\rho)^2(z)} \Delta \log(f^*\rho)(z) \\ &= -\frac{1}{\rho^2(f(z)) |f'(z)|^2} \Delta \log(\rho(f(z)) |f'(z)|) \\ &= -\frac{1}{\rho^2(f(z)) |f'(z)|^2} (\Delta \log(\rho(f(z))) + \Delta \log(|f'(z)|)) \\ &= -\frac{1}{\rho^2(f(z))} (\Delta \log(\rho)(f(z))) . \end{aligned}$$

7.3 AHLFORS' LEMMA

Theorem 7.3.1 (Ahlfors' Lemma) *Let ρ be a pseudometric in a domain U . Assume that $K_\rho \leq -1$ when defined. Then*

$$\rho \leq \lambda_{\mathbb{D}}.$$

Proof. We fix $0 < r < 1$. Denote by

$$\lambda_r(z) = \frac{2r}{r^2 - |z|^2}$$

the hyperbolic metric on $D_r(0)$. We will show that $\rho \leq \lambda_r$. Let $h(z) = \frac{\rho(z)}{\lambda_r(z)}$. This function is non-negative and continuous on $D_r(0)$. Moreover, for $|z| \rightarrow r$, $|h(z)| \rightarrow 0$. Thus h attains a maximum value $M = h(p)$ at some point $p \in D_r(0)$. We have to show that $M \leq 1$. In case $\rho(p) = 0$, we have $M = 0$ and are done. Otherwise, h is C^2 near p , and we can compute at p

$$\begin{aligned} 0 &\geq \Delta \log h \\ &\geq \Delta \log \rho - \Delta \log \lambda_r \\ &= -K_\rho(p) \rho^2(p) + K_{\lambda_r}(p) \lambda_r^2(p) \\ &\geq \rho^2(p) - \lambda_r^2(p) \end{aligned}$$

so that indeed $M \leq 1$. 

Corollary 7.3.2 *Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic. Then*

$$\frac{|f'(z)|}{1 - |f(z)|^2} \leq \frac{1}{1 - |z|^2}.$$

Proof. Let

$$\rho = f^* \lambda_{\mathbb{D}} = \frac{2|f'(z)|}{1 - |f(z)|^2}.$$

Then ρ satisfies the assumptions of **Ahlfors' Lemma 7.3.1**, and the claim follows. 

In particular if $f(0) = 0$, we obtain $|f'(0)| \leq 1$ as in **Schwarz' Lemma 2.5.9**.

This allows us to reprove **Liouville's Theorem 2.3.2**.

Theorem 7.3.3 *An entire bounded function is constant.*

Proof. Suppose $f : \mathbb{C} \rightarrow \mathbb{D}$ is an entire, non-constant function. Then $\rho = f^* \rho_{\mathbb{D}}$ is a Hermitian metric with $K_\rho = -1$, except for isolated points. For fixed $p \in \mathbb{C}$ and $r > 0$, restrict ρ to the disk $D_r(p)$ with its hyperbolic metric

$$\lambda_{D_r(p)}(z) = \frac{2r}{r^2 - |z - p|^2}.$$

By **Ahlfors' Lemma 7.3.1**, we have $\rho \leq \lambda_{D_r(p)}$ so that in particular $\rho(p) \leq \frac{2}{r}$. With $r \rightarrow \infty$, we obtain $\rho(p) = 0$. This implies $f'(p) = 0$. As this holds for all p , f is constant. 

Note that this proof makes no use of the Cauchy integral theorem. In fact, we only need the Cauchy-Riemann equations, or that f is a conformal map.

In the next section, we will need a version of **Ahlfors' Lemma 7.3.1** where the Hermitian metric is only continuous. The condition of curvature is implemented through a *supporting metric*:

Definition 7.3.4 An upper semicontinuous Hermitian metric $\rho : U \rightarrow \mathbb{R}^{\geq 0}$ on a domain U is called *ultrahyperbolic* if for each $p \in U$ with $\rho(p) > 0$ there is a Hermitian metric $\rho_p : U_p \rightarrow \mathbb{R}^{> 0}$ of class C^2 defined in a neighborhood U_p of p such that for $z \in U_p$

1. $K_{\rho_p}(z) \leq -1$;
2. $\rho_p(z) \leq \rho(z)$;
3. $\rho_p(p) = \rho(p)$.

The metric ρ_p is called a *supporting metric* of ρ at p .

We can now formulate a refined version of **Ahlfors' Lemma 7.3.1**:

Theorem 7.3.5 (Ahlfors' Lemma) Let ρ be an ultrahyperbolic metric in $\mathbb{D}$. Then

$$\rho \leq \lambda_{\mathbb{D}}.$$

Proof. We argue as in the previous proof. We fix $0 < r < 1$. Denote by

$$\lambda_r(z) = \frac{2r}{r^2 - |z|^2}$$

the hyperbolic metric on $D_r(0)$. We will again show that $\rho \leq \lambda_r$. Let $h(z) = \frac{\rho(z)}{\lambda_r(z)}$. This function is non-negative and continuous on $D_r(0)$. Moreover, for $|z| \rightarrow r$, $|h(z)| \rightarrow 0$. Thus h attains a maximum value M at some point $p \in D_r(0)$. We have to show that $M \leq 1$. In case that $\rho(p) = 0$, we have $M = 0$ and are done.

Otherwise, let ρ_p be a supporting metric for ρ at p , defined in a neighborhood U_p of p . Let

$$h_p(z) = \frac{\rho_p(z)}{\lambda_r(z)} \leq \frac{\rho(z)}{\lambda_r(z)} = h(z)$$

with equality for $z = p$, the function h_p still assumes a local maximum at p but now is C^2 in U . By the curvature condition on ρ_p we obtain

$$\begin{aligned} 0 &\geq \Delta \log h_p \\ &\geq \Delta \log \rho_p - \Delta \log \lambda_r \\ &= -K_{\rho_p}(p) \rho_p^2(p) + K_{\lambda_r}(p) \lambda_r^2(p) \\ &\geq \rho_p^2(p) - \lambda_r^2(p) \\ &= \rho^2(p) - \lambda_r^2(p) \end{aligned}$$

so that indeed $M \leq 1$.



7.4 LANDAU-SCHOTTKY THEORY

The goal of this section is to prove the following two theorems:

Theorem 7.4.1 (Landau) *There is a universal constant L such that for $f \in O(\mathbb{D})$ which is normalized so that $f(0) = 0$ and $f'(0) = 1$, $f(\mathbb{D})$ contains a disk of radius at least L . The smallest such L is called Landau's constant.*

Theorem 7.4.2 (Bloch) *There is a universal constant B such that for $f \in O(\mathbb{D})$ which is normalized so that $f(0) = 0$ and $f'(0) = 1$, $f(\mathbb{D})$ contains a disk of radius at least L that is the injective image of a subset of $\mathbb{D}$.*

Clearly, we have $0 \leq B \leq L \leq 1$. The optimal values for L and B are not known. The Ahlfors-Grunsky conjecture from 1936 states that

$$B = \sqrt{\frac{\sqrt{3}-1}{2}} \cdot \frac{\Gamma(\frac{1}{3})\Gamma(\frac{11}{12})}{\Gamma(\frac{1}{4})} = 0.4719 \dots$$

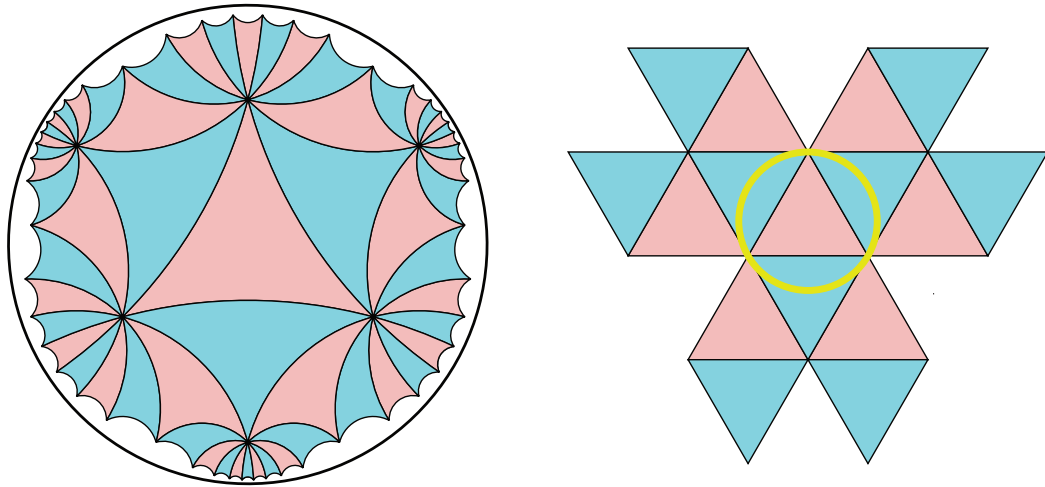


Figure 7.1 The Ahlfors-Grunsky map

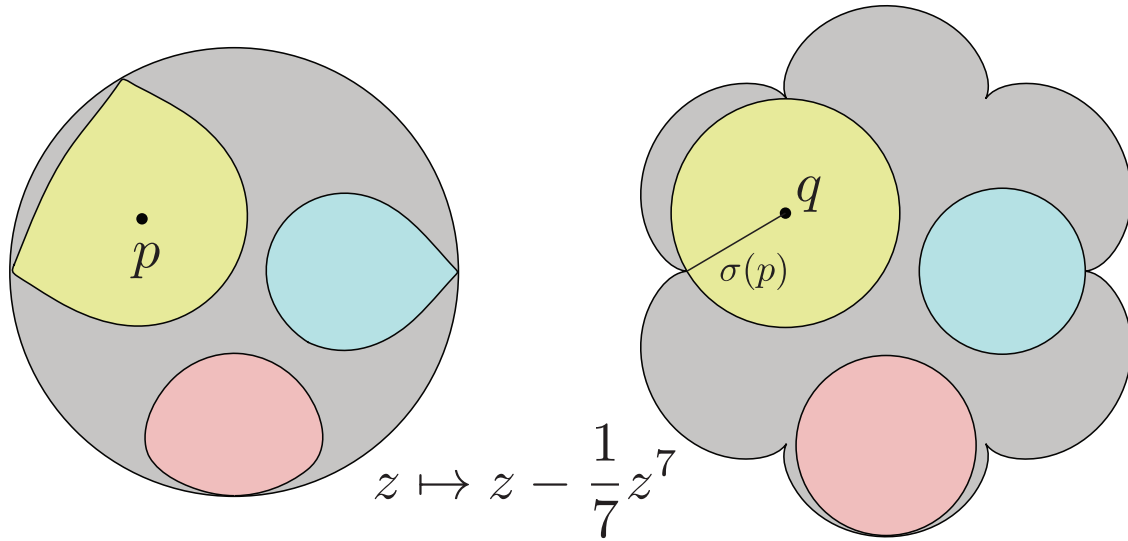
This strange number belongs to the following function: Consider a regular circular triangle Δ_1 with angles $\pi/6$ and center at 0. Scale the triangle so that it tiles the unit disk. Now let $f : \Delta_1 \rightarrow \Delta_2$ be a biholomorphic map onto an equilateral Euclidean triangle Δ_2 . Continue this map by using the Schwarz reflection principle to a map $f : \mathbb{D} \rightarrow \mathbb{C}$, and normalize f so that $|f'(0)| = 1$.

We will now show the following estimate of Ahlfors, which implies both Bloch's and Landau's theorems.

Theorem 7.4.3 (Ahlfors) *For a function $f \in O(\mathbb{D})$ with $f(0) = 0$ and $f'(0) = 1$ we have $B_f > \frac{\sqrt{3}}{4}$.*

Proof. The proof consists of seven steps.

1. Definition of the radius function

Figure 7.2 Definition of σ

Let $f : \mathbb{D} \rightarrow U$ be a surjective holomorphic map. For $p \in \mathbb{D}$, let $q = f(p)$, and define $\sigma(p)$ to be the largest number such that $D_{\sigma(p)}(q)$ is biholomorphic image of an open neighborhood U_p of p in $\mathbb{D}$.

In case that $f'(p) = 0$, we set $\sigma(p) = 0$.

We clearly have $\sigma(p) \leq B_f$. Let $g : D_{\sigma(p)}(q) \rightarrow U_p$ be the inverse of f .

2. The boundary of U_p We claim that the boundary of U_p contains points of $\partial\mathbb{D}$ or points with $f'(z) = 0$. Assume the contrary, and let $w \in \partial D_{\sigma(p)}(q)$. Choose a sequence $w_n \in D_{\sigma(p)}(q)$ with $w_n \rightarrow w$, and let $z_n = g(w_n)$. After choosing a subsequence, we can assume that $z_n \rightarrow z \in \partial U_p \subset \mathbb{D}$. We then have $f(z) = w$ and $f'(z) \neq 0$. Choose a small disk $D_\epsilon(w)$ on which f^{-1} exists with $f^{-1}(w) = z$. On $D_{\sigma(p)}(q) \cap D_\epsilon(w)$ we then have $f^{-1} = g$, so we have a holomorphic extension of g to $D_{\sigma(p)}(q) \cup D_\epsilon(w)$. Now cover $\partial D_{\sigma(p)}(q)$ with finitely many such disks, and extend g holomorphically to all of them. This way we find a disk $D_s(q)$ with $s > \sigma(p)$ on which f^{-1} is defined and extends g , a contradiction to the definition of σ .

3. Continuity of σ We claim that $\sigma : \mathbb{D} \rightarrow [0, B_f]$ is a continuous function. Moreover, near points p with $f'(p) = 0$, we have explicitly that

$$\sigma(z) = |f(z) - f(p)|.$$

Suppose first that $\sigma(p) \neq 0$, and let $q = f(p)$. For $w \in D_{\sigma(p)}(q)$ and $z = g(w) \in U_p$ we have

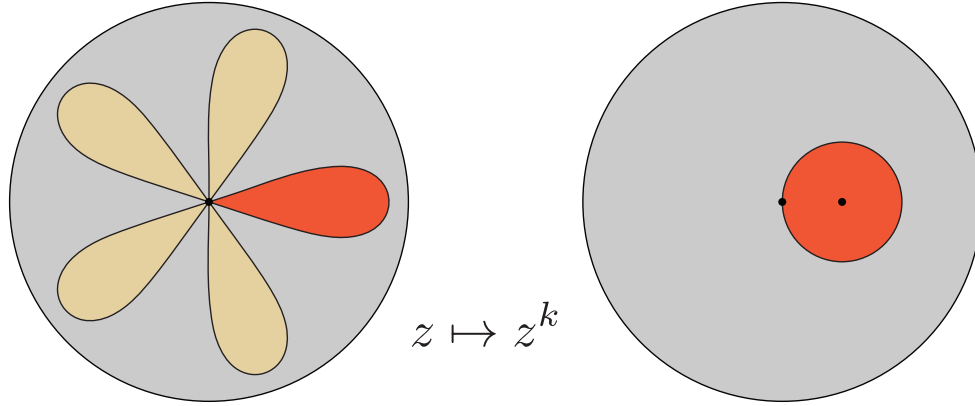
$$\sigma(p) - |w - q| \leq \sigma(z) \leq \sigma(p) + |w - q|$$

by triangle inequality and the definition of σ . This shows

$$\lim_{z \rightarrow p} \sigma(z) = \sigma(p).$$

Now suppose that $\sigma(p) = 0$ so that also $f'(p) = 0$. Then clearly $\sigma(z)$ cannot be larger than $|f(z) - f(p)|$, because f stops being injective at p . To see that we have equality for z sufficiently close to p , choose a disk $D_\epsilon(q)$ and a neighborhood U of p in which we can write

$$f(z) = q + h(z)^k$$

Figure 7.3 σ near a branched point

with $k \geq 2$ and holomorphic function h that maps U biholomorphically onto a disk centered at 0. Let $U_\epsilon = f^{-1}(D_{\epsilon/2}(q)) \subset U$. For $z \in U_\epsilon$, we then have $s = |f(z) - f(p)| < \frac{\epsilon}{2}$ so that f is invertible on $D_s(f(z))$. Thus for $z \in U_\epsilon$ we have $\sigma(z) = |f(z) - f(p)|$ as claimed.

4. Definition of a pseudometric Define a pseudometric on $\mathbb{D}$ by

$$\lambda(z) = \frac{\sqrt{A}|f'(z)|}{2\sqrt{\sigma(z)}(A - \sigma(z))},$$

where $A > B_f$ will be chosen later. We can assume that $B_f < \infty$, because otherwise there is nothing to show. We pause for a moment to motivate this expression. Near points where $\sigma(z)$ can be written as $\sigma(z) = |f(z) - q|$, this metric is the pullback of $\lambda_{\mathbb{D}}$ under the map

$$F_p(z) = \frac{1}{\sqrt{A}} \sqrt{f(z) - q},$$

because

$$\begin{aligned} F_p^* \lambda_{\mathbb{D}}(z) &= \frac{2}{1 - \frac{|f(z) - q|}{A}} \frac{|f'(z)|}{2\sqrt{A} \sqrt{|f(z) - q|}} \\ &= \frac{\sqrt{A}|f'(z)|}{(A - |f(z) - q|) \sqrt{|f(z) - q|}} \\ &= \frac{\sqrt{A}|f'(z)|}{(A - \sigma(z)) \sqrt{\sigma(z)}}. \end{aligned}$$

This construction provides us with a metric of curvature -1 that depends on σ and will allow us eventually to estimate B_f . The square root attempts to correct for a suspected presence of a branched point of f at p . The constant A will need to be chosen as small as possible so that this map maps into the unit disk.

We now claim that λ has a continuous extension to points p where $f'(p) = 0$. Write

$$f(z) = q + (z - p)^k h(z),$$

where $k \geq 2$ and h is holomorphic with $h(p) \neq 0$. For z close enough to p we then have

$$\lambda = F_p^* \lambda_{\mathbb{D}}(z)$$

with

$$F_p(z) = \frac{1}{\sqrt{A}} (z - p)^{\frac{k}{2}} \sqrt{h(z)} .$$

As $h(p) \neq 0$, $\lambda(p) = 0$ is a continuous extension for $k > 2$, while for $k = 2$,

$$\begin{aligned} \lambda(p) &= \lim_{z \rightarrow p} \sqrt{A} \frac{|z - p| \cdot |2h(z) + (z - p)h'(z)|}{2\sqrt{|z - p|^2 |h(z)| (A - |z - p|^2 |h(z)|)}} \\ &= \sqrt{A} \frac{|h(p)|}{\sqrt{|h(p)| A}} \\ &= \left(\frac{|h(p)|}{A} \right)^{1/2} \neq 0 \end{aligned}$$

is even a differentiable extension.

5. Existence of Supporting Metrics Let $p \in \mathbb{D}$ be point with $\lambda(p) \neq 0$. If $f'(p) = 0$, then λ is already regular at p , as then $k = 2$ in the discussion above, so we can use λ to support itself. It remains to construct a supporting metric for points p with $f'(p) \neq 0$. Let $q = f(p)$, and choose $q' \in \partial D_{\sigma(p)}(q)$ so that there is no $p' \in \partial U_p \cap \mathbb{D}$ with $f(p') = q'$ and $f'(p') \neq 0$. In other words, q' prevents f from being invertible on an even larger disk. Such a point exists by the argument in Step 2. Then, define on U_p the smooth function

$$\sigma_p(z) = |f(z) - q'| .$$

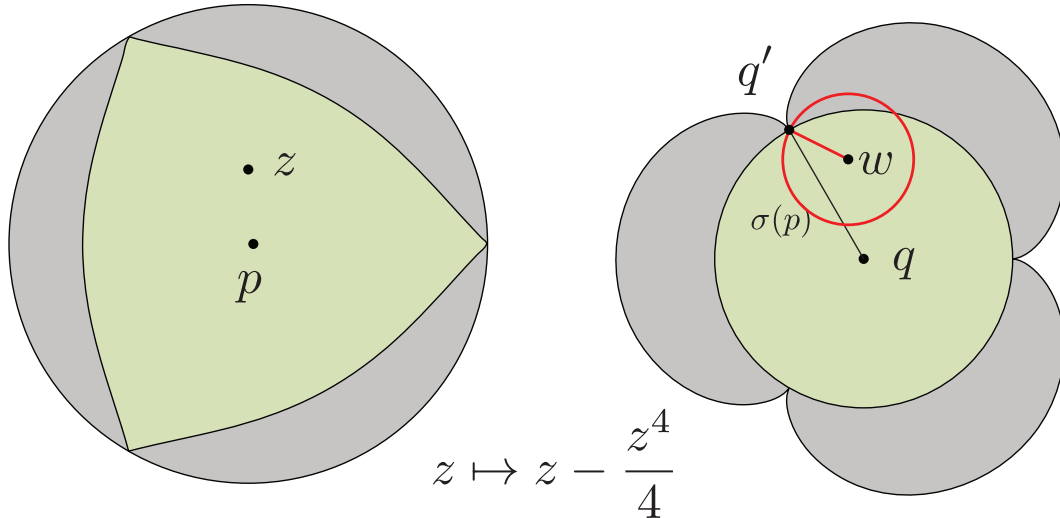


Figure 7.4 $\sigma_p(z) \geq \sigma(z)$

For a point $z \in U_p$, let $w = f(z)$. Then f cannot be invertible on disk of radius larger than $|w - q'|$ centered at w , so that $\sigma_p(z) \geq \sigma(z)$ with equality at p . Now choose $A > 3B_f$. Then $\sqrt{t}(A - t)$ is monotone increasing on the interval $[0, B_f]$ as

$$\frac{d}{dt} \sqrt{t}(A - t) = \frac{A - 3t}{2\sqrt{t}} .$$

Thus the metric

$$\lambda_p = \sqrt{A} \frac{|f'(z)|}{\sqrt{\sigma_p(z)(A - \sigma_p(z))}}$$

satisfies $\lambda_p \leq \lambda$ and $\lambda_p(p) = \lambda(p)$ and is therefore a supporting metric of λ at p .

6. Curvature Computation Let $p \in \mathbb{D}$ be a point with $\lambda(p) \neq 0$. In Step 4, we defined in case that $f'(p) = 0$ (entailing $f''(p) \neq 0$, as $\lambda(p) \neq 0$)

$$F_p(z) = \frac{1}{\sqrt{A}} \sqrt{f(z) - q}.$$

Similarly, for points $p \in \mathbb{D}$ with $f'(p) \neq 0$ we define

$$F_p(z) = \frac{1}{\sqrt{A}} \sqrt{f(z) - q'},$$

where q' is chosen as in Step 5. As $A > 3B_f$ and $|f(z) - q'| < 2B_f$, we have $|F_p(z)| < 1$ in either case. With the same computation as in Step 4,

$$F_p^* \lambda_{\mathbb{D}} = \lambda_p.$$

This implies that $K_{\sigma_p} = K_{\sigma} = -1$.

7. Conclusion By [Alfors' Lemma 7.3.5](#), we have in all of $\mathbb{D}$ that $\lambda \leq \lambda_{\mathbb{D}}$:

$$\sqrt{A} \frac{|f'(z)|}{\sqrt{\sigma(z)(A - \sigma(z))}} \leq \frac{2}{1 - |z|^2}.$$

As $\sigma(z) \leq B_f < \frac{A}{3}$, we can replace $\sigma(z)$ by B_f :

$$\sqrt{A} \frac{|f'(z)|}{\sqrt{B_f(A - B_f)}} \leq \frac{2}{1 - |z|^2}.$$

Taking the limit $A \rightarrow 3B_f$ evaluating at $z = 0$ shows

$$\frac{1}{4} \sqrt{3} \leq B_f.$$

7.5 UNIVALENT FUNCTION

Definition 7.5.1 A function $f \in O(U)$ is called *univalent* or *schlicht* if it is injective.

By the [Open Mapping Theorem 2.5.6](#), a univalent function is a biholomorphic map $f : U \rightarrow f(U)$. We also note that necessarily $f'(z) \neq 0$ in U .

In this section, we will discuss univalent functions $f \in O(\mathbb{D})$. Denote by

$$\mathcal{S} = \{f \in O(\mathbb{D}) : f(0) = 0, f'(0) = 1, \text{ and } f \text{ is univalent}\}$$

Example 7.5.1 Define the *Koebe function* as

$$\begin{aligned}
 k(z) &= \frac{z}{(1-z)^2} \\
 &= \frac{1}{4} \left(\frac{1+z}{1-z} \right)^2 - \frac{1}{4} \\
 &= z \frac{d}{dz} \frac{1}{1-z} \\
 &= \sum_{n=1}^{\infty} n z^n .
 \end{aligned}$$

Because

$$z \mapsto i \frac{1+z}{1-z}$$

is a biholomorphic map from the unit disk to the upper half plane,

$$z \mapsto \left(\frac{1+z}{1-z} \right)^2$$

maps the unit disk biholomorphically to $\mathbb{C} - \mathbb{R}^{\leq 0}$ so that k maps the unit disk biholomorphically to $\mathbb{C} - \mathbb{R}^{\leq -\frac{1}{4}}$.

Theorem 7.5.2 *Let $f \in \mathcal{S}$. Then there is a unique odd function $g \in \mathcal{S}$ such that*

$$g(z)^2 = f(z^2) .$$

Proof. The function $f(z)^2$ has a double order 0 at 0 and does not vanish elsewhere in $\mathbb{D}$. Thus we can write $f(z)^2 = z^2 h(z)$ with $h(0) = 1$ and $h(z) \neq 0$ in all of $\mathbb{D}$. Thus h has a square root in $\mathbb{D}$, whose sign we choose so that $\sqrt{h(0)} = +1$. Then we define $g(z) = z\sqrt{h(z)}$. Next, we show that g is odd. As $g(z)^2 = g(-z)^2$, we either have $g(z) = g(-z)$ or $g(z) = -g(-z)$ throughout $\mathbb{D}$ by the **Identity Principle 2.5.5**. As $g'(0) = 1$, g must be odd. It remains to show that g is univalent. Suppose $g(z_1) = g(z_2)$. Then $f(z_1^2) = f(z_2^2)$. As f is univalent, $z_1^2 = z_2^2$. If $z_1 = -z_2$, then $-g(z_1) = g(-z_1) = g(z_1)$, so that $g(z_1) = 0$. As g is only 0 at 0, $z_1 = 0 = z_2$. Thus $z_1 = z_2$, and g is univalent. 

Definition 7.5.3 *Let $\Delta = \mathbb{C} - \overline{\mathbb{D}}$ be the complement of the closed unit disk. We define*

$$\begin{aligned}
 \Sigma = \{f \in O(\mathbb{D}) : f \text{ is univalent and } f(1/z) \text{ has a simple pole at } 0 \\
 \text{with residue } 1\}
 \end{aligned}$$

Example 7.5.2 Let $f \in \mathcal{S}$. Then

$$g(\zeta) = \frac{1}{f(1/\zeta)}$$

is in Σ . We denote the subset of Σ that arises this way by Σ' . Note that functions in Σ' do never vanish.

An example of particular importance is the function $1/k(1/z)$, where $k(z)$ is the Koebe function. It maps the complement of the closed unit disk biholomorphically onto the complex plane with the slit $[-4, 1]$ removed.

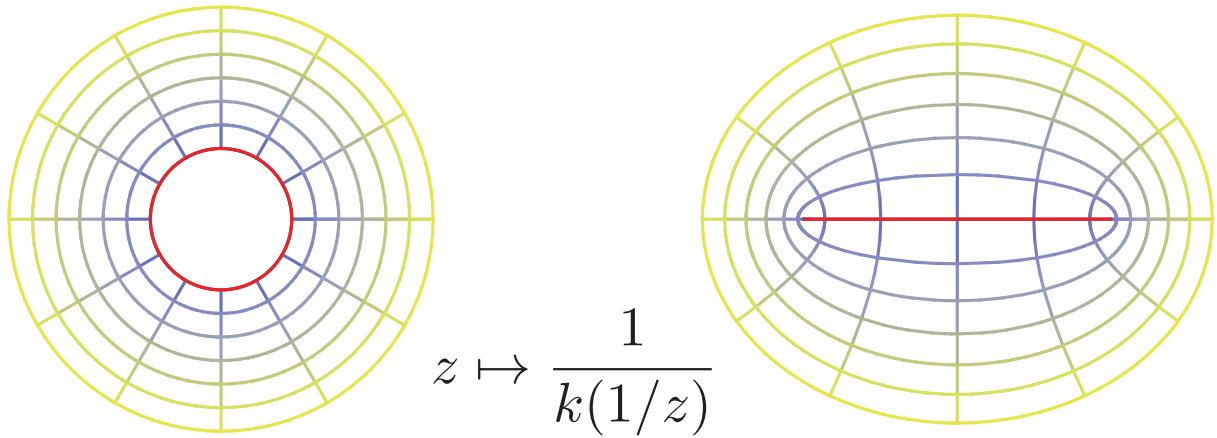


Figure 7.5 Squeezing a disk to a slit

Functions in Σ have a Laurent expansion

$$g(\zeta) = \zeta + \sum_{k=0}^{\infty} b_k \zeta^{-k}.$$

Theorem 7.5.4 (Grönwall) *Let $g \in \Sigma$. Then*

$$\sum_{n=1}^{\infty} n |b_n|^2 \leq 1.$$

Proof. For $r > 1$, let $C_r = g(|z| = r)$. As g is univalent, C_r is a real analytic simple closed curve that bounds the open set

$$A_r = \mathbb{C} - g(|z| \geq r).$$

We compute the area of A_r , using Stokes' theorem and the transformation formula.

$$\begin{aligned} \text{area } A_r &= \int_{A_r} dx dy \\ &= \frac{i}{2} \int_{A_r} dw \overline{dw} \\ &= \frac{i}{2} \int_{A_r} d(w \overline{dw}) \\ &= \frac{i}{2} \int_{C_r} w \overline{dw} \\ &= \frac{i}{2} \int_{|z|=r} g(\zeta) \overline{g'(\zeta)} \overline{d\zeta}. \end{aligned}$$

The integral on the right is now computed using the Laurent series of g : We have

$$g'(\zeta) = 1 - \sum_{n=1}^{\infty} n b_n \zeta^{-n-1}.$$

Using $\zeta = re^{it}$, we obtain

$$\text{area } A_r = \frac{i}{2} \int_0^{2\pi} \left(re^{it} + \sum_{m=0}^{\infty} b_m r^{-m} e^{-imt} \right) \cdot \left(1 - \sum_{n=1}^{\infty} \overline{b_n} n r^{-n-1} e^{i(n+1)t} \right) \cdot (-ire^{-it}) dt$$

Note that the integrals $\int_0^{2\pi} e^{kit} dt$ vanish for all k except when $k = 0$. Thus

$$\text{area } A_r = \pi \left(r^2 - \sum_{n=1}^{\infty} n |b_n|^2 r^{-2n} \right).$$

As $\text{area } A_r \geq 0$ for all $r > 1$, we obtain

$$\sum_{n=1}^N n |b_n|^2 r^{-2n} \leq r^2$$

for all N and all $r > 1$. Letting $r \rightarrow 1$ for fixed N and then $N \rightarrow \infty$ proves the claim. 

Corollary 7.5.5 For $g \in \Sigma$ we have $|b_1| \leq 1$. If $|b_1| = 1$, then

$$g(\zeta) = \zeta + b_0 + \frac{b_1}{\zeta}.$$

Proof. This follows immediately from **Grönwall's Theorem 7.5.4**. 

As an application we prove about functions in $\mathcal{S}$:

Theorem 7.5.6 (Bieberbach) Let

$$f(z) = z + \sum_{n=2}^{\infty} a_n z^n$$

univalent. Then $|a_2| \leq 2$, with equality if and only if f is a rotation of the Koebe function:

$$f(z) = e^{-it} k(e^{it} z).$$

Proof. Let

$$h(z) = \sqrt{f(z^2)} = z + \sum_{n=1}^{\infty} c_{2n+1} z^{2n+1}.$$

Then we have

$$2c_3 = a_2.$$

Consider

$$g(\zeta) = \frac{1}{h(1/\zeta)} = \zeta + \sum_{n=1}^{\infty} b_n \zeta^{-n}.$$

Comparing Laurent series shows $b_1 = -\frac{1}{2}a_2$. By the corollary, we obtain $|a_2| \leq 2$. In case of equality, we get

$$g(\zeta) = \zeta + \frac{b_1}{\zeta}$$

so that

$$h(z) = \frac{z}{1 + b_1 z^2}$$

and therefore

$$f(z^2) = h(z)^2 = \frac{z^2}{(1 + b_1 z^2)^2}$$

so that

$$f(z) = \frac{z}{(1 + b_1 z)^2}$$

If we write $b_1 = -e^{it}$, the claim follows. 

In 1916, Bieberbach went on to conjecture that $|a_n| \leq n$ for all n . This was the (infamous) Bieberbach conjecture.

- 1916: $|a_2| \leq 2$ (Bieberbach)
- 1923: $|a_3| \leq 3$ (Löwner)
- 1955: $|a_4| \leq 4$ (Garabedian and Schiffer)
- 1969: $|a_6| \leq 6$ (Ozawa)
- 1972: $|a_5| \leq 5$ (Pederson)

Then, in 1985, De Branges proved the general case. This was politically charged: De Branges had earlier attempted to prove the Bieberbach conjecture, but his proof was found to be incorrect. Nobody in the West was willing to read his new complicated manuscript. So at the height of the Cold War, he went to (then) Leningrad and discussed the proof in detail with Russian experts. They believed him. Only then were experts in the West willing to follow his arguments.

Corollary 7.5.7 (1/4-Theorem) *Let $f \in \mathcal{S}$. Then $f(\mathbb{D})$ contains the open disk $D_{1/4}(0)$. This is sharp, as the Koebe function shows.*

Proof. Let $b \notin f(\mathbb{D})$, and write

$$f(z) = z + \sum_{n=2}^{\infty} a_n z^n.$$

Then $|a_2| \leq 2$. Now let

$$g(z) = \frac{f(z)}{1 - \frac{1}{b}f(z)} = z + \left(a_2 + \frac{1}{b}\right)z^2 + \dots \in \mathcal{S}$$

Therefore

$$\left|a_2 + \frac{1}{b}\right| \leq 2,$$

as well, and by the triangle inequality

$$2 \geq \frac{1}{|b|} - |a_2| \geq \frac{1}{|b|} - 2,$$

which implies $|b| \geq \frac{1}{4}$. 

As an application, we prove Koebe's

Theorem 7.5.8 (Distortion Theorem) *Let $f \in \mathcal{S}$. Then for $z \in \mathbb{D}$,*

$$\frac{1 - |z|}{(1 + |z|)^3} \leq |f'(z)| \leq \frac{1 + |z|}{(1 - |z|)^3}$$

$$\frac{|z|}{(1 + |z|)^2} \leq |f(z)| \leq \frac{|z|}{(1 - |z|)^2}$$

Proof. We begin by transferring the information $|a_2| \leq 2$ from **Bieberbach's Theorem 7.5.6** to other points of $\mathbb{D}$. For $w \in \mathbb{D}$, the function

$$f\left(\frac{z+w}{1+\bar{w}z}\right) = f(w) + (1 - |w|^2) f'(w) z + \frac{1}{2} ((1 - |w|^2)^2 f''(w) - 2\bar{w}(1 - |w|^2) f'(w)) z^2 + \dots$$

is holomorphic in $\mathbb{D}$ and univalent. Its normalization

$$h(z) = \frac{f\left(\frac{z+w}{1+\bar{w}z}\right) - f(w)}{(1 - |w|^2) f'(w)}$$

$$= w + \left(\frac{1}{2}(1 - |w|^2) \frac{f''(w)}{f'(w)} - \bar{w}\right) z^2 + \dots$$

is therefore also in $\mathcal{S}$. By **Bieberbach's Theorem 7.5.6**, the second coefficient has absolute value ≤ 2 . This implies (writing z instead of w)

$$\left| z \frac{f''(z)}{f'(z)} - \frac{2|z|^2}{1 - |z|^2} \right| \leq \frac{4|z|}{1 - |z|^2}.$$

This shows that

$$\left| \frac{\partial}{\partial r} \log((1 - r^2) f'(re^{it})) \right| = \left| -\frac{2r}{1 - r^2} + e^{it} \frac{f''(re^{it})}{f'(re^{it})} \right| \leq \frac{4}{1 - r^2}.$$

Integrating and using that $f'(0) = 1$ gives

$$|\log((1 - |z|^2) f'(z))| \leq \int_0^{|z|} \left| \frac{\partial}{\partial r} \log((1 - r^2) f'(re^{it})) \right| dr \leq 2 \log \frac{1 + |z|}{1 - |z|}.$$

Taking the real part on the left shows

$$2 \log \frac{1 - |z|}{1 + |z|} \leq \log((1 - |z|^2) |f'(z)|) \leq 2 \log \frac{1 + |z|}{1 - |z|}.$$

This is equivalent to the first inequality. We now show the upper estimate in the second inequality. Using that $f(0) = 0$ and the upper estimate from the first inequality,

$$|f(z)| \leq \int_0^{|z|} |f'(re^{it})| dr \leq \int_0^{|z|} \frac{1 + r}{(1 - r)^3} dr = \frac{|z|}{(1 - |z|)^2}.$$

To prove the lower estimate, fix $0 < r < 1$, and let z_0 be a point on the circle $|z| = r_0$ such that $f(z_0)$ is closest to 0. Then, as f is univalent, the preimage γ of the segment from 0 to $f(z_0)$ is a simple analytic arc from 0 to z_0 contained in $|z| \leq r$. We conclude

$$\begin{aligned}
|f(z_0)| &= \int_{f(\gamma)} |dw| \\
&= \int_{\gamma} |f'(z)| |dz| \\
&\geq \int_{\gamma} \frac{1-|z|}{(1+|z|)^3} |dz| \\
&\geq \int_{\gamma} \frac{1-|z|}{(1+|z|)^3} d|z| \\
&= \frac{r}{(1+r)^2}.
\end{aligned}$$

An important consequence of the distortion theorem is that univalent functions in the unit disk are locally uniformly bounded. We will make use of this in the next section.

7.6 CARATHÉODORY KERNEL CONVERGENCE

Definition 7.6.1 Let F_n be a sequence of domains in $\mathbb{C}$ with $0 \in F_n$. The kernel of the sequence F_n is the set F consisting of the point 0 and all points z with the property such that there is a domain U containing 0 and z that is contained in all F_n for sufficiently large n :

$$\{0, z\} \subset U \subset F_n \quad \text{for } n \text{ large enough.}$$

If there are no such points, the kernel F is just the set $\{0\}$. Otherwise, each set U is contained in F , which is therefore a domain containing 0. We say that F_n converges to F if all subsequences of F_n have the same kernel.

Example 7.6.1 Let F_n be a decreasing sequence of domains, and let F° be the set of interior points of the intersection $\bigcup F_n$. Then F_n converges to the component F of F° that contains 0, and to $F = \{0\}$ if F° does not contain 0.

Proof. As taking a subsequence doesn't change F° , it suffices to show that the kernel of F_n is F . If $0 \in F^\circ$, then clearly F is contained in the kernel. Let z be a point in the kernel. Then there is a domain U containing 0 and z that is contained in all F_n by monotonicity. Therefore $U \subset F^\circ$. As U contains 0 and is connected, $U \subset F$. If $0 \notin F^\circ$, then no point $z \neq 0$ can be in the kernel, as it would then contain 0 as an interior point, and thus $0 \in F^\circ$. 

Example 7.6.2 Let

$$F_n = \mathbb{C} - \{1 + it : |t| \geq \frac{1}{n}\}$$

be the plane with two approaching vertical slits removed. Then F_n converges to $\text{Re } z < 1$.

The main reason to introduce kernel convergence is the following characterization of locally uniform convergence of univalent functions:

Theorem 7.6.2 (Carathéodory) Let $f_n \in O(\mathbb{D})$ be univalent and normalized so that $f_n(0) = 0$ and $f'_n(0) > 0$. Let $F_n = f_n(\mathbb{D})$. Then the sequence f_n converges locally uniformly if and only if the sequence of domains F_n converges to its kernel $F \neq \mathbb{C}$. In that case, F is the image domain of the limit function f .

Proof. First, suppose that $f_n \rightarrow f$ locally uniformly in $\mathbb{D}$, and let F be the kernel of F_n . We claim that $F = f(\mathbb{D})$. By **Hurwitz' Theorem 4.3.3**, f is either univalent or constant.

We show first that $f(\mathbb{D}) \subset F$. If f is identically 0, this is obvious. Hence, we can assume that f is injective. Let $f(z_0) = w_0 \in f(\mathbb{D})$. If $w_0 = 0$, then clearly also $w_0 \in F$, so let's assume $w_0 \neq 0$. Choose r such that $0 < |z_0| < r < 1$. Then the domain $U = f(|z| < r)$ contains w_0 and 0. We claim that $U \subset F_n$ for n sufficiently large. If this is false, then there are sequences $n_k \rightarrow \infty$ and $w_k \in U$ with $w_k \notin F_{n_k}$. After another subsequence, we can assume that $w_k \rightarrow w$ with $w \in \overline{U}$. As $f_{n_k}(z) \neq w_k$ for all $z \in \mathbb{D}$, we also have $f_{n_k}(z) - w_k + w \neq w$ for all $z \in \mathbb{D}$. By **Hurwitz' Theorem 4.3.3**, we have $f(z) \neq w$ in $\mathbb{D}$, as f is not constant. But this contradicts $w \in \overline{U} \subset f(\mathbb{D})$.

Now we show that $F \subset f(\mathbb{D})$. Let $0 \neq w_0 \in F$. Then there is an open set $U \subset F_n$ for large n that contains 0 and w_0 . Let $g_n : U \rightarrow \mathbb{D}$ the inverse of f_n . By **Montel's Theorem 5.1.6**, there is a subsequence g_{n_k} that converges in U locally uniformly to a limit g . This limit satisfies $g(0) = 0$ and $|g(w)| \leq 1$, and hence $|g(w)| < 1$ by the **Open Mapping Theorem 2.5.6**. Thus $f_{n_k}(z)$ converges uniformly in a neighborhood of $z_0 = g(w_0)$. As $g_{n_k}(w_0) \rightarrow g(w_0) = z_0$, we obtain $w_0 = f_{n_k}(g_{n_k}(w_0)) \rightarrow f(z_0)$ so that $w_0 \in f(\mathbb{D})$.

This shows that $F = f(\mathbb{D})$. As f is injective or constant, it follows also that $F \neq \mathbb{C}$.

Now suppose that F_n converges to its kernel $F \neq \mathbb{C}$. By Koebe's **One Quarter Theorem 7.5.7**,

$$\{w : |w| \leq \frac{1}{4} f'_n(0)\} \subset F_n.$$

If $f'_n(0)$ is unbounded, a subsequence of F_n will have $\mathbb{C}$ as its kernel, contradicting $F \neq \mathbb{C}$. Hence, we can assume that $f'_n(0) \leq C$. By the **Distortion Theorem 7.5.8**,

$$|f_n(z)| \leq C \frac{|z|}{(1 - |z|)^2}$$

for all $z \in \mathbb{D}$, so that f_n is locally uniformly bounded. Thus f_n has a locally uniform convergent subsequence. In order to show that f_n itself already converges locally uniformly, it suffices to show that any two locally uniform convergent subsequences have the same limit. Suppose that is not the case, and we have two such subsequences f_{m_k} and f_{n_k} with different limits f and g . By the first part of the theorem (which we have already shown above), the domains F_{m_k} and F_{n_k} converge to the images $f(\mathbb{D})$ and $g(\mathbb{D})$. By assumption, both domain sequences have the same kernel, so that $f(\mathbb{D}) = g(\mathbb{D})$. By the uniqueness of the Riemann mapping function, $f = g$. 

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